



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Ampere's Law

Ampere's circuit law states that the line integral of the tangential component around a closed path is the same as the net current I_{enc} enclosed by the path or the circulation of $\mathbf{H}$ equals I_{enc} is such that:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc}$$

Ampere's law is a special case of Biot-Savart's law; the former may be derived from the latter. By applying Stokes's theorem to the left-hand side of the above equation:

$$I_{enc} = \oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

but

$$I_{enc} = \int_S \mathbf{J} \cdot d\mathbf{S}$$

comparing the surface integrals reveals:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

This is the third Maxwell's equation to be derived; it is essentially Ampere's law in point form. Observe that"

$$\nabla \times \mathbf{H} = \mathbf{J} \neq 0$$

Hence, a magnetostatic field is not conservative. Ampere's circuit law can now applied to determine $\mathbf{H}$ for some symmetrical current distributions as Gauss's law. An infinite line current, an infinite current sheet, and an infinitely long coaxial transmission line are considered.

Infinite Line Current

Consider an infinitely long filamentary current I along the z -axis as in Fig. (1) to determine $\mathbf{H}$ at an observation point P , we allow a closed path to pass through P .

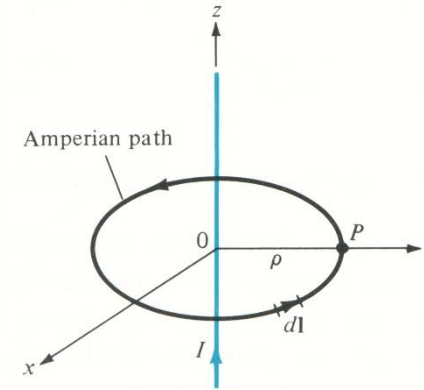


Fig. (1): Ampere's law applied to an infinite line current.

This path, on which Ampere's law is to be applied, is known as an Amperian path (analogous to Gaussian surface). A concentric circle is chosen as the Amperian path in which $\mathbf{H}$ is constant provided ρ is constant. Since this path encloses the whole current I according to Ampere's law:

$$I = \int \mathbf{H}_\phi \cdot \mathbf{a}_\phi \cdot \rho d\phi \mathbf{a}_\phi = H_\phi \int \rho d\phi = H_\phi \cdot 2\pi\rho$$

or

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

Infinitely Long Coaxial Transmission Line

Consider an infinitely long transmission line consisting of two concentric cylinders with axes along the z -axis. The cross-section of the line is shown in Fig. (2), where the z -axis is out of the page.

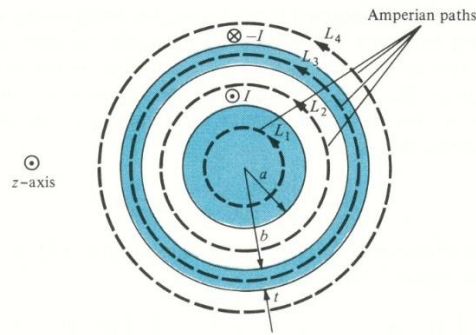


Fig. (2): Cross-section of the transmission line; the positive z direction is out of the page.

The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness t and carries return current I . Now, $\mathbf{H}$ may be determined everywhere assuming that current is uniformly distributed in both conductors. Since the current distribution is symmetrical, Ampere's law is applied along the Amperian path for each of the four possible regions: $0 \leq \rho \leq a$, $a \leq \rho \leq b$, $b \leq \rho \leq b+t$, and $\rho \geq b+t$. For region $0 \leq \rho \leq a$, Ampere's law is applied to path L_1 , giving:

$$\oint_{L_1} \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{S}$$

Since the current is uniformly distributed over the cross section,

$$\mathbf{J} = \frac{I}{\pi a^2} \mathbf{a}_z$$

and because:

$$d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$$

then:

$$I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{S} = \frac{I}{\pi a^2} \iint \rho d\phi d\rho = \frac{I}{\pi a^2} \pi \rho^2 = \frac{I \rho^2}{a^2}$$

hence

$$H_\phi \int dl = H_\phi 2\pi \rho = \frac{I \rho^2}{a^2}$$

or

$$H_\phi = \frac{I \rho}{2\pi a^2}$$

For region $a \leq \rho \leq b$, we use path L_2 as the Amperian path:

$$\oint_{L_2} \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = I$$

$$H_\phi 2\pi \rho = I$$

or

$$H_{\phi} = \frac{I}{2\pi\rho}$$

Since the whole current, I is enclosed by L_2 . For region $b \leq \rho \leq b+t$, we use path L_3 :

$$\oint \mathbf{H} \cdot d\mathbf{l} = H_{\phi} \cdot 2\pi\phi = I_{\text{enc}}$$

where

$$I_{\text{enc}} = I + \int \mathbf{J} \cdot d\mathbf{S}$$

and $\mathbf{J}$ in this case is the current density (current per unit area) of the outer conductor and is along $-\mathbf{a}_z$:

$$\mathbf{J} = -\frac{I}{\pi[(b+t)^2 - b^2]} \mathbf{a}_z$$

thus

$$I_{\text{enc}} = I - \frac{I}{\pi[(b+t)^2 - b^2]} \mathbf{a}_z \int_{\phi=0}^{2\pi} \int_{\rho=b}^{\rho} \rho d\rho d\phi$$

$$I_{\text{enc}} = I \left[1 - \frac{\rho^2 - b^2}{t^2 + 2bt} \right]$$

substituting:

$$H_{\phi} = \frac{I}{2\pi\rho} \left[1 - \frac{\rho^2 - b^2}{t^2 + 2bt} \right]$$

For region $\rho \geq b+t$, along path L_4 getting:

$$\oint_{L_4} \mathbf{H} \cdot d\mathbf{l} = I - I = 0$$

or

$$H_{\phi} = 0$$

putting the equations together gives:

$$\mathbf{H} = \begin{cases} \frac{I \rho}{2 \pi a^2} \mathbf{a}_\phi & 0 \leq \rho \leq a \\ \frac{I}{2 \pi \rho} \mathbf{a}_\phi & a \leq \rho \leq b \\ \frac{I}{2 \pi \rho} \left[1 - \frac{\rho^2 - b^2}{t^2 + 2 b t} \right] \mathbf{a}_\phi & b \leq \rho \leq b + t \\ 0 & \rho \geq b + t \end{cases}$$

The magnitude of $\mathbf{H}$ is sketched in Fig. (3).

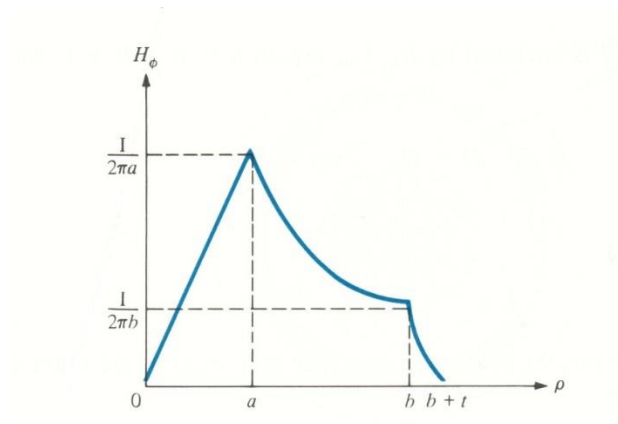


Fig. (3): Plot of $\mathbf{H}$ against $\mathbf{a}_\phi$.

Notice from these examples that the ability to take $\mathbf{H}$ from under the integral sign is the key to using Ampere's law to determine $\mathbf{H}$. In other words; Ampere's law can only be used to find $\mathbf{H}$ due to symmetric current distributions for which it is possible to find a closed path over which $\mathbf{H}$ is constant in magnitude.

Example

A toroid with a circular cross section is as shown in Fig. (4) has N turn and carries a current I . Determine $\mathbf{H}$ inside and outside the toroid?

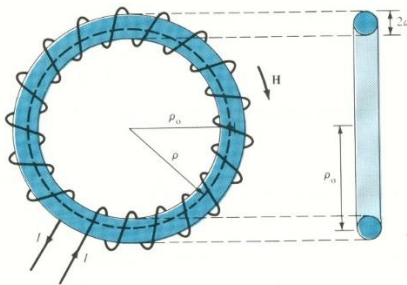


Fig. (4): A toroid with a circular cross section.

Apply Ampere's law with an Amperian path as a circle of radius ρ as shown in Fig. (4). Since N wires cut through this path each carrying current I , the net current enclosed by the Amperian path is $N I$, Hence:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc} \rightarrow H \cdot 2\pi\rho$$

or

$$\mathbf{H} = \frac{N I}{2\pi\rho} \quad \text{for } \rho_0 - a < \rho < \rho_0 + a$$

where ρ_0 is the mean radius of the toroid. An approximate value for $\mathbf{H}$ is:

$$\mathbf{H} = \frac{N I}{2\pi\rho_0} = \frac{N I}{l}$$

For the region outside the toroid, $\mathbf{H}$ is zero.



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Electromagnetics I
Boundary Conditions

If the field exists in a region of two different media, the conditions the field must satisfy at the interface separating the media are called *boundary conditions*. These conditions help determine the field on one side of the boundary if the field on the other side is known. The types of material will dictate the conditions the media are made of. We shall consider the boundary conditions at an interface separating:

- Dielectric ϵ_{r1} and dielectric ϵ_{r2} .
- Conductor and dielectric.
- Conductor and free space.

To determine the boundary conditions, we need to use Maxwell's equations:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

and

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q$$

Also, we need to decompose the electric field intensity $\mathbf{E}$ into two components:

$$\mathbf{E} = \mathbf{E}_t + \mathbf{E}_n$$

where $\mathbf{E}_t$ and $\mathbf{E}_n$ are, respectively, the tangential and normal components of $\mathbf{E}$ to the interface of interest. A similar decomposition can be done for the electric flux density $\mathbf{D}$.

Dielectric-Dielectric Boundary Conditions

Consider the $\mathbf{E}$ field in a region of two different dielectrics characterized by $\epsilon_1 = \epsilon_o \epsilon_{r1}$ and $\epsilon_2 = \epsilon_o \epsilon_{r2}$ as shown in Fig. (1). $\mathbf{E}_1$ and $\mathbf{E}_2$ in the media 1 and 2, respectively, can be decomposed as:

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}$$

Apply to the closed path $abcd$ assuming that the path is very small with respect to the variation of $\mathbf{E}$, then:

$$E_{1t}\Delta w - E_{1n}\frac{\Delta h}{2} - E_{2n}\frac{\Delta h}{2} - E_{2t}\Delta w + E_{2n}\frac{\Delta h}{2} + E_{1n}\frac{\Delta h}{2} = 0$$

where

$$E_t = |\mathbf{E}_t|$$

and

$$E_n = |\mathbf{E}_n|$$

As Δh goes to zero:

$$E_{1t} = E_{2t}$$

Thus, the tangential components of $\mathbf{E}$ are the same on the two sides of the boundary. In other words, $\mathbf{E}_t$ undergoes no change on the boundary, and it is said to be continuous across the boundary. Since

$$\mathbf{D} = \epsilon \mathbf{E} = \mathbf{D}_t + \mathbf{D}_n$$

which can be written as

$$\frac{D_{1t}}{\epsilon_1} = E_{1t} = E_{2t} = \frac{D_{2t}}{\epsilon_2}$$

or

$$\frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

$\mathbf{D}_t$ undergoes some change across the interface. Hence $\mathbf{D}_t$ is said to be *discontinuous* across the interface.

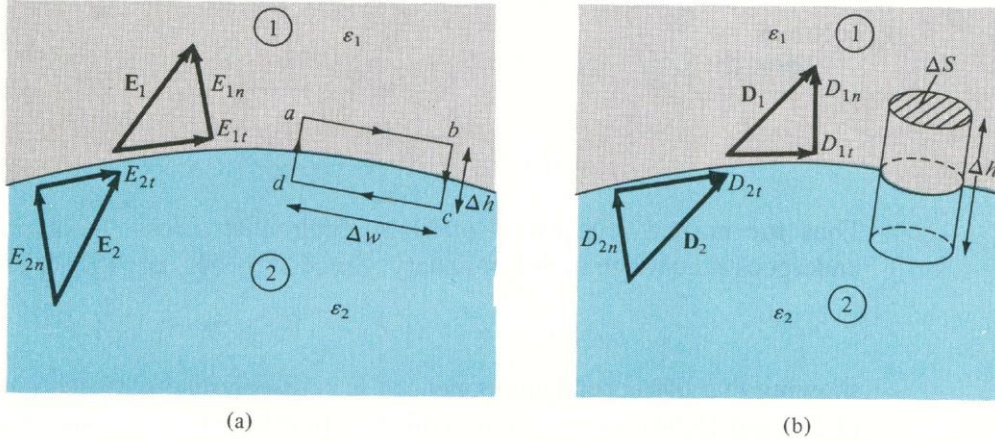


Fig. (1): Dielectric-dielectric boundary.

Similarly, consider the pillbox (Gaussian surface), allowing Δh to go zero gives:

$$\Delta Q = \rho_s \Delta S = D_{1n} \Delta S - D_{2n} \Delta S$$

or

$$D_{1n} - D_{2n} = \rho_s$$

where ρ_s is the free density placed deliberately at the boundary. It should be borne in mind that the last equation is based on the assumption that $\mathbf{D}$ is directed from region 2 to region 1. If no free charges at the interface, $\rho_s = 0$. Thus, the normal component of $\mathbf{D}$ is continuous across the interface, and $\mathbf{D}_n$ undergoes no change at the boundary, and the normal component of $\mathbf{E}$ is discontinuous at the boundary. The boundary conditions are usually applied in finding the electric field on one side of the boundary given the field on the other side, as illustrated in Fig. (2):

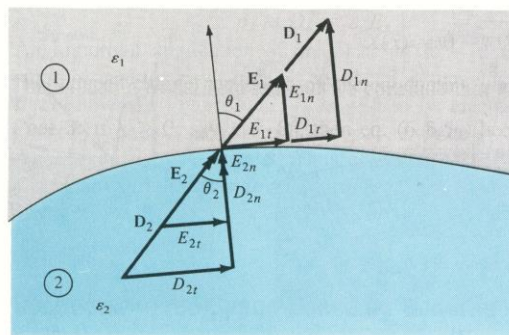


Fig. (2): Refraction of $\mathbf{D}$ or $\mathbf{E}$ at the dielectric-dielectric boundary.

$$E_1 \sin \theta_1 = E_{1t} = E_{2t} = E_2 \sin \theta_2$$

or

$$E_1 \sin \theta_1 = E_2 \sin \theta_2$$

Similarly:

$$\varepsilon_1 E_1 \cos \theta_1 = D_{1n} = D_{2n} = \varepsilon_2 E_2 \cos \theta_2$$

or

$$\varepsilon_1 E_1 \cos \theta_1 = \varepsilon_2 E_2 \cos \theta_2$$

dividing gives:

$$\frac{\tan \theta_1}{\varepsilon_1} = \frac{\tan \theta_2}{\varepsilon_2}$$

since

$$\varepsilon_1 = \varepsilon_0 \varepsilon_{r1}$$

and

$$\varepsilon_2 = \varepsilon_0 \varepsilon_{r2}$$

hence

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$$

This is the law of refraction of the electric field at a boundary-free charge. Thus, in general, an interface between two dielectrics produces bending of the flux lines due to unequal polarization charges that accumulate on the sides of the interface.

Conductor- Dielectric Boundary Conditions

If conductors such as copper and silver are regarded as though they were perfect conductors, then for a conductor-dielectric interface, consider the closed path *abcd* in Fig. (3).

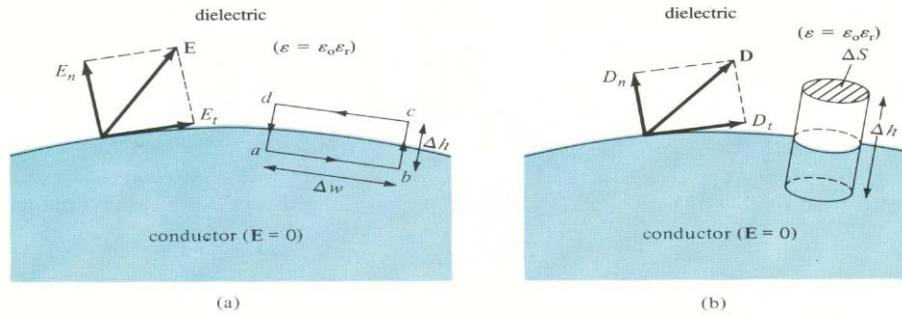


Fig. (3): Conductor –dielectric boundary.

$$0 = 0 \cdot \Delta w + 0 \cdot \Delta h / 2 + E_n \cdot \Delta h / 2 - E_t \cdot \Delta w - E_n \cdot \Delta h / 2 - 0 \cdot \Delta h / 2$$

$$E_t = 0$$

Similarly, the pillbox gives:

$$\Delta h \rightarrow 0$$

we get

$$\Delta Q = D_n \cdot \Delta S - 0 \cdot \Delta S$$

because

$$\mathbf{D} = \epsilon \mathbf{E}$$

equals zero inside the conductor. Thus:

$$D_n = \Delta Q / \Delta S = \rho_s$$

or

$$D_n = \rho_s$$

Thus, the following conclusions can be made about a perfect conductor under static conditions. First, no electric field may exist within a conductor. That is:

$$\rho_s = 0$$

$$\mathbf{E} = 0$$

since

$$\mathbf{E} = -\nabla V = 0$$

there can be no potential differences between any two points in the conductor; a conductor is an equipotential body. Second, the electric field $\mathbf{E}$ can be external to the conductor and normal to its surface; that is

$$D_t = \varepsilon_0 \varepsilon_r E_t = 0$$

$$D_n = \varepsilon_0 \varepsilon_r E_n = \rho_S$$

Conductor–Free Space Boundary Conditions

The boundary conditions at the interface between a conductor and free space can be obtained by replacing ε_r by 1. The electric field $\mathbf{E}$ external to the conductor is expected to be normal to its surface. Thus the boundary conditions are:

$$D_t = \varepsilon_0 E_t = 0$$

$$D_n = \varepsilon_0 E_n = \rho_S$$



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Coordinate Systems

Physical quantities are functions of space and time, and hence their spatial variations must be described by defining all points uniquely in a suitable manner. This requires using an appropriate coordinate system, mainly Cartesian (x, y, z) , polar (cylindrical) (ρ, ϕ, z) and spherical (r, θ, ϕ) . The coordinates are mutually perpendicular in these orthogonal systems. In Cartesian coordinates, a vector $\mathbf{A}$ can be represented as in Fig. (1).

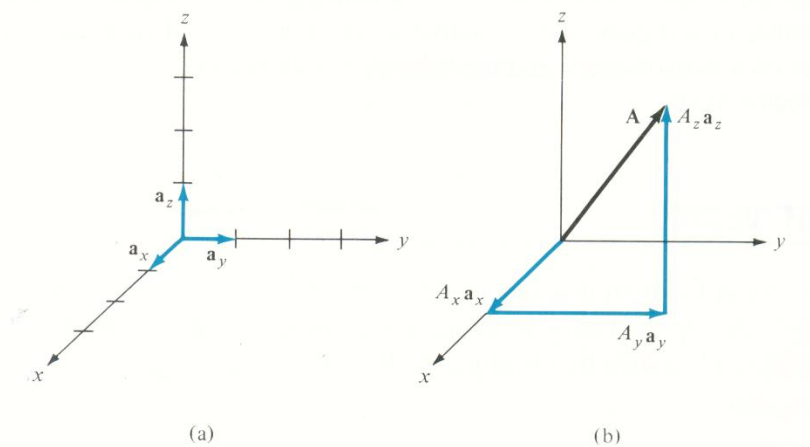


Fig. (1): Vector $\mathbf{A}$ in representation in Cartesian coordinates.

The circular cylindrical coordinate system is very convenient whenever there is a cylindrical symmetry. A point P in cylindrical coordinate system is represented as (ρ, ϕ, z) and is as shown in Fig. (2). Here ρ is the radius of the cylinder and ϕ is azimuthal angle, and:

$$0 \leq \rho \leq \infty$$

$$0 \leq \phi \leq 2\pi$$

$$-\infty \leq z \leq \infty$$

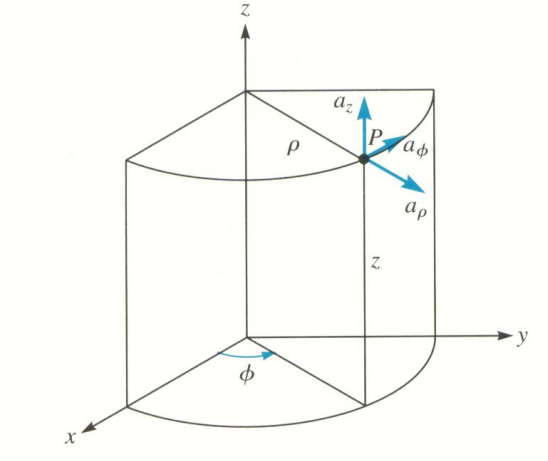


Fig. (2): Point P and unit vectors in the cylindrical coordinate system.

A vector $\mathbf{A}$ in cylindrical coordinates can be written as:

$$\mathbf{A} = A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$

with a magnitude:

$$|\mathbf{A}| = \sqrt{A_\rho^2 + A_\phi^2 + A_z^2}$$

The following also holds:

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$$

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$$

The relationships between the variables (x, y, z) and (ρ, ϕ, z) are hence:

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \frac{y}{x}$$

$$z = z$$

or

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

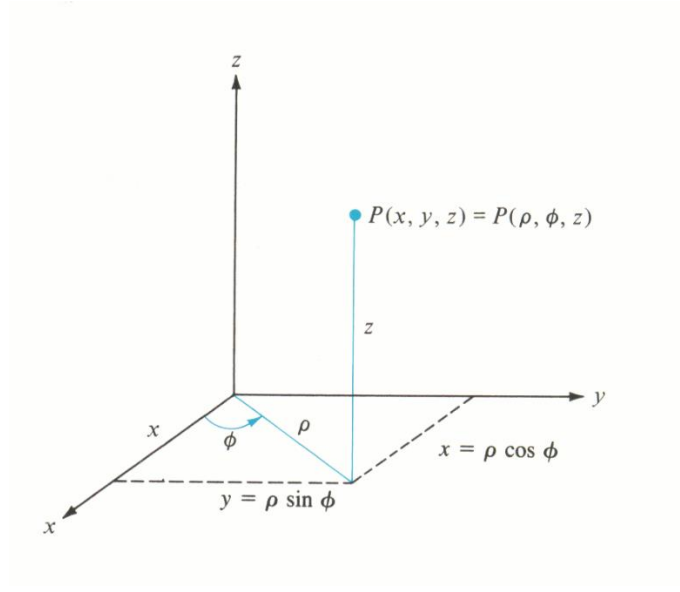


Fig. (3:) Relationship between (x, y, z) and (ρ, ϕ, z) .

The relationships between $(\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z)$ and $(\mathbf{a}_\rho, \mathbf{a}_\phi, \mathbf{a}_z)$ are obtained geometrically from Fig. (4):

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \mathbf{a}_z$$

or

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\mathbf{a}_z = \mathbf{a}_z$$

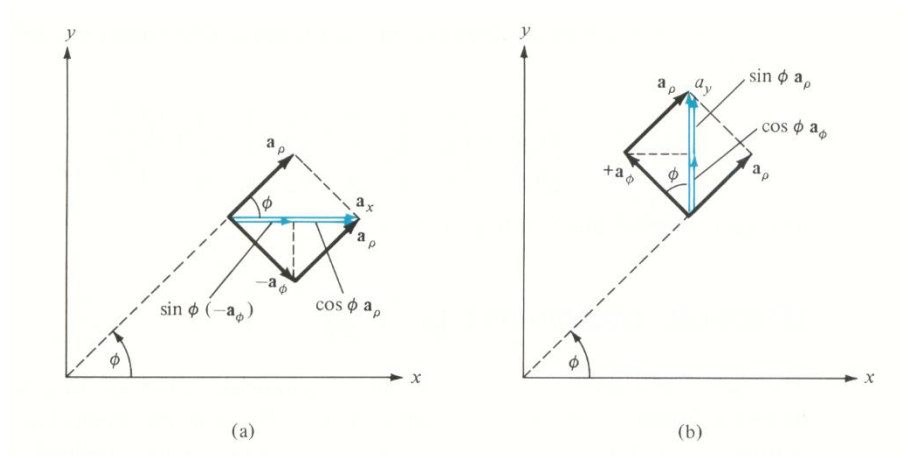


Fig. (4:) Unit vector transformation for $\mathbf{a}_x$ and $\mathbf{a}_y$ in cylindrical coordinates.

The vector relationships between (A_x, A_y, A_z) and (A_ρ, A_ϕ, A_z) are obtained by simple substitution:

$$\mathbf{A} = (A_x \cos \phi + A_y \sin \phi) \mathbf{a}_\rho + (-A_x \sin \phi + A_y \cos \phi) \mathbf{a}_\phi + A_z \mathbf{a}_z$$

$$A_\rho = A_x \cos \phi + A_y \sin \phi$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_z$$

In matrix form, the transformation of $\mathbf{A}$ from $(A_x, A_y, A_z) \rightarrow (A_\rho, A_\phi, A_z)$ is thus:

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

The inverse of the transformation $(A_\rho, A_\phi, A_z) \rightarrow (A_x, A_y, A_z)$ is obtained as:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

or directly

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

An alternative way is to use the dot product:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \mathbf{a}_x \cdot \mathbf{a}_\rho & \mathbf{a}_x \cdot \mathbf{a}_\phi & \mathbf{a}_x \cdot \mathbf{a}_z \\ \mathbf{a}_y \cdot \mathbf{a}_\rho & \mathbf{a}_y \cdot \mathbf{a}_\phi & \mathbf{a}_y \cdot \mathbf{a}_z \\ \mathbf{a}_z \cdot \mathbf{a}_\rho & \mathbf{a}_z \cdot \mathbf{a}_\phi & \mathbf{a}_z \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

In spherical coordinates (r, θ, ϕ) , r is the radius of the sphere as shown in Fig.(5), and θ is the colatitude between the z-axis and point P , angle, and ϕ is measured from x-axis which makes the same as in cylindrical coordinates, hence:

$$0 \leq r \leq \infty$$

$$0 \leq \theta \leq \pi$$

$$0 \leq \phi \leq 2\pi$$

Consequently, $\mathbf{A}$ can be written as

$$\mathbf{A} = A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$$

with a magnitude:

$$|\mathbf{A}| = \sqrt{A_r^2 + A_\theta^2 + A_\phi^2}$$

The following also holds:

$$\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

$$\mathbf{a}_r \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0$$

$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$

$$\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$$

$$\mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$

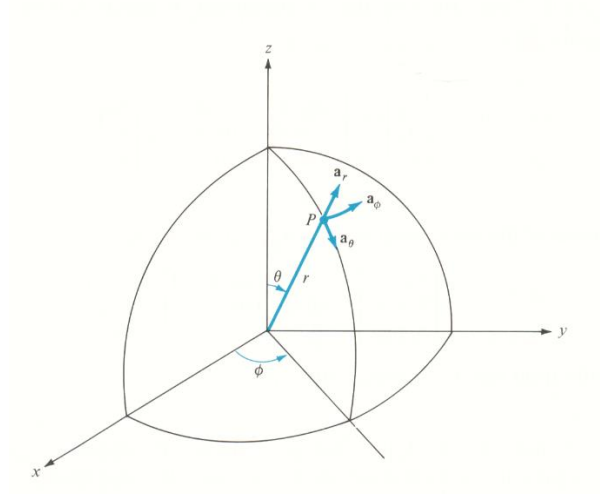


Fig. (5): Point P and unit vectors in the spherical coordinates.

The relationships between the space variables in Cartesian coordinates (x, y, z) can be related to (r, θ, ϕ) :

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}$$

$$\phi = \tan^{-1} \frac{y}{x}$$

or

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

The unit vectors are also related, thus:

$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

or

$$\mathbf{a}_r = \sin\theta \cos\phi \mathbf{a}_x + \sin\theta \sin\phi \mathbf{a}_y + \cos\theta \mathbf{a}_z$$

$$\mathbf{a}_\theta = \cos\theta \cos\phi \mathbf{a}_x + \cos\theta \sin\phi \mathbf{a}_y - \sin\theta \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\sin\phi \mathbf{a}_x + \cos\phi \mathbf{a}_y$$

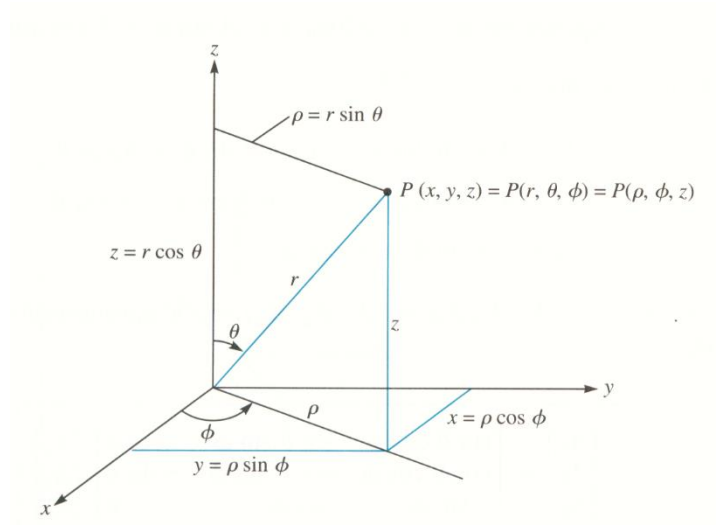


Fig. (6): Relationships between space variables (x, y, z) , (ρ, ϕ, z) and (r, θ, ϕ) .

The vector relationships between (A_x, A_y, A_z) and (A_r, A_θ, A_ϕ) are obtained by simple substitution:

$$\mathbf{A} = (A_x \sin\theta \cos\phi + A_y \sin\theta \sin\phi + A_z \cos\theta) \mathbf{a}_r + (A_x \cos\theta \cos\phi + A_y \cos\theta \sin\phi - A_z \sin\theta) \mathbf{a}_\theta + (-A_x \sin\phi + A_y \cos\phi) \mathbf{a}_\phi$$

thus:

$$A_r = A_x \sin\theta \cos\phi + A_y \sin\theta \sin\phi + A_z \cos\theta$$

$$A_\theta = A_x \cos\theta \cos\phi + A_y \cos\theta \sin\phi - A_z \sin\theta$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

In matrix form, the transformation of $\mathbf{A}$ from $(A_x, A_y, A_z) \rightarrow (A_r, A_\theta, A_\phi)$ is thus:

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ -\cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

The inverse of the transformation $(A_r, A_\theta, A_\phi) \rightarrow (A_x, A_y, A_z)$ is obtained as

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

An alternative way is to use the dot product:

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \mathbf{a}_r \cdot \mathbf{a}_x & \mathbf{a}_r \cdot \mathbf{a}_y & \mathbf{a}_r \cdot \mathbf{a}_z \\ \mathbf{a}_\theta \cdot \mathbf{a}_x & \mathbf{a}_\theta \cdot \mathbf{a}_y & \mathbf{a}_\theta \cdot \mathbf{a}_z \\ \mathbf{a}_\phi \cdot \mathbf{a}_x & \mathbf{a}_\phi \cdot \mathbf{a}_y & \mathbf{a}_\phi \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Constant-coordinate surfaces are generated by keeping one of the coordinate variables constant and allowing the other two to vary. In the Cartesian system, if x is kept constant and y and z are varied, then an infinite plane is generated such that:

$$x = \text{constant}$$

$$y = \text{constant}$$

$$z = \text{constant}$$

The intersection of two planes is a line, such as the line RPQ in Fig. (7), and the intersection of three orthogonal infinite planes is a point P .

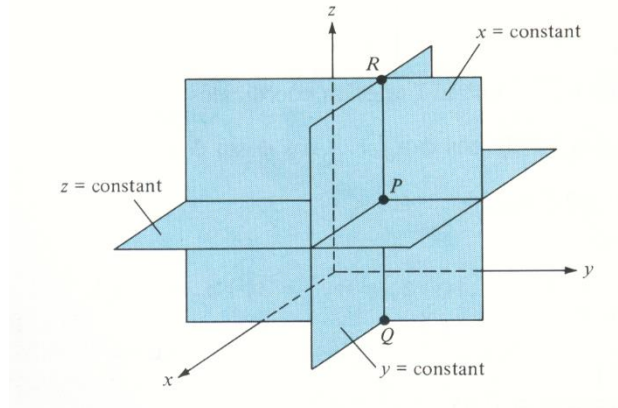


Fig. (7): Constant x , y , z surfaces.

Orthogonal surfaces in cylindrical coordinates are also generated as in Fig. (8), hence:

$$\rho = \text{constant}$$

$$\phi = \text{constant}$$

$$z = \text{constant}$$

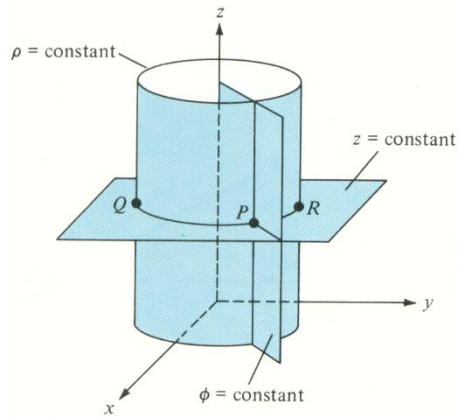


Fig. (8): Constant ρ , ϕ , z surfaces.

and in spherical coordinates as in Fig.(9), hence:

$$r = \text{constant}$$

$$\theta = \text{constant}$$

$$\phi = \text{constant}$$

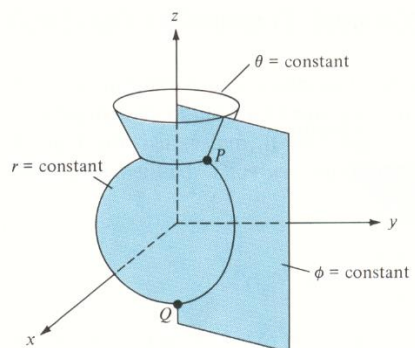


Fig. (9): Constant r , θ , ϕ surfaces.



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Coordinate Systems-Examples

Example 1

Given point $P(-2, 6, 3)$ and vector $\mathbf{A} = y\mathbf{a}_x + (x + z)\mathbf{a}_y$, express P and $\mathbf{A}$ in cylindrical and spherical coordinates. Evaluate $\mathbf{A}$ at P in the Cartesian, cylindrical, and spherical systems.

Solution

At point P : $x = -2, y = 6, z = 3$. Hence,

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 36} = 6.32$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{6}{-2} = 108.43^\circ$$

$$z = 3$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 36 + 9} = 7$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{\sqrt{40}}{3} = 64.62^\circ$$

Thus,

$$P(-2, 6, 3) = P(6.32, 108.43^\circ, 3) = P(7, 64.62^\circ, 108.43^\circ)$$

In the Cartesian system, $\mathbf{A}$ at P is

$$\mathbf{A} = 6\mathbf{a}_x + \mathbf{a}_y$$

For vector $\mathbf{A}$, $A_x = y, A_y = x + z, A_z = 0$. Hence, in the cylindrical system

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$\begin{aligned}
A_\rho &= y \cos \phi + (x + z) \sin \phi \\
A_\phi &= -y \sin \phi + (x + z) \cos \phi \\
A_z &= 0
\end{aligned}$$

But $x = \rho \cos \phi$, $y = \rho \sin \phi$, and substituting these yields

$$\begin{aligned}
\mathbf{A} = (A_\rho, A_\phi, A_z) &= [\rho \cos \phi \sin \phi + (\rho \cos \phi + z) \sin \phi] \mathbf{a}_\rho \\
&\quad + [-\rho \sin^2 \phi + (\rho \cos \phi + z) \cos \phi] \mathbf{a}_\phi
\end{aligned}$$

At P

$$\rho = \sqrt{40}, \quad \tan \phi = \frac{6}{-2}$$

Hence,

$$\begin{aligned}
\cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}} \\
\mathbf{A} &= \left[\sqrt{40} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\rho \\
&\quad + \left[-\sqrt{40} \cdot \frac{36}{40} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\
&= \frac{-6}{\sqrt{40}} \mathbf{a}_\rho - \frac{38}{\sqrt{40}} \mathbf{a}_\phi = -0.9487 \mathbf{a}_\rho - 6.008 \mathbf{a}_\phi
\end{aligned}$$

Similarly, in the spherical system

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$\begin{aligned}
A_r &= y \sin \theta \cos \phi + (x + z) \sin \theta \sin \phi \\
A_\theta &= y \cos \theta \cos \phi + (x + z) \cos \theta \sin \phi \\
A_\phi &= -y \sin \phi + (x + z) \cos \phi
\end{aligned}$$

But $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$. Substituting these yields

$$\begin{aligned}
\mathbf{A} &= (A_r, A_\theta, A_\phi) \\
&= r[\sin^2 \theta \cos \phi \sin \phi + (\sin \theta \cos \phi + \cos \theta) \sin \theta \sin \phi] \mathbf{a}_r \\
&\quad + r[\sin \theta \cos \theta \sin \phi \cos \phi + (\sin \theta \cos \phi + \cos \theta) \cos \theta \sin \phi] \mathbf{a}_\theta \\
&\quad + r[-\sin \theta \sin^2 \phi + (\sin \theta \cos \phi + \cos \theta) \cos \phi] \mathbf{a}_\phi
\end{aligned}$$

At P

$$r = 7, \quad \tan \phi = \frac{6}{-2}, \quad \tan \theta = \frac{\sqrt{40}}{3}$$

Hence,

$$\begin{aligned}
 \cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}}, \quad \cos \theta = \frac{3}{7}, \quad \sin \theta = \frac{\sqrt{40}}{7} \\
 \mathbf{A} &= 7 \cdot \left[\frac{40}{49} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{\sqrt{40}}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_r \\
 &\quad + 7 \cdot \left[\frac{\sqrt{40}}{7} \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \cdot \frac{-2}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\theta \\
 &\quad + 7 \cdot \left[\frac{-\sqrt{40}}{7} \cdot \frac{36}{40} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\
 &= \frac{-6}{7} \mathbf{a}_r - \frac{18}{7\sqrt{40}} \mathbf{a}_\theta - \frac{38}{\sqrt{40}} \mathbf{a}_\phi \\
 &= -0.8571 \mathbf{a}_r - 0.4066 \mathbf{a}_\theta - 6.008 \mathbf{a}_\phi
 \end{aligned}$$

Note that $|\mathbf{A}|$ is the same in the three systems; that is,

$$|\mathbf{A}(x, y, z)| = |\mathbf{A}(\rho, \phi, z)| = |\mathbf{A}(r, \theta, \phi)| = 6.083$$

Example 2

Two uniform vector fields are given by $\mathbf{E} = -5\mathbf{a}_\rho + 10\mathbf{a}_\phi + 3\mathbf{a}_z$ and $\mathbf{F} = \mathbf{a}_\rho + 2\mathbf{a}_\phi - 6\mathbf{a}_z$. Calculate

- $|\mathbf{E} \times \mathbf{F}|$
- The vector component of $\mathbf{E}$ at $P(5, \pi/2, 3)$ parallel to the line $x = 2, z = 3$
- The angle that $\mathbf{E}$ makes with the surface $z = 3$ at P

Solution

$$\begin{aligned}
 \text{(a) } \mathbf{E} \times \mathbf{F} &= \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ -5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix} \\
 &= (-60 - 6)\mathbf{a}_\rho + (3 - 30)\mathbf{a}_\phi + (-10 - 10)\mathbf{a}_z \\
 &= (-66, -27, -20) \\
 |\mathbf{E} \times \mathbf{F}| &= \sqrt{66^2 + 27^2 + 20^2} = 74.06
 \end{aligned}$$

- (b) Line $x = 2, z = 3$ is parallel to the y -axis, so the component of $\mathbf{E}$ parallel to the given line is

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y$$

But at $P(5, \pi/2, 3)$

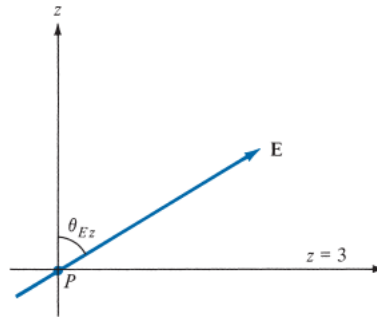
$$\begin{aligned}
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 &= \sin \pi/2 \mathbf{a}_\rho + \cos \pi/2 \mathbf{a}_\phi = \mathbf{a}_\rho
 \end{aligned}$$

Therefore,

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y = (\mathbf{E} \cdot \mathbf{a}_\rho) \mathbf{a}_\rho = -5\mathbf{a}_\rho \quad (\text{or } -5\mathbf{a}_y)$$

- (c) Since the z -axis is normal to the surface $z = 3$, we can use the dot product to find the angle between the z -axis and $\mathbf{E}$, as shown

$$\begin{aligned}
 \mathbf{E} \cdot \mathbf{a}_z &= |\mathbf{E}|(1) \cos \theta_{Ez} \rightarrow 3 = \sqrt{134} \cos \theta_{Ez} \\
 \cos \theta_{Ez} &= \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98^\circ
 \end{aligned}$$



Hence, the angle between $z = 3$ and E is

$$90^\circ - \theta_{Ez} = 15.02^\circ$$



Princess Sumaya University for Technology
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Electromagnetics I
Coordinate Systems-Examples

Example 1

Given point $P(-2, 6, 3)$ and vector $\mathbf{A} = y\mathbf{a}_x + (x + z)\mathbf{a}_y$, express P and $\mathbf{A}$ in cylindrical and spherical coordinates. Evaluate $\mathbf{A}$ at P in the Cartesian, cylindrical, and spherical systems.

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$$\rho = \sqrt{40}, \quad \tan \phi = \frac{6}{-2}$$

Hence,

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or

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 &\quad + 7 \cdot \left[\frac{\sqrt{40}}{7} \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \cdot \frac{-2}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\theta \\
 &\quad + 7 \cdot \left[\frac{-\sqrt{40}}{7} \cdot \frac{36}{40} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\
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Example 2

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- $|\mathbf{E} \times \mathbf{F}|$
- The vector component of $\mathbf{E}$ at $P(5, \pi/2, 3)$ parallel to the line $x = 2, z = 3$
- The angle that $\mathbf{E}$ makes with the surface $z = 3$ at P

Solution

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 |\mathbf{E} \times \mathbf{F}| &= \sqrt{66^2 + 27^2 + 20^2} = 74.06
 \end{aligned}$$

- (b) Line $x = 2, z = 3$ is parallel to the y -axis, so the component of $\mathbf{E}$ parallel to the given line is

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y$$

But at $P(5, \pi/2, 3)$

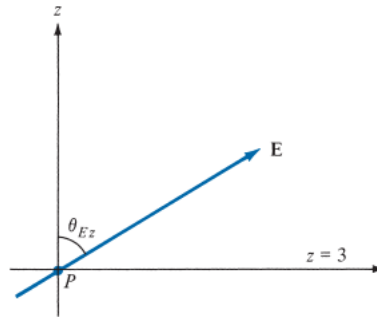
$$\begin{aligned}
 \mathbf{a}_y &= \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi \\
 &= \sin \pi/2 \mathbf{a}_\rho + \cos \pi/2 \mathbf{a}_\phi = \mathbf{a}_\rho
 \end{aligned}$$

Therefore,

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y = (\mathbf{E} \cdot \mathbf{a}_\rho) \mathbf{a}_\rho = -5\mathbf{a}_\rho \quad (\text{or } -5\mathbf{a}_y)$$

- (c) Since the z -axis is normal to the surface $z = 3$, we can use the dot product to find the angle between the z -axis and $\mathbf{E}$, as shown

$$\begin{aligned}
 \mathbf{E} \cdot \mathbf{a}_z &= |\mathbf{E}|(1) \cos \theta_{Ez} \rightarrow 3 = \sqrt{134} \cos \theta_{Ez} \\
 \cos \theta_{Ez} &= \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98^\circ
 \end{aligned}$$



Hence, the angle between $z = 3$ and $\mathbf{E}$ is

$$90^\circ - \theta_{Ez} = 15.02^\circ$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Coulomb's Law and Field Intensity

An electrostatic field is produced by a static charge distribution. The devices used in solid-state electronics are based on electrostatics. These include resistors, capacitors, and active devices such as bipolar and field effect transistors, which are based on control of electron motion by electrostatic fields. Almost all computer peripheral devices are based on electrostatic fields. Touch pads capacitance keyboards, cathode-ray tubes, liquid crystal displays, and electrostatic printers are typical examples. In medical work, diagnosis is often carried out with the aid of electrostatics, as incorporated in electrocardiograms. In industry, electrostatics is applied in a variety of forms such as paint spraying, electrodeposition, electrochemical machining, and separation of fine particles. Electrostatics is used in agriculture to sort seeds, direct sprays to plants, measure the moisture content of crops, spin cotton, and speed baking of bread and smoking of meat. There are two fundamental laws governing electrostatic fields, which are based on experimental studies, Coulomb's law, and Gauss's law.

Although Coulomb's law is applicable in finding the electric field due to any charge configuration, it is easier to use Gauss's law when charge distribution is symmetrical. Based on Coulomb's law, the concept of electric field intensity will be introduced and applied to cases involving point, line, surface, and volume charges. Special problems that can be solved with much effort using Coulomb's law will be solved with ease by applying Gauss's law. It is assumed here that the electric field is in a vacuum or free space.

Coulomb's Law

Coulomb's law is an experimental law formulated in 1785 by the French colonel, Charles Augustine de Coulomb. It deals with the force a point charge exerts on another point charge. By a point charge, we mean a charge that is located on a body whose dimensions are much smaller than other relevant dimensions. For example, a collection of electric charges on a pinhead may be regarded as a point charge. Charges are generally measured in coulombs (C). One coulomb is approximately equivalent to 6×10^{18} electrons. It is a very large unit of charge because one electron charge (e) is 1.6019×10^{-19} C. Coulomb's law states that the force F between two point charges Q_1 and Q_2 is:

- Along the line joining them,
- Directly proportional to the product $Q_1 Q_2$ of the charges,
- Inversely proportional to the square of the distance R between them.

Expressed mathematically:

$$\mathbf{F} = \frac{k Q_1 Q_2}{R^2}$$

where k is the proportionality constant. In SI units, charges Q_1 and Q_2 are in coulombs (C), the distance R is in meters (m), and the force $\mathbf{F}$ is in Newton (N), so that $k = \frac{1}{4\pi \epsilon_0}$.

The constant ϵ_0 is known as the permittivity of free space has the value:

$$\epsilon_0 = 8.854 \times 10^{-12} \cong \frac{10^{-9}}{36\pi} \text{ F / m}$$

thus:

$$\mathbf{F} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2}$$

If point charges Q_1 and Q_2 are located at points having position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, then the vector force $\mathbf{F}_{12}$ on Q_2 due to Q_1 as shown in Fig. (1), is given by:

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2} \mathbf{a}_{R_{12}}$$

where

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1$$

$$R = |\mathbf{R}_{12}|$$

$$\mathbf{a}_{R_{12}} = \frac{\mathbf{R}_{12}}{R}$$

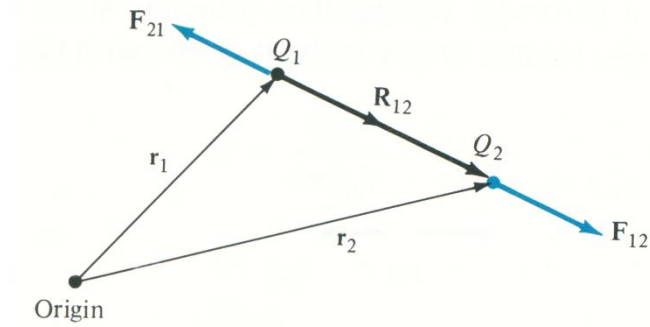


Fig. (1): Coulomb vector force on the point charges Q_1 and Q_2 .

The equations are rewritten, thus:

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2} \mathbf{R}_{12}$$

or

$$\mathbf{F}_{12} = \frac{Q_1 Q_2 (\mathbf{r}_2 - \mathbf{r}_1)}{4\pi \epsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|^3}$$

It is worthwhile to note that the force $\mathbf{F}_{21}$ on Q_1 due to Q_2 is given by:

$$\mathbf{F}_{21} = |\mathbf{F}_{12}| \mathbf{a}_{R_{21}} = |\mathbf{F}_{12}| (-\mathbf{a}_{R_{12}})$$

or

$$\mathbf{F}_{21} = -\mathbf{F}_{12}$$

since

$$\mathbf{a}_{R_{21}} = -\mathbf{a}_{R_{12}}$$

Like charges repel each other while unlike charges attract as illustrated in Fig. (2).

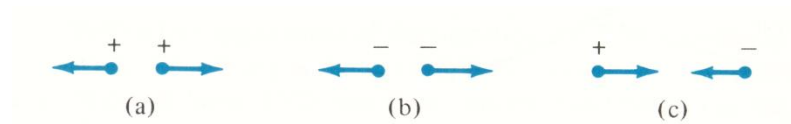


Fig. (2) A-like charges repel, (a and b); (c) unlike charges attract.

The distance $\mathbf{R}$ between the charged bodies Q_1 and Q_2 must be large compared with the linear dimensions of the bodies; that is, Q_1 and Q_2 must be point charges. Q_1 and Q_2 must be static (at rest). The signs of Q_1 and Q_2 must be taken into account. If more than two

point charges are present, one can use the principle of superposition to determine the force on a particular charge. The principle states that if there are N charges $Q_1, Q_2, Q_3, \dots, Q_N$ then:

$$\mathbf{F} = \frac{Q Q_1 (\mathbf{r} - \mathbf{r}_1)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_1|^3} + \frac{Q Q_2 (\mathbf{r} - \mathbf{r}_2)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{Q Q_N (\mathbf{r} - \mathbf{r}_N)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_N|^3}$$

or

$$\mathbf{F} = \frac{Q}{4\pi \epsilon_0} \sum_{k=1}^N \frac{Q_k (\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$

Electric Field Intensity $\mathbf{E}$

The electric field intensity or electric field strength $\mathbf{E}$ is the force per unit charge when placed in the electric field, thus:

$$\mathbf{E} = \lim_{Q \rightarrow 0} \frac{\mathbf{F}}{Q}$$

or simply

$$\mathbf{E} = \frac{\mathbf{F}}{Q}$$

The electric field intensity at point $\mathbf{r}$ due to a point charge located at $\mathbf{r}'$ is readily obtained as:

$$\mathbf{E} = \frac{Q}{4\pi \epsilon_0 R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

For N point charges $Q_1, Q_2, \dots, Q_N$ located at $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ and $\mathbf{r}_N$, the electric field intensity at point $\mathbf{r}$ is obtained as:

$$\mathbf{E} = \frac{Q_1 (\mathbf{r} - \mathbf{r}_1)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_1|^3} + \frac{Q_2 (\mathbf{r} - \mathbf{r}_2)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{Q_N (\mathbf{r} - \mathbf{r}_N)}{4\pi \epsilon_0 |\mathbf{r} - \mathbf{r}_N|^3}$$

or

$$\mathbf{E} = \frac{1}{4\pi \epsilon_0} \sum \frac{Q_k (\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$

Electric Fields Due To Continuous Charge Distribution

It is possible to have continuous charge distribution along a line, on a surface, or a volume

as illustrated in Fig. (3):

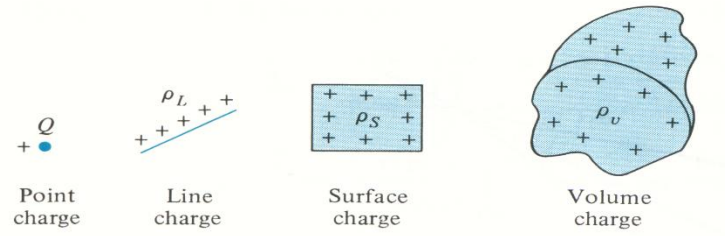


Fig. (3) Various charge distributions.

It is customary to denote the line charge density, surface charge density, and volume charge density by ρ_L , ρ_S and ρ_V respectively. The electric field intensity due to each of the charge distributions may be regarded as the summation of the field contributed by the numerous point charges making up the charge distribution. Thus:

$$E = \int \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{line charge})$$

$$\mathbf{E} = \int \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{surface charge})$$

$$\mathbf{E} = \int \frac{\rho_V dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad (\text{volume charge})$$

Line Charge

Consider a line charge with uniform charge density ρ_L extending from A to B along the z-axis as shown in Fig. (4).

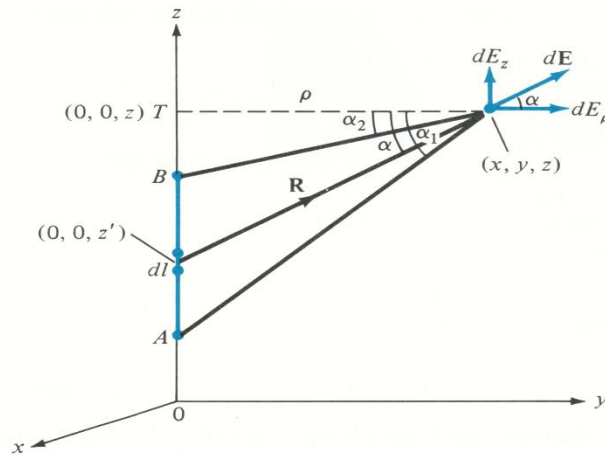


Fig. (4): Evaluation of the E field due to a line charge.

The charge element dQ associated with the element $dl = dz$ of the line is:

$$dQ = \rho dl = \rho_L dz$$

and hence the total charge Q is:

$$Q = \int_{z_A}^{z_B} \rho_L dz$$

The electric field intensity $\mathbf{E}$ at an arbitrary point $P(x, y, z)$ can be found by first assuming that the field point is denoted by (x, y, z) and the source point by (x', y', z') . Thus from Fig. (4):

$$dl = dz'$$

$$\mathbf{R} = (x, y, z) - (0, 0, z') = x\mathbf{a}_x + y\mathbf{a}_y + (z - z')\mathbf{a}_z$$

or

$$\mathbf{R} = \rho\mathbf{a}_\rho + (z - z')\mathbf{a}_z$$

$$R^2 = |\mathbf{R}|^2 = x^2 + y^2 + (z - z')^2 = \rho^2 + (z - z')^2$$

$$\frac{\mathbf{a}_R}{R^2} = \frac{\mathbf{R}}{|\mathbf{R}|^3} = \frac{\rho\mathbf{a}_\rho + (z - z')\mathbf{a}_z}{[\rho^2 + (z - z')^2]^{3/2}}$$

hence:

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0} \int \frac{\rho\mathbf{a}_\rho + (z - z')\mathbf{a}_z}{[\rho^2 + (z - z')^2]^{3/2}} dz'$$

$$R = [\rho^2 + (z - z')^2]^{1/2} = \rho \sec \alpha$$

$$z' = 0T - \rho \tan \alpha,$$

$$dz' = -\rho \sec^2 \alpha d\alpha$$

so $\mathbf{E}$ becomes:

$$\mathbf{E} = \frac{-\rho_L}{4\pi\epsilon_o} \int_{\alpha_1}^{\alpha_2} \int \frac{\rho \sec^2 \alpha [\cos \alpha \mathbf{a}_\rho + \sin \alpha \mathbf{a}_z] d\alpha}{\rho^2 \sec^2 \alpha}$$

Thus for a finite line charge:

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_o \rho} \left[-(\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\rho + (\cos \alpha_2 - \cos \alpha_1) \mathbf{a}_z \right]$$

As a special case, for an infinite line charge, point B is at (0, 0, ∞) and A at (0, 0, $-\infty$) so that $\alpha_1 = \pi/2$, $\alpha_2 = -\pi/2$; the z-component vanishes and $\mathbf{E}$ becomes:

$$\mathbf{E} = \frac{\rho_L}{2\pi \epsilon_o \rho} \mathbf{a}_\rho$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Electric Dipole and Energy Density

Example 1

Two dipoles with dipole moments $-5\mathbf{a}_z \text{ nC} \cdot \text{m}$ and $9\mathbf{a}_z \text{ nC} \cdot \text{m}$ are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.

Solution:

$$V = \sum_{k=1}^2 \frac{\mathbf{p}_k \cdot \mathbf{r}_k}{4\pi\epsilon_0 r_k^3}$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{\mathbf{p}_1 \cdot \mathbf{r}_1}{r_1^3} + \frac{\mathbf{p}_2 \cdot \mathbf{r}_2}{r_2^3} \right]$$

where

$$\mathbf{p}_1 = -5\mathbf{a}_z, \quad \mathbf{r}_1 = (0, 0, 0) - (0, 0, -2) = 2\mathbf{a}_z, \quad r_1 = |\mathbf{r}_1| = 2$$

$$\mathbf{p}_2 = 9\mathbf{a}_z, \quad \mathbf{r}_2 = (0, 0, 0) - (0, 0, 3) = -3\mathbf{a}_z, \quad r_2 = |\mathbf{r}_2| = 3$$

Hence,

$$V = \frac{1}{4\pi \cdot \frac{10^{-9}}{36\pi}} \left[\frac{-10}{2^3} - \frac{27}{3^3} \right] \cdot 10^{-9}$$

$$= -20.25 \text{ V}$$

PRACTICE EXERCISE 4.13

An electric dipole of $100 \mathbf{a}_z \text{ pC} \cdot \text{m}$ is located at the origin. Find V and $\mathbf{E}$ at points

(a) $(0, 0, 10)$

(b) $(1, \pi/3, \pi/2)$

Answer: (a) 9 mV , $1.8\mathbf{a}_z \text{ mV/m}$, (b) 0.45 V , $0.9\mathbf{a}_r + 0.7794\mathbf{a}_\theta \text{ V/m}$.

Example 2

The point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, and $(1, 0, 0)$, respectively. Find the energy in the system.

Solution:

Method 1:

$$\begin{aligned}W &= W_1 + W_2 + W_3 \\&= 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32}) \\&= Q_2 \cdot \frac{Q_1}{4\pi\epsilon_0 |(0, 0, 1) - (0, 0, 0)|} \\&\quad + \frac{Q_3}{4\pi\epsilon_0} \left[\frac{Q_1}{|(1, 0, 0) - (0, 0, 0)|} + \frac{Q_2}{|(1, 0, 0) - (0, 0, 1)|} \right] \\&= \frac{1}{4\pi\epsilon_0} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right) \\&= \frac{1}{4\pi \cdot \frac{10^{-9}}{36\pi}} \left(-4 - 3 + \frac{12}{\sqrt{2}} \right) \cdot 10^{-18} \\&= 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ}\end{aligned}$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Electric Flux Density and Gauss's Law

Surface Charge

Consider an infinite sheet of charge in the xy -plane with uniform charge density ρ_S . The charge associated with an elemental area dS is:

$$dQ = \rho_S dS$$

and hence the total charge is:

$$Q = \int \rho_S \cdot dS$$

The contribution to the $\mathbf{E}$ field at point $P(0, 0, h)$ by the elemental surface 1 shown in Fig. (1) is:

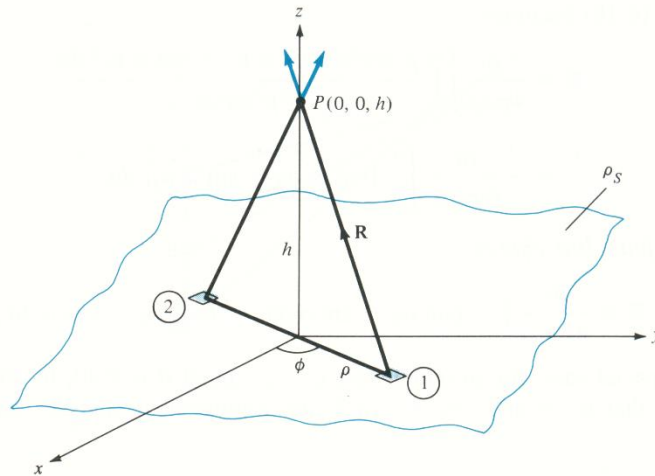


Fig. (1) Evaluation of the $\mathbf{E}$ field due to an infinite sheet of charge.

$$d\mathbf{E} = \frac{dQ}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

From Fig. (1):

$$\mathbf{R} = \rho(-\mathbf{a}_\rho) + h\mathbf{a}_z$$

$$R = |\mathbf{R}| = [\rho^2 + h^2]^{1/2}$$

$$\mathbf{a}_R = \frac{\mathbf{R}}{R},$$

$$dQ = \rho_s dS = \rho_s \rho d\phi d\rho$$

Substitution of these terms into $\mathbf{E}$ gives:

$$d\mathbf{E} = \frac{\rho_s \rho d\phi d\rho [-\rho \mathbf{a}_\rho + h \mathbf{a}_z]}{4\pi\epsilon_o [\rho^2 + h^2]^{3/2}}$$

Due to the symmetry of the charge distribution, for every element 1, there is a corresponding element 2 whose contribution along $\mathbf{a}_\rho$ cancels that of element 1, as illustrated in Fig. (1). Thus, the contributions to E_ρ add up to zero so that $\mathbf{E}$ has only z-component. This can also be shown mathematically by replacing $\mathbf{a}_\rho$ with $\cos\phi \mathbf{a}_x + \sin\phi \mathbf{a}_y$. Integration of $\cos\phi$, or $\sin\phi$ over $0 < \phi < 2\pi$ gives zero. Therefore:

$$\mathbf{E} = \int d\mathbf{E}_z = \frac{\rho_s}{4\pi\epsilon_o} \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\infty} \frac{h\rho d\rho d\phi}{[\rho^2 + h^2]^{3/2}} \mathbf{a}_z$$

$$\mathbf{E} = \frac{\rho_s h}{4\pi\epsilon_o} 2\pi \int_0^{\infty} [\rho^2 + h^2]^{-3/2} \frac{1}{2} d(\rho^2) \mathbf{a}_z$$

$$\mathbf{E} = \frac{\rho_s h}{2\epsilon_o} \left\{ -[\rho^2 + h^2]^{-1/2} \right\}_0^{\infty} \mathbf{a}_z$$

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_o} \mathbf{a}_z$$

that is, $\mathbf{E}$ has only z-component if the charge is in the xy-plane. In general, for an infinite sheet of charge:

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_o} \mathbf{a}_n$$

where $\mathbf{a}_n$ is a unit vector normal to the sheet.

The electric field is normal to the sheet and it is surprisingly independent of the distance between the sheet and the point of observation P . In a parallel plate capacitor, the electric field existing between the two plates having equal and opposite charges is given by:

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_o} \mathbf{a}_n + \frac{-\rho_s}{2\epsilon_o} (-\mathbf{a}_n) = \frac{\rho_s}{\epsilon_o} \mathbf{a}_n$$

Electric Flux Density

The electric field intensity is dependent on the medium in which the charge is placed. Suppose a new vector field $\mathbf{D}$ independent of the medium is defined by:

$$\mathbf{D} = \epsilon_o \mathbf{E}$$

We define *electric flux* ψ in terms of $\mathbf{D}$ hence,

$$\psi = \int \mathbf{D} \cdot d\mathbf{S}$$

In SI units, one line of electric flux emanates from ± 1 C and terminates on -1 C. Therefore, the electric flux is measured in coulombs. Hence, the vector field $\mathbf{D}$ is called the *electric flux density* and is measured in coulombs per square meter. For historical reasons, the electric flux density is also called *electric displacement*. It is apparent that all the formulas derived for $\mathbf{E}$ from Coulomb's law can be used in calculating $\mathbf{D}$, except that we have to multiply those formulas by ϵ_o . For example, for an infinite sheet of charge:

$$\mathbf{D} = \frac{\rho_s}{2} \mathbf{a}_n$$

Note that $\mathbf{D}$ is a function of charge and position only; it is independent of the medium.

Gauss's Law— Maxwell's Equation

Gauss's law constitutes one of the fundamental laws of electromagnetism. Essentially, it states that the total electric flux ψ through any *closed* surface is equal to the total charge enclosed by that surface. Thus:

$$\psi = Q_{enc}$$

that is:

$$\psi = \oint d\psi = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

$$Q = \int \rho_v dv$$

or

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \rho_V dv$$

By applying divergence theorem to the middle term:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{D} dv$$

Comparing the two volume integrals results in:

$$\rho_V = \nabla \cdot \mathbf{D}$$

which is the first of the four *Maxwell's equations* to be derived. It states that the volume charge density is the same as the divergence of the electric flux density. Note that Gauss's law is an alternative statement of Coulomb's law; proper application of the divergence theorem to Coulomb's law results in Gauss's law.

Gauss's law provides an easy means of finding $\mathbf{E}$ or $\mathbf{D}$ for symmetrical charge distributions such as a point charge, an infinite line charge, an infinite cylindrical surface charge, and a spherical distribution of charge. A continuous charge distribution has rectangular symmetry if it depends only on x (or y or z), cylindrical symmetry if it depends only on ρ , or spherical symmetry if it depends only on r (independent of θ and Φ). It must be stressed that whether the charge distribution is symmetric or not, Gauss's law always holds. For example, consider the charge distribution in Fig. (2) where v_1 and v_2 are closed surfaces (or volumes). The total flux leaving v_1 is $10 - 5 = 5$ nC because only 10 nC and -5 nC charges are enclosed by v_1 . Although charges 20 nC and 15 nC outside v_1 do contribute to the flux crossing v_1 , the net flux crossing v_1 , according to Gauss's law, is irrespective of those charges outside v_1 . Similarly, the total flux leaving v_2 is zero because no charge is enclosed by v_2 . Thus we see that Gauss's law, $\Psi = Q_{\text{enclosed}}$, is still obeyed even though the charge distribution is not symmetric. However, we cannot use the law to determine $\mathbf{E}$ or $\mathbf{D}$ when the charge distribution is not symmetric; we must resort to Coulomb's law to determine $\mathbf{E}$ or $\mathbf{D}$ in that case.

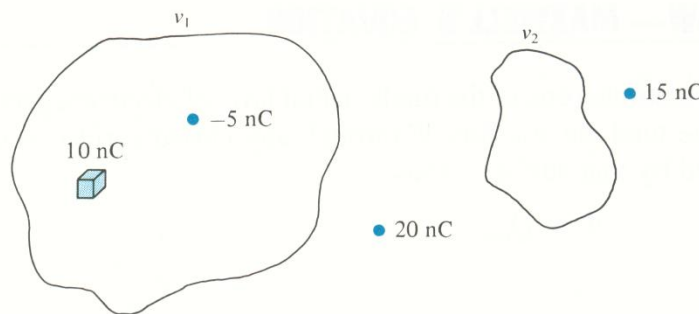


Fig. (2) Illustration of Gauss's law; flux leaving v_1 is 5 nC and that leaving v_2 is 0 C.

Applications of Gauss's Law

The procedure for applying Gauss's law to calculate the electric field involves first knowing whether symmetry exists. Once symmetric charge distribution exists, we construct a mathematical closed surface (known as a **Gaussian surface**). The surface is chosen such that $\mathbf{D}$ is normal or tangential to the Gaussian surface. When $\mathbf{D}$ is normal to the surface, $\mathbf{D} \cdot d\mathbf{S} = D dS$ because $\mathbf{D}$ is constant on the surface. When $\mathbf{D}$ is tangential to the surface, $\mathbf{D} \cdot d\mathbf{S} = 0$. Thus we must choose a surface that has some of the symmetry exhibited by the charge distribution. We shall now apply these basic ideas to the following cases.

A. Point Charge

Suppose a point charge Q is located at the origin. To determine $\mathbf{D}$ at a point P , it is easy to see that choosing a spherical surface containing P will satisfy symmetry conditions. Thus, a spherical surface centered at the origin is the Gaussian surface, in this case and is shown in Fig. (3).

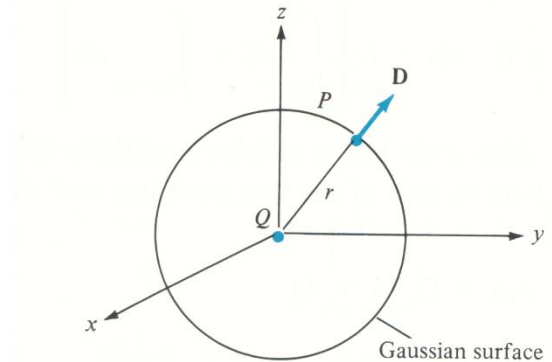


Fig. (3) Gaussian Surface about a point charge.

Since $\mathbf{D}$ is everywhere normal to the Gaussian surface:

$$\mathbf{D} = D_r \mathbf{a}_r$$

applying Gauss's law gives:

$$Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_r \oint dS = D_r 4\pi r^2$$

where:

$$\oint dS = \int_0^{2\pi} \int_0^\pi r^2 \sin \theta d\theta d\phi = 4\pi r^2$$

is the surface area of the Gaussian surface. Thus:

$$\mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r$$

Infinite Line Charge

Suppose the infinite line of uniform charge ρ_L C/m lies along the z-axis. To determine $\mathbf{D}$ at a point P , we choose a cylindrical surface containing P to satisfy symmetry condition as shown on Fig. (4).

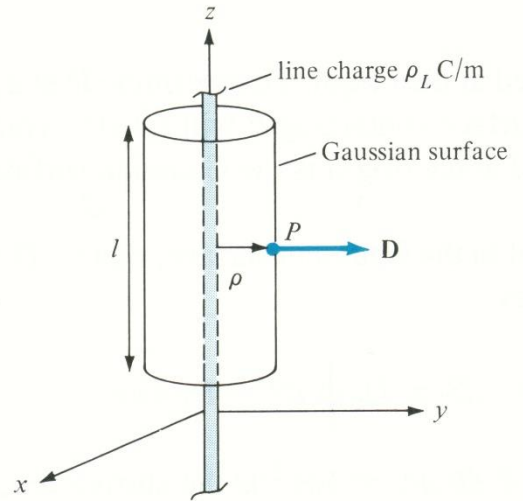


Fig. (4) A Gaussian surface about an infinite line charge.

$\mathbf{D}$ is constant on and normal to the cylindrical Gaussian surface; that is, $\mathbf{D} = D_\rho \mathbf{a}_\rho$. If we apply Gauss's law to an arbitrary length ℓ of the line:

$$\rho_L \ell = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_\rho \oint dS = D_\rho 2\pi \rho \ell$$

where:

$$\int dS = 2\pi \rho \ell$$

is the surface area of the Gaussian surface. Note that $\int \mathbf{D} \cdot d\mathbf{S}$ evaluated on the top and bottom surfaces of the cylinder is zero since $\mathbf{D}$ has no z-component; that means that $\mathbf{D}$ is tangential to those surfaces. Thus:

$$\mathbf{D} = \frac{\rho_L}{2\pi \rho} \mathbf{a}_\rho$$

as expected.

Infinite Sheet of Charge

Consider the infinite sheet of uniform charge ρ_s C/m² lying on the $z=0$ plane. To determine $\mathbf{D}$ at point P , we choose a rectangular box that is cut symmetrically by the sheet of charge and has two of its faces parallel to the sheet as shown in Fig. (5). As $\mathbf{D}$ is normal to the sheet, $\mathbf{D} = D_z \mathbf{a}_z$, and applying Gauss's law gives:

$$\rho_s \int dS = Q = \oint \mathbf{D} \cdot d\mathbf{S} = D_z \left[\int_{top} dS + \int_{bottom} dS \right]$$

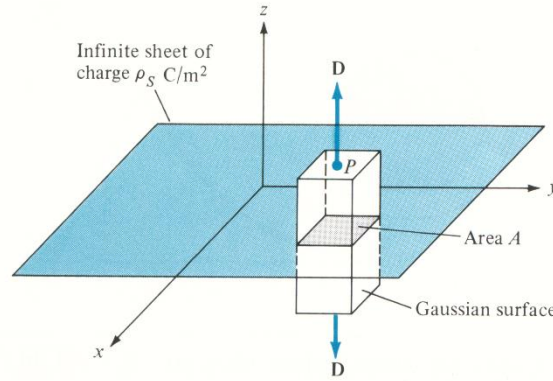


Fig. (5) Gaussian surface about the infinite sheet of charge.

Note that $\mathbf{D} \cdot d\mathbf{S}$ evaluated on the sides of the box is zero because $\mathbf{D}$ has no components along $\mathbf{a}_x$ and $\mathbf{a}_y$. If the top and bottom area of the box each has area A , then:

$$\rho_s A = D_z (A + A)$$

and thus:

$$\mathbf{D} = \frac{\rho_s}{2} \mathbf{a}_z$$

Uniformly Charged Sphere

Consider a sphere of radius a with a uniform charge ρ_v C/m³. To determine $\mathbf{D}$ everywhere, we construct Gaussian surfaces for cases $r \leq a$ and $r \geq a$ separately. Since the charge has spherical symmetry, it is obvious that a spherical surface is an appropriate Gaussian surface. For $r \leq a$ the total charge enclosed by the spherical surface of radius r , as shown in Fig. (6) is:

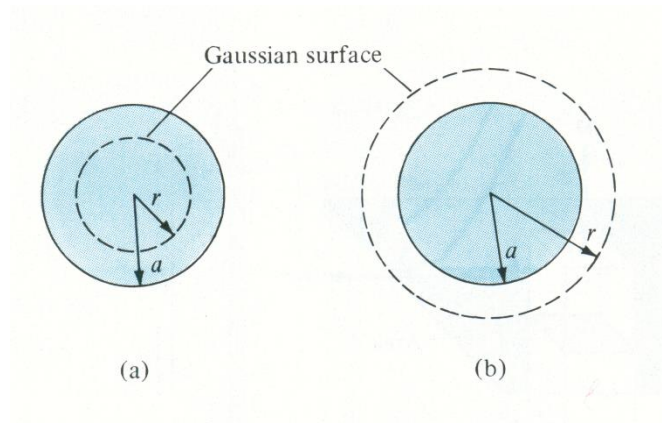


Fig. (6) Gaussian surface about a charged sphere.

$$Q_{enc} = \int \rho_v dv = \rho_v \int dv = \rho_v \int_0^{2\pi} \int_0^{\pi} \int_0^r r^2 \sin \theta dr d\theta d\phi$$

$$Q_{enc} = \rho_v \frac{4}{3} \pi r^3$$

and

$$\Psi = \int \mathbf{D} \cdot d\mathbf{S} = D_r \int dS = D_r \int_0^{2\pi} \int_0^{\pi} r^2 \sin \theta d\theta d\phi$$

$$\Psi = D_r 4\pi r^2$$

hence ,

$$\Psi = Q_{enc}$$

gives:

$$D_r 4\pi r^2 = \frac{4\pi r^3}{3} \rho_v$$

or

$$D = \frac{r}{3} \rho_v \mathbf{a}_r$$

for $0 < r \leq a$

For a charge enclosed by the surface,

$$Q_{enc} = \int \rho_v dv = \rho_v \int dv = \rho_v \int_0^{2\pi} \int_0^\pi \int_0^a r^2 \sin \theta dr d\theta d\phi$$

$$Q_{enc} = \rho_v \frac{4}{3} \pi a^3$$

while

$$\psi = \oint \mathbf{D} \cdot d\mathbf{S} = D_r 4\pi r^2$$

hence

$$D_r 4\pi r^2 = \frac{4}{3} \pi a^3 \rho_v$$

or

$$\mathbf{D} = \frac{a^3}{3r^2} \rho_v \mathbf{a}_r$$

for $r \geq a$. Thus, $\mathbf{D}$ everywhere is given by:

$$\mathbf{D} = \begin{cases} \frac{r}{3} \rho_v \mathbf{a}_r & 0 < r \leq a \\ \frac{a^3}{3r^2} \rho_v \mathbf{a}_r & r \geq a \end{cases}$$

and $|\mathbf{D}|$ is as sketched in Fig. (7):

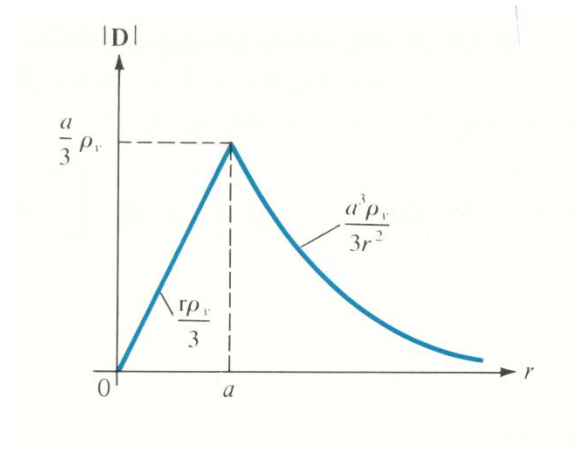


Fig. (7) Sketch of $\mathbf{D}$.

The ability to take $\mathbf{D}$ out of the integral sign is the key to finding $\mathbf{D}$ using Gaussian's law. In other words, $\mathbf{D}$ must be constant on the Gaussian surface.



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Electrostatic Dipole and Equipotential Lines

An electric dipole is formed when two point charges of equal magnitude but opposite sign are separated by a small distance. Consider the dipole shown in Fig. (1). The potential at point $P(r, \theta, \phi)$ is given by:

$$V = \frac{Q}{4\pi\epsilon} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = \frac{Q}{4\pi\epsilon} \left[\frac{r_2 - r_1}{r_1 r_2} \right]$$

Where r_1 and r_2 are the distances between P and $+Q$ and P and $-Q$ respectively. If $r \gg d$, $r_2 - r_1 \cong d \cos \theta$, $r_1 r_2 \cong r^2$, then:

$$V = \frac{Q}{4\pi\epsilon} \frac{d \cos \theta}{r^2}$$

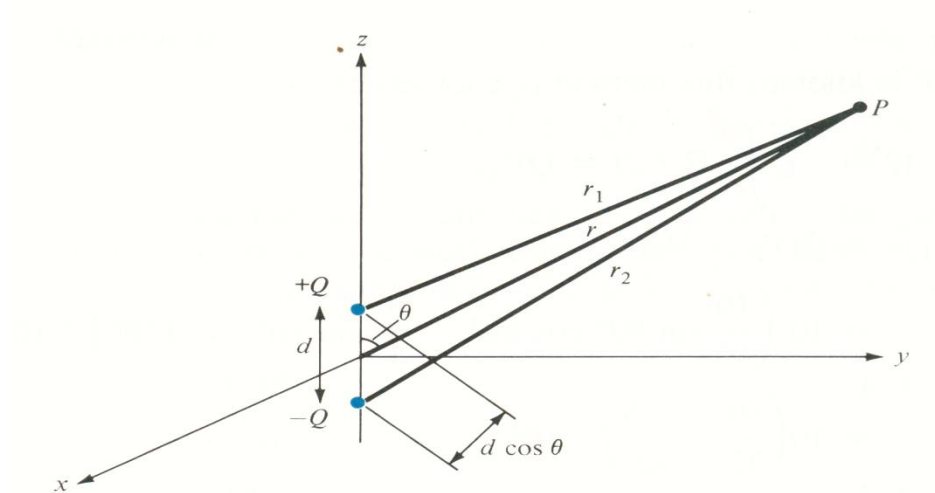


Fig. (1) Dipole at point P.

Since $d \cos \theta = \mathbf{d} \cdot \mathbf{a}_r$, where $\mathbf{d} = d\mathbf{a}_z$, if we define

$$\mathbf{p} = Q\mathbf{d}$$

as the *dipole moment*, eq. (4.78) may be written as

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_r}{4\pi\epsilon_0 r^2}$$

Note that the dipole moment $\mathbf{p}$ is directed from 2Q to 1Q. If the dipole center is not at the origin but at $\mathbf{r}'$, eq. (4.80) becomes

$$V(\mathbf{r}) = \frac{\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

The electric field due to the dipole with center at the origin, shown in Figure 4.20, can be obtained

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta \right] \\ &= \frac{Qd \cos \theta}{2\pi\epsilon_0 r^3} \mathbf{a}_r + \frac{Qd \sin \theta}{4\pi\epsilon_0 r^3} \mathbf{a}_\theta \end{aligned}$$

$$\mathbf{E} = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)$$

A point charge is a *monopole*, and its electric field varies inversely as r^2 while its potential field varies inversely as r . Notice that a point charge is a monopole and its electric field varies inversely as r^2 , while its potential field varies inversely as r . We see that the electric field due to a dipole varies inversely with r^3 , while its potential varies inversely with r^2 . The electric fields due to successive higher-order multipoles (such as a quadrupole consisting of two dipoles or an octupole consisting of two quadrupoles) vary inversely as r^4 , r^5 , r^6 , while their corresponding potentials vary inversely as r^3 , r^4 , r^5 .

Electric Flux Lines and Equipotential Surfaces

An electric flux line is an imaginary path or line drawn in such a way that its direction at any point is the direction of the electric field at that point. They are the lines to which the electric field density $\mathbf{D}$ is tangential at every point.

The following properties of electric flux lines should be noted:

- The lines always start at positive charges and terminate at negative charges. Positive charges are thus regarded as *sources* of electric flux lines, and negative charges as *sinks*.
- No two-flux lines can intersect except at singular or equilibrium points at which the resultant $\mathbf{D}$ is zero.
- The electric flux density $\mathbf{D}$ is tangential to the flux lines at every point.

To determine the analytic expression for flux lines, we let:

$$\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y$$

It is evident from Fig. (2) that at any point (x, y) :

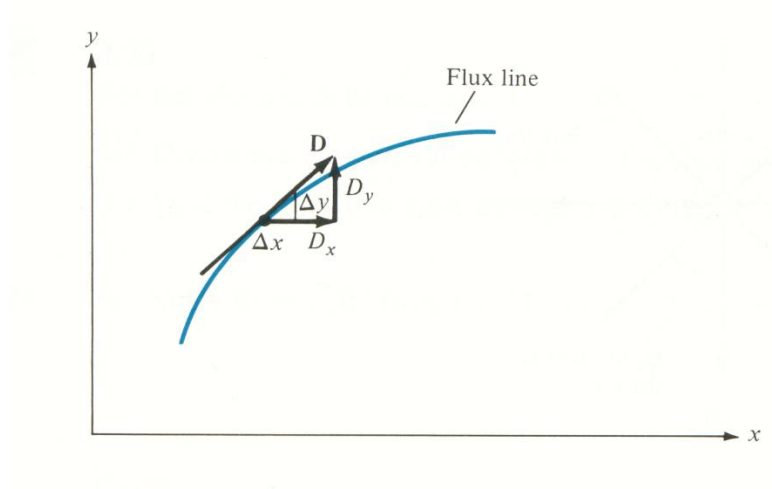


Fig. (2) Flux density tangential to flux lines.

$$\frac{\Delta y}{\Delta x} = \frac{D_y}{D_x}$$

for a three-dimensional field:

$$\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z$$

so in Cartesian coordinates, the flux lines are given by:

$$\frac{dx}{D_x} = \frac{dy}{D_y} = \frac{dz}{D_z}$$

Similarly, it can be shown that in cylindrical coordinates, where:

$$\mathbf{D} = (D_\rho, D_\theta, D_z)$$

the flux lines are given by (cylindrical):

$$\frac{d\rho}{D_\rho} = \frac{\rho d\phi}{D_\theta} = \frac{dz}{D_z}$$

and in spherical coordinates:

$$\mathbf{D} = (D_r, D_\theta, D_\phi)$$

so the flux lines satisfy (spherical):

$$\frac{dr}{D_r} = \frac{r d\theta}{D_\theta} = \frac{r \sin\theta d\theta}{D_\phi}$$

Any surface on which the potential is the same throughout is known as an *equipotential surface*. The intersection of an equipotential surface and a plane results in a path or line known as an *equipotential line*. No work is done in moving a charge from one point to another along an equipotential line or surface ($V_A - V_B = 0$), hence:

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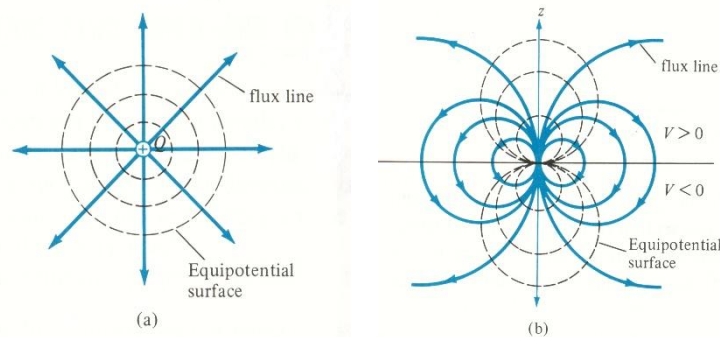


Fig. (3) Flux lines normal to equipotential surfaces.

A typical application of field mapping (flux lines and equipotential surfaces) is found in the diagnosis of the human heart. The human heart beats in response to an electric field potential difference across it. The heart can be characterized as a dipole, with a field map similar to that shown in Fig. (3b). Such a field map is useful for detecting abnormal heart position.



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Electrostatic Dipole and Equipotential Lines

An electric dipole is formed when two point charges of equal magnitude but opposite sign are separated by a small distance. Consider the dipole shown in Fig. (1). The potential at point $P(r, \theta, \phi)$ is given by:

$$V = \frac{Q}{4\pi\epsilon} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = \frac{Q}{4\pi\epsilon} \left[\frac{r_2 - r_1}{r_1 r_2} \right]$$

Where r_1 and r_2 are the distances between P and $+Q$ and P and $-Q$ respectively. If $r \gg d$, $r_2 - r_1 \cong d \cos \theta$, $r_1 r_2 \cong r^2$, then:

$$V = \frac{Q}{4\pi\epsilon} \frac{d \cos \theta}{r^2}$$

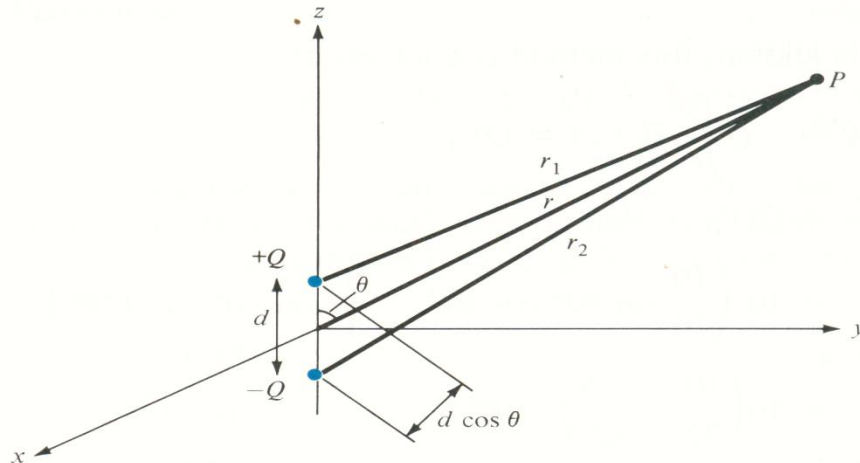


Fig. (1) Dipole at point P.

Since $d \cos \theta = \mathbf{d} \cdot \mathbf{a}_r$, where $\mathbf{d} = d\mathbf{a}_z$, if we define

$$\mathbf{p} = Q\mathbf{d}$$

as the *dipole moment*, eq. (4.78) may be written as

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_r}{4\pi\epsilon_0 r^2}$$

Note that the dipole moment $\mathbf{p}$ is directed from 2Q to 1Q. If the dipole center is not at the origin but at $\mathbf{r}'$, eq. (4.80) becomes

$$V(\mathbf{r}) = \frac{\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

The electric field due to the dipole with center at the origin, shown in Figure 4.20, can be obtained

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta \right] \\ &= \frac{Qd \cos \theta}{2\pi\epsilon_0 r^3} \mathbf{a}_r + \frac{Qd \sin \theta}{4\pi\epsilon_0 r^3} \mathbf{a}_\theta \end{aligned}$$

$$\mathbf{E} = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)$$

A point charge is a *monopole*, and its electric field varies inversely as r^2 while its potential field varies inversely as r . Notice that a point charge is a monopole and its electric field varies inversely as r^2 , while its potential field varies inversely as r . We see that the electric field due to a dipole varies inversely with r^3 , while its potential varies inversely with r^2 . The electric fields due to successive higher-order multipoles (such as a quadrupole consisting of two dipoles or an octupole consisting of two quadrupoles) vary inversely as r^4 , r^5 , r^6 , while their corresponding potentials vary inversely as r^3 , r^4 , r^5 .

Electric Flux Lines and Equipotential Surfaces

An electric flux line is an imaginary path or line drawn in such a way that its direction at any point is the direction of the electric field at that point. They are the lines to which the electric field density $\mathbf{D}$ is tangential at every point.

The following properties of electric flux lines should be noted:

- The lines always start at positive charges and terminate at negative charges. Positive charges are thus regarded as *sources* of electric flux lines, and negative charges as *sinks*.
- No two-flux lines can intersect except at singular or equilibrium points at which the resultant $\mathbf{D}$ is zero.
- The electric flux density $\mathbf{D}$ is tangential to the flux lines at every point.

To determine the analytic expression for flux lines, we let:

$$\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y$$

It is evident from Fig. (2) that at any point (x, y) :

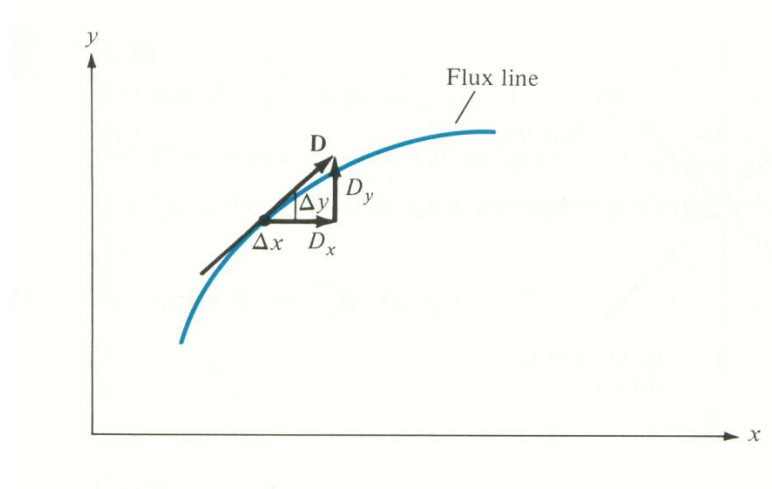


Fig. (2) Flux density tangential to flux lines.

$$\frac{\Delta y}{\Delta x} = \frac{D_y}{D_x}$$

for a three-dimensional field:

$$\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z$$

so in Cartesian coordinates, the flux lines are given by:

$$\frac{dx}{D_x} = \frac{dy}{D_y} = \frac{dz}{D_z}$$

Similarly, it can be shown that in cylindrical coordinates, where:

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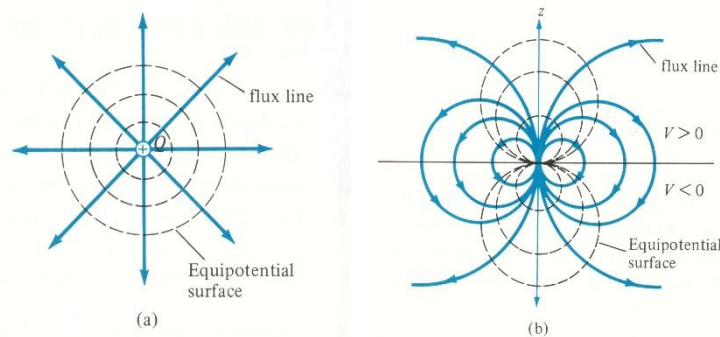


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Electromagnetics I
Electrostatic Potential

The electric field intensity $\mathbf{E}$ due to a charge distribution can be obtained from Coulomb's law in general or from Gauss's law when the charge distribution is symmetric. Another way of obtaining $\mathbf{E}$ is from the electric scalar potential V to be defined in this section. In a sense, this way of finding $\mathbf{E}$ is easier because it is easier to handle scalars than vectors. Suppose we wish to move a point charge Q from point A to point B in an electric field $\mathbf{E}$ as shown in Figure (1). From Coulomb's law, the force on Q is:

$$\mathbf{F} = Q \mathbf{E}$$

so that the *work done* in displacing the charge by $d\mathbf{l}$ is:

$$dW = -\mathbf{F} \cdot d\mathbf{l} = -Q \mathbf{E} \cdot d\mathbf{l}$$

The negative sign indicates that the work is being done by an external agent. Thus the total work done or the potential energy required in moving Q from A to B is:

$$W = -Q \int_B^A \mathbf{E} \cdot d\mathbf{l}$$

Dividing W by Q gives the potential energy per unit charge. This quantity, denoted by V_{AB} is known as the *potential difference* between points A and B, thus:

$$V_{AB} = \frac{W}{Q} = -\int_B^A \mathbf{E} \cdot d\mathbf{l}$$

Note that:

- In determining V_{AB} , A is the initial point while B is the final point
- If V_{AB} is negative, there is a loss in potential energy in moving Q from A to B; this implies that the work is being done by the field. However, if V_{AB} is positive, there is a gain in potential energy in the movement; an external agent performs the work.
- V_{AB} is independent of the path taken.

○ V_{AB} is measured in joules per coulomb, commonly referred to as volts (V).

As an example, if the $\mathbf{E}$ field in Fig. (1) is due to a point charge Q located at the origin, then:

$$V_{AB} = -\int_{r_B}^{r_A} \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \cdot d\mathbf{a}_r$$

$$V_{AB} = \frac{Q}{4\pi\epsilon} \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$$

or

$$V_{AB} = V_B - V_A$$

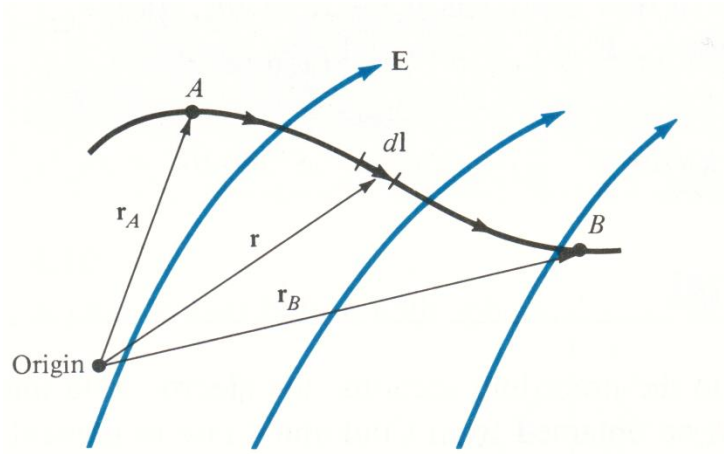


Fig. (1) Electric field due to a point charge at the origin.

where V_B and V_A , are the *potentials (or absolute potentials)* at B and A respectively. Thus, the potential difference V_{AB} may be regarded as the potential at B with reference to A. It is customary to choose infinity as reference; that is, we assume the potential at infinity is zero. Thus if $V_A = 0$ as $r_A \rightarrow \infty$, the potential at any point ($r_B \rightarrow r$) due to a point charge Q located at the origin:

$$V = \frac{Q}{4\pi\epsilon r}$$

Because $\mathbf{E}$ points in the radial direction, any contribution from a displacement in the θ or ϕ direction is wiped out by the dot product $\mathbf{E} \cdot d\mathbf{l} = E \cos \theta dl = \mathbf{E} \cdot d\mathbf{r}$. Hence, the potential difference V_{AB} is independent of the path as asserted earlier. In general, the potential at any point is defined as the potential difference between that point and a chosen point at which the potential is zero. By assuming zero potential at infinity, the potential at a distance r

from the point charge is the work done per unit charge by an external agent in transferring a test charge from infinity to that point. Thus:

$$V = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l}$$

If the point charge Q is not located at the origin but at a point whose position vector is $\mathbf{r}'$, the potential $V(x, y, z)$ or simply $V(r)$ at $\mathbf{r}$ becomes:

$$V(r) = \frac{Q}{4\pi\epsilon|\mathbf{r} - \mathbf{r}'|}$$

We have considered the electric potential due to a point charge. The same basic ideas apply to other types of charge distribution because any charge distribution can be regarded as consisting of point charges. The superposition principle, which we applied to electric fields, applies to potentials. For n point charges $Q_1, Q_2, \dots, Q_n$, located at points with position vectors $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ the potential at $\mathbf{r}$ is:

$$V(r) = \frac{Q}{4\pi\epsilon|\mathbf{r} - \mathbf{r}_1|} + \frac{Q}{4\pi\epsilon|\mathbf{r} - \mathbf{r}_2|} + \dots + \frac{Q}{4\pi\epsilon|\mathbf{r} - \mathbf{r}_n|}$$

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$$V(r) = \frac{Q}{4\pi\epsilon} \sum_{k=1}^n \frac{Q_k}{|\mathbf{r} - \mathbf{r}_k|}$$

For continuous charge distributions, we replace Q_k with charge element $\rho_l dl$, $\rho_s dS$ or $\rho_v dv$ and the summation becomes integration, so the potential at $\mathbf{r}$ becomes:

$$V(r) = \frac{1}{4\pi\epsilon} \int_L \frac{\rho_l(r') dl'}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{Line charge})$$

$$V(r) = \frac{1}{4\pi\epsilon} \int_s \frac{\rho_s(r') dS'}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{Surface charge})$$

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where the primed coordinates are used customarily to denote source point location and the unprimed coordinates refer to field point (the point at which V is to be determined). The following points should be noted:

1. the zero potential (reference) point has been chosen arbitrarily to be at infinity. If any other point is chosen as reference, then:

$$V = \frac{Q}{4\pi \epsilon r} + C$$

where C is a constant that is determined at the chosen point of reference. The potential at a point can be determined in two ways depending on whether the charge distribution or $\mathbf{E}$ is known. If $\mathbf{E}$ is known, we simply use:

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The potential difference V_{AB} can be found generally from:

$$V_{AB} = V_B - V_A = \frac{W}{Q} = -\int_A^B \mathbf{E} \cdot d\mathbf{l}$$

Relationship between $\mathbf{E}$ and V -Maxwell's Equation

The potential difference between points A and B is independent of the path taken. Hence:

$$V_{AB} = -V_{BA}$$

or

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This shows that the line integral of $\mathbf{E}$ along a closed path as shown in Fig. (2) must be zero. Physically, this implies that no network is done in moving a charge along a closed path in an electrostatic field. Applying Stokes's theorem gives:

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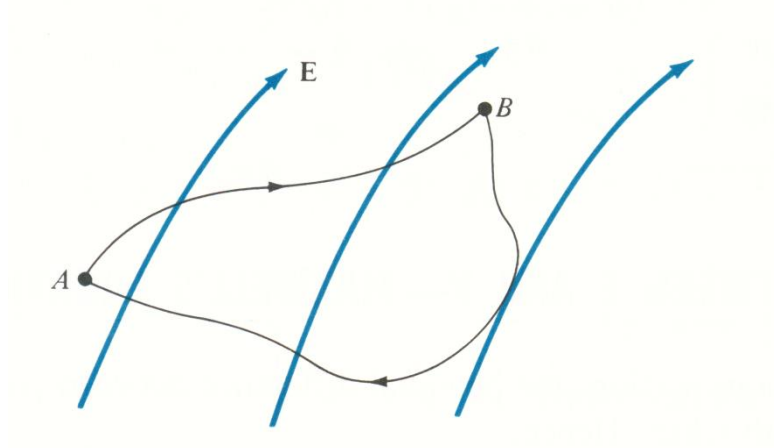


Fig. (2) Line integral of E on a closed path.

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Electromagnetics I
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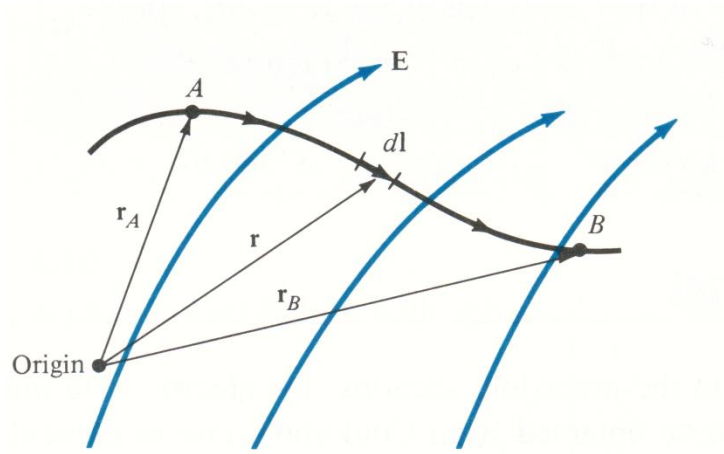


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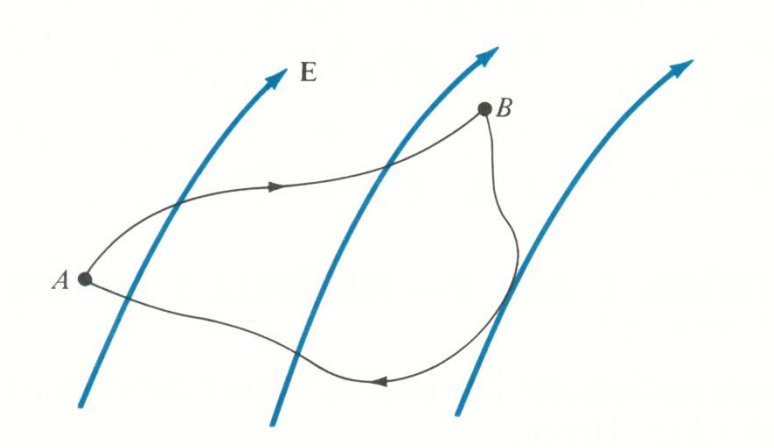


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Electromagnetics I
Energy Density in Electrostatic Fields

To determine the energy present in an assembly of charges, the amount of work necessary to assemble them must first be determined. If three point charges Q_1 , Q_2 and Q_3 are positioned in an initially empty space shown shaded in Fig. (1). No work is required to transfer Q_1 from infinity to P_1 , because the space is initially charge free and there is no electric field. The work done in transferring Q_2 from infinity to P_2 is equal to the product of Q_2 and the potential V_{21} , at P_2 due to Q_1 . Similarly, the work done in positioning Q_3 at P_3 is equal to $Q_3(V_{31} + V_{32})$.

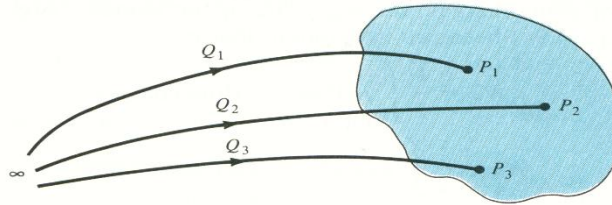


Fig. (1) Moving 3 charges from infinity to an initially empty space.

Where Q_3 and V_{31} , are the potentials at P_3 due to Q_2 and Q_1 respectively. Hence the total work done in positioning the three charges is:

$$W_E = W_1 + W_2 + W_3$$

$$W_E = 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32})$$

If the charges were positioned in reverse order:

$$W_E = W_3 + W_2 + W_1$$

$$W_E = 0 + Q_2 V_{23} + Q_1 (V_{12} + V_{13})$$

where V_{23} is the potential at P_2 due to Q_3 , V_{12} and V_{13} are respectively the potentials at P_1 due to Q_2 and Q_3 . Thus:

$$2W_E = Q_1 (V_{12} + V_{13}) + Q_2 (V_{21} + V_{23}) + Q_3 (V_{31} + V_{32})$$

$$W_E = \frac{1}{2}(Q_1V_1 + Q_2V_2 + Q_3V_3)$$

where V_1 , V_2 and V_3 are total potentials at P_1 , P_2 , and P_3 respectively. In general, if there are n point charges:

$$W_E = \frac{1}{2} \sum_{k=1}^n Q_k V_k$$

If, instead of point charges, the region has a continuous charge distribution, then:

$$W_E = \frac{1}{2} \int \rho_l V dl \quad (\text{Line charge})$$

$$W_E = \frac{1}{2} \int \rho_s V ds \quad (\text{Surface charge})$$

$$W_E = \frac{1}{2} \int \rho_v V dv \quad (\text{Volume charge})$$

Since

$$\rho_v = \nabla \cdot \mathbf{D}$$

then:

$$W_E = \frac{1}{2} \int_V (\nabla \cdot \mathbf{D}) V dv$$

But for any vector $\mathbf{A}$ and scalar V , the identity:

$$\nabla \cdot V\mathbf{A} = \mathbf{A} \cdot \nabla V + V(\nabla \cdot \mathbf{A})$$

or

$$(\nabla \cdot \mathbf{A})V = \nabla \cdot V\mathbf{A} - \mathbf{A} \cdot \nabla V$$

hence:

$$W_E = \frac{1}{2} \int_V (\nabla \cdot V\mathbf{D}) dv - \frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv$$

Applying the divergence theorem to the first term on the right-hand side of this equation, yields:

$$W_E = \frac{1}{2} \oint_S (V\mathbf{D}) \cdot d\mathbf{S} - \frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv$$

Recall that V varies as $1/r$ and $\mathbf{D}$ as $1/r^2$ for point charges; V varies as $1/r^2$ and $\mathbf{D}$ as $1/r^3$ for dipoles; and so on. Hence, VD in the first term on the right-hand side, the variable must vary at least as $1/r^3$ while dS varies as $1/r^2$. Consequently, the integral must tend to zero as the surface S becomes large. Hence:

$$W_E = \frac{1}{2} \int_V (\mathbf{D} \cdot \nabla V) dv = \frac{1}{2} \int_V (\mathbf{D} \cdot \mathbf{E}) dv$$

In addition, since:

$$\mathbf{E} = -\nabla V$$

and

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$W_E = \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \int_V \epsilon E^2 dv$$

Hence, the electrostatic energy density W_E is defined as:

$$w_E = \frac{dW_E}{dV} = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon E^2 = \frac{D^2}{2\epsilon}$$

So:

$$W_E = \int w_E dv$$



Princess Sumaya University for Technology
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Electromagnetics I
Faraday's Law

After Oersted's experimental discovery (upon which Biot-Savart and Ampere based their laws) that a steady current produces a magnetic field, it seemed logical to determine if magnetism would produce electricity. In 1831, about 11 years after Oersted's discovery, Michael Faraday in London and Joseph Henry in New York discovered that a time-varying magnetic field would produce an electric current. According to Faraday's experiments, a static magnetic field produces no current flow. Still, a time-varying field produces an induced voltage (called *electromotive force* or simply emf) in a closed circuit, which causes a flow of current. Faraday discovered that the induced emf, in any closed circuit, is equal to the time rate of change of the magnetic flux linkage by the circuit. This is called *Faraday's law*, and it can be expressed as:

$$V_{emf} = -\frac{d\lambda}{dt} = -N \frac{d\Psi}{dt}$$

where N is the number of turns in the circuit and Ψ is the flux through each turn. The negative sign shows that the induced voltage acts in such a way as to oppose the flux producing it. This is known as *Lenz's law*, which emphasizes that the direction of current flow in the circuit is such that the induced magnetic field produced by the induced current will oppose the original magnetic field.

Electric charges cause the electric fields considered so far; the flux lines begin and end on the charges in such fields. However, other kinds of electric fields are not directly caused by electric charges. These are emf-produced fields. Sources of emf include electric generators, batteries, thermocouples, fuel cells, and photovoltaic cells, which all convert non-electrical energy into electrical energy.

Consider the electric circuit in Fig. (1) where the battery is a source of emf. The electrochemical action of the battery results in an emf-produced field $\mathbf{E}_f$. Due to the charge accumulation at the battery terminals, an electrostatic field $\mathbf{E}_e = -\nabla V$ also exists.

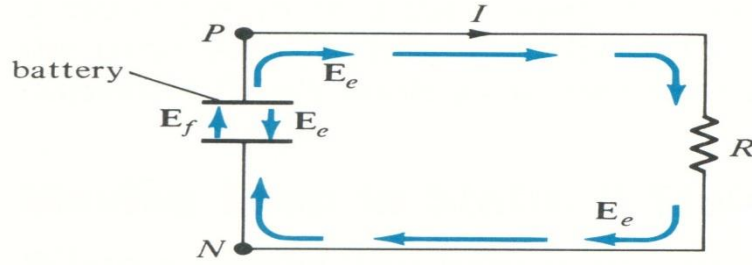


Fig. (1): A circuit showing an emf-producing and electrostatic field $\mathbf{E}_e$.

The total electric field at any point is:

$$\mathbf{E} = \mathbf{E}_f + \mathbf{E}_e$$

$\mathbf{E}_f$ is zero outside the battery, $\mathbf{E}_f$ and $\mathbf{E}_e$ have opposite directions in the battery, and the direction inside the battery is opposite to that outside. If we integrate over the closed circuit:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \oint_L \mathbf{E}_f \cdot d\mathbf{l} + 0 = \int_N^P \mathbf{E}_f \cdot d\mathbf{l}$$

where

$$\oint \mathbf{E}_e \cdot d\mathbf{l} = 0$$

because $\mathbf{E}_e$ is conservative. The emf of the battery is the line integral of the emf-produced field; that is,

$$V_{emf} = \int_N^P \mathbf{E}_f \cdot d\mathbf{l} = - \int_N^P \mathbf{E}_e \cdot d\mathbf{l} = IR$$

since $\mathbf{E}_f$ and $\mathbf{E}_e$ are equal but opposite within the battery. It may also be regarded as the potential difference ($V_P - V_N$) between the battery's open-circuit terminals.

Transform and Motional emfs

For a circuit with a single turn ($N = 1$):

$$V_{emf} = - \frac{d\Psi}{dt}$$

In terms of $\mathbf{E}$ and $\mathbf{B}$:

$$V_{emf} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}$$

It is evident that in a time-varying situation, both electric and magnetic fields are present and are interrelated. The variation of flux with time may be caused in three ways:

- by having a stationary loop in a time-varying $\mathbf{B}$ field.
- by having a time-varying loop area in a static $\mathbf{B}$ field.
- by having a time-varying loop area in a time-varying $\mathbf{B}$ field.

Each of these will be considered separately.

A. Stationary Loop in Time-Varying $\mathbf{B}$ Field (Transformer emf)

In the case of a stationary conducting loop in a time-varying magnetic $\mathbf{B}$ field, shown in Fig.(2):

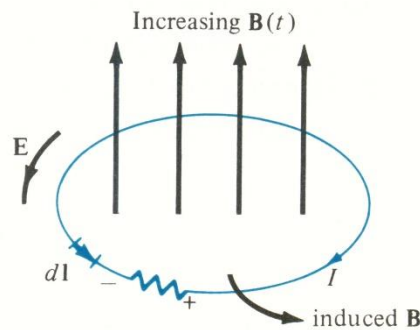


Fig. (2): Induced emf due to a stationary loop in a time-varying $\mathbf{B}$ field.

$$V_{emf} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{d\mathbf{B}}{dt} \cdot d\mathbf{S}$$

This emf induced by the time-varying current (producing the time-varying $\mathbf{B}$ field) in a stationary loop is often referred to as transformer emf in power analysis due to transformer action. By applying Stokes's theorem to the middle term:

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

For the two integrals to be equal, their integrands must be equal; that is,

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

This is one of the Maxwell's equations for time-varying fields. It shows that the time-varying $\mathbf{E}$ field is not conservative:

$$\nabla \times \mathbf{E} \neq 0$$

This does not imply that the principles of energy conservation are violated. For example, the work done in taking charge of a closed path in a time-varying electric field is due to the energy from the time-varying magnetic field.

Moving Loop in Static $\mathbf{B}$ Field (Motional emf)

When a conducting loop moves in a static $\mathbf{B}$ field, an emf is induced in the loop. The force on a charge moving with uniform velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is:

$$\mathbf{F}_m = Q\mathbf{u} \times \mathbf{B}$$

We define the motional electric field $\mathbf{E}_m$ as

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{Q} = \mathbf{u} \times \mathbf{B}$$

If we consider a conducting loop, moving with uniform velocity $\mathbf{u}$ as consisting of a large number of free electrons, the emf induced in the loop is:

$$V_{emf} = \oint_L \mathbf{E}_m \cdot d\mathbf{l} = \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

This type of emf is called *motional emf* or *flux-cutting emf* because it is due to motional action. It is the EMF found in electrical machines such as motors, generators, and alternators. Fig. (3) illustrates a two-pole DC machine with one armature coil and a two-bar commutator. The voltage is generated as the coil rotates within the magnetic field.

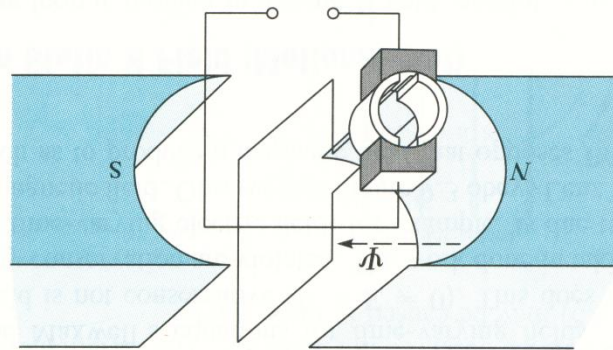


Fig. (3): A direct-current machine.

Another example of motional emf is illustrated in Fig. (4), where a rod moves between a pair of rails.

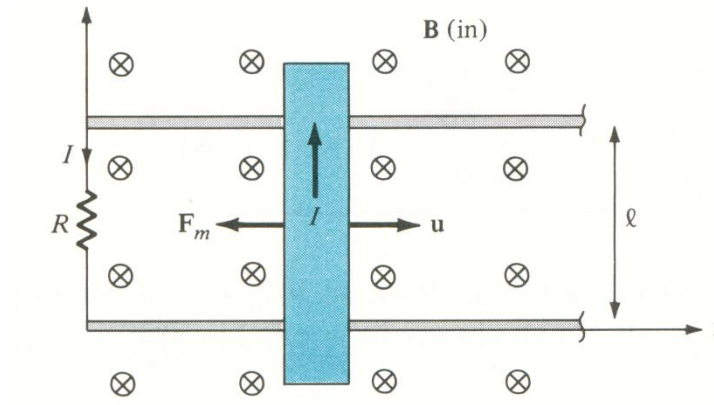


Fig. (4): Induced emf due to a moving loop in a static $\mathbf{B}$ field.

In this example, $\mathbf{B}$ and $\mathbf{u}$ are perpendicular:

$$\mathbf{F}_m = I \mathbf{l} \times \mathbf{B}$$

or

$$F_m = I \ell B$$

and

$$V_{emf} = u B \ell$$

By applying Stokes's theorem to:

$$\int_S (\nabla \times \mathbf{E}_m) \cdot d\mathbf{S} = \int_S \nabla \times (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{S}$$

or

$$\nabla \times \mathbf{E}_m = \nabla \times (\mathbf{u} \times \mathbf{B})$$

Moving Loop in Time-Varying Field

This is the general case in which a moving conducting loop is in a time-varying magnetic field. Both transformer emf and motional emf are present. Combining the above equations gives the total emf as:

$$V_{emf} = \oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} + \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

or

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{u} \times \mathbf{B})$$

Displacement Current

We shall now reconsider Maxwell's curl equation for magnetic fields (Ampere's circuit law) for time-varying conditions. For static EM fields, we recall that:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

However, the divergence of the curl of any vector field is identically zero. Hence,

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

The continuity of current requires that:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \neq 0$$

Thus, the above two equations are incompatible for time-varying conditions. We must modify these equations by adding a term to such that:

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d$$

where $\mathbf{J}_d$ is to be determined and defined. Again, the divergence of the curl of any vector is zero. Hence:

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{J}_d$$

thus

$$\nabla \cdot \mathbf{J}_d = -\nabla \cdot \mathbf{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t} (\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$$

or

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

Hence

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

This is Maxwell's equation (based on Ampere's circuit law) for the time-varying field. The term

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

is known as displacement current density, and $\mathbf{J}$ is the conduction current density ($\mathbf{J} = \sigma \mathbf{E}$). The insertion of $\mathbf{J}_d$ was one of Maxwell's major contributions. Without the term $\mathbf{J}_d$, electromagnetic wave propagation (radio or TV waves, for example) would be impossible. At low frequencies, $\mathbf{J}_d$ is usually neglected compared with $\mathbf{J}$. However, at radio frequencies, the two terms are comparable. At the time of Maxwell, high-frequency sources were unavailable, and displacement current density could not be verified experimentally. Years later, Hertz succeeded in generating and detecting radio waves, thereby verifying displacement current density. This is one of the rare situations where mathematical arguments paved the way to experimental investigation. Based on the displacement current density, we define the *displacement current* as:

$$I_d = \int \mathbf{J}_d \cdot d\mathbf{S} = \int \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$



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Electromagnetics I
Inductors and Inductances

A circuit (or closed conducting path) carrying current I produces a magnetic field B , which causes a flux to pass through each turn of the circuit, as shown in Fig. 1.

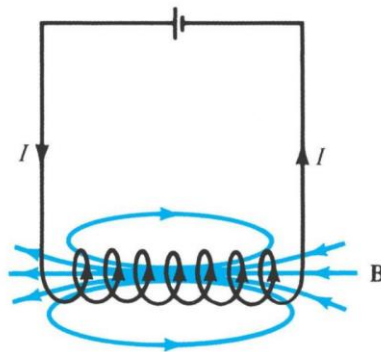


Fig. 1: Magnetic field B produced by a circuit.

If the circuit has N identical turns, we define the flux linkage λ to be:

$$\lambda = N\Psi$$

In addition, if the medium surrounding the circuit is linear, the flux linkage λ is proportional to the current I producing it. That is:

$$\lambda \propto I$$

or

$$\lambda = LI$$

where L is a constant of proportionality called the inductance of the circuit. The inductance L is a property of the physical arrangement of the circuit. An inductor is a circuit or part of a circuit with an inductance. Thus, we may define the inductance L of an inductor as the ratio of the magnetic flux linkage λ to the current I through the inductor:

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

The unit of inductance is the Henry (H), which is the same as Weber/ampere. Since Henry is a large unit, inductances are usually expressed in millehenries (mH). The inductance is commonly referred to as *self-inductance* since the inductor itself produces the linkages. Like capacitance, we may regard inductance as a measure of how magnetic energy is stored in an inductor. The magnetic energy (in joules) stored in an inductor is expressed in circuit theory as:

$$L = \frac{2 W_m}{I^2}$$

Thus, the self-inductance of a circuit may be defined or calculated based on energy considerations. If we have two circuits carrying current I_1 and I_2 instead of having a single circuit, as shown in Fig. 2, a magnetic interaction exists between the circuits. Four component fluxes ψ_{11} , ψ_{12} , ψ_{21} , and ψ_{22} are produced. The flux ψ_{12} , for example, is the flux passing through circuit 1 due to current I_2 in circuit 2.

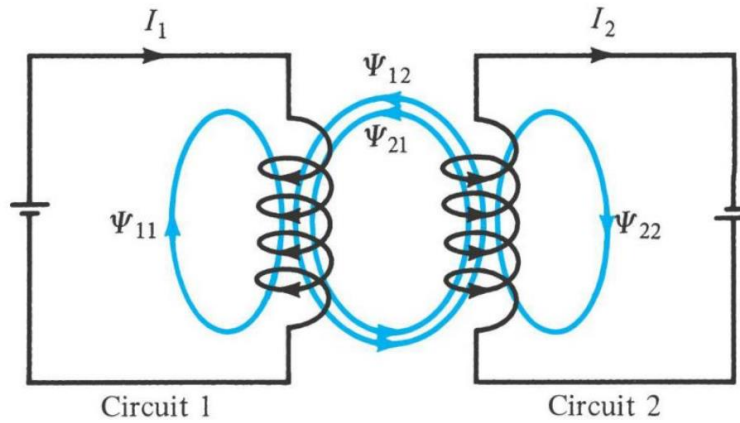


Fig. 2: Magnetic interaction between two circuits.

If B_2 in the field due to I_2 and S_1 is the area of circuit 1, then:

$$\psi_{12} = \int_{S_1} \mathbf{B}_2 \cdot d\mathbf{S}$$

then we define the mutual inductance M_{12} as the ratio of the flux linkage $\lambda_{12} = N_1 \psi_{12}$ on circuit 1 due to current I_2 , that is:

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \psi_{12}}{I_2}$$

Similarly, the mutual inductance M_{21} is defined as the flux linkages of circuit 2 per unit current I_1 ; that is,

$$M_{21} = \frac{\lambda_{21}}{I_1} = \frac{N_2 \psi_{21}}{I_1}$$

It can be shown by using energy concepts that if the medium surrounding the circuits is linear (i.e., in the absence of ferromagnetic material),

$$M_{12} = M_{21}$$

The mutual inductance M_{12} or M_{21} is expressed in Henrys and should not be confused with the magnetization vector M expressed in Amperes/meter. We define the self-inductance of circuits 1 and 2, respectively, as:

$$L_1 = \frac{\lambda_{11}}{I_1} = \frac{N_1 \Psi_1}{I_1}$$

and

$$L_2 = \frac{\lambda_{22}}{I_2} = \frac{N_2 \Psi_2}{I_2}$$

where

$$\Psi_1 = \Psi_{11} + \Psi_{12}$$

$$\Psi_2 = \Psi_{21} + \Psi_{22}$$

The total energy in the magnetic field is the sum of the energies due to L_1 , L_2 , and M_{12} (or M_{21})

$$W_m = W_1 + W_2 + W_{12}$$

$$W_m = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 \pm M_{12} I_1 I_2$$

The positive sign is taken if currents I_1 and I_2 flow such that the magnetic fields of the two circuits strengthen each other. The negative sign is taken if the currents flow so their magnetic fields oppose each other. An inductor is a conductor arranged appropriately to store magnetic energy. Typical examples of inductors are toroids, solenoids, coaxial transmission lines, and parallel-wire transmission lines. The inductance of each inductor can be determined by following a procedure similar to that for determining a capacitor's capacitance. For a given inductor, we find the self-inductance L by taking these steps:

1. Choose a suitable coordinate system.
2. Let the inductor carry current I

3. Determine $\mathbf{B}$ from Biot-Savart's law (or from Ampere's law if symmetry exists) and calculate Ψ from:

$$\Psi = \int \mathbf{B} \cdot d\mathbf{S}$$

4. Finally, find L from

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

The mutual inductance between two circuits may be calculated using a similar procedure. In an inductor such as a coaxial or a parallel-wire transmission line, the inductance produced by the flux internal to the conductor is called the internal inductance L_{in} . In contrast, that produced by the flux external to it is called external inductance L_{ext} . The total inductance L is:

$$L = L_{in} + L_{ext}$$

Just as it was shown that for capacitors:

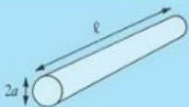
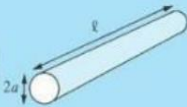
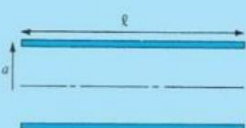
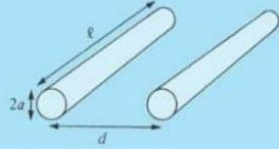
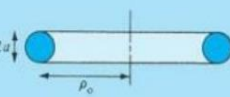
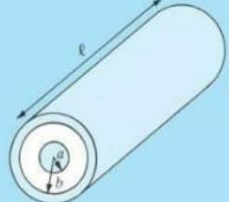
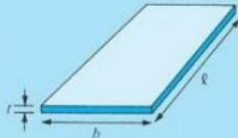
$$RC = \frac{\varepsilon}{\sigma}$$

It can be shown that for an inductor:

$$L_{ext}C = \mu\varepsilon$$

Thus, L_{ext} may be calculated if C is known.

A Collection of Formulas for Inductance of Common Elements

1. Wire $L = \frac{\mu_0 \ell}{8\pi}$ 	5. Circular loop $L = \frac{\mu_0 \ell}{2\pi} \left(\ln \frac{4\ell}{d} - 2.45 \right)$ $\ell = 2\pi\rho_0, \rho_0 \gg d$ 
2. Hollow cylinder $L = \frac{\mu_0 \ell}{2\pi} \left(\ln \frac{2\ell}{a} - 1 \right)$ $\ell \gg a$ 	6. Solenoid $L = \frac{\mu_0 N^2 S}{\ell}$ $\ell \gg a$ 
3. Parallel wires $L = \frac{\mu_0 \ell}{\pi} \ln \frac{d}{a}$ $\ell \gg d, d \gg a$ 	7. Torus (of circular cross section) $L = \mu_0 N^2 [\rho_0 - \sqrt{\rho_0^2 - a^2}]$ 
4. Coaxial conductor $L = \frac{\mu_0 \ell}{\pi} \ln \frac{b}{a}$ 	8. Sheet $L = \mu_0 2\ell \left(\ln \frac{2\ell}{b+t} + 0.5 \right)$ 

Magnetic Energy

Just as the potential energy in an electrostatic field was derived as:

$$W_E = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \int \epsilon E^2 dv$$

We would like to derive a similar expression for the energy in a magneto-static field. A simple approach is to use the magnetic energy in the field of an inductor.

$$W_m = \frac{1}{2} LI^2$$

The energy is stored in the magnetic field. $\mathbf{B}$ of the inductor. We want to express the above equation in terms of $\mathbf{B}$ or $\mathbf{H}$. Consider a differential volume in a magnetic field, as shown in Fig. 3.

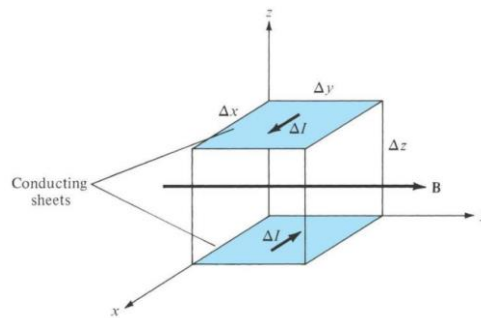


Fig. 3: A differential volume in a magnetic field.

Let the volume be covered with the conducting sheets at the top and bottom surfaces with current ΔI . We assume the whole region is filled with such differential volumes, each volume has an inductance:

$$\Delta L = \frac{\Delta \Psi}{\Delta I} = \frac{\mu H \Delta x \Delta z}{\Delta I}$$

where

$$\Delta I = H \Delta y$$

Substituting, we have:

$$\Delta W_m = \frac{1}{2} \Delta L \Delta I^2 = \frac{1}{2} \mu H^2 \Delta x \Delta y \Delta z$$

Or

$$\Delta W_m = \frac{1}{2} \mu H^2 \Delta v$$

The magneto-static energy density w_m (in J/m³) defined as:

$$w_m = \lim_{\Delta v \rightarrow 0} \frac{\Delta W_m}{\Delta v} = \frac{1}{2} \mu H^2$$

Hence:

$$w_m = \frac{1}{2} B \cdot H = \frac{B^2}{2\mu}$$

Thus, the energy in a magnetostatic field in a linear medium is:

$$W_m = \int w_m dv$$

or

$$W_m = \frac{1}{2} \int \mathbf{B} \cdot \mathbf{H} dv = \frac{1}{2} \int \mu H^2 dv$$

which is similar to the electrostatic field.

Magnetic Circuits

The concept of a magnetic circuit is based on solving some magnetic field problems using the circuit approach. Magnetic devices such as toroids, transformers, motors, generators, and relays may be considered magnetic circuits. The analysis of such circuits is made simple if an analogy between magnetic circuits and electric circuits is exploited. Once this is done, we can directly apply concepts in electric circuits to solve analogous magnetic circuits. The analogy between magnetic and electric circuits is summarized in Table 1 and portrayed in Fig. 4.

Table 1: Analogy between electric and magnetic circuits.

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $J = \frac{I}{S} = \sigma E$	Flux density $B = \frac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \frac{1}{R}$	Permeance $\mathcal{P} = \frac{1}{\mathcal{R}}$
Ohm's law $R = \frac{V}{I} = \frac{\ell}{\sigma S}$	Ohm's law $\mathcal{R} = \frac{\mathcal{F}}{\Psi} = \frac{\ell}{\mu S}$
or $V = E\ell = IR$	or $\mathcal{F} = H\ell = \Psi\mathcal{R} = NI$
Kirchoff's laws:	Kirchoff's laws:
$\sum I = 0$	$\sum \Psi = 0$
$\sum V - \sum RI = 0$	$\sum \mathcal{F} - \sum \mathcal{R} \Psi = 0$

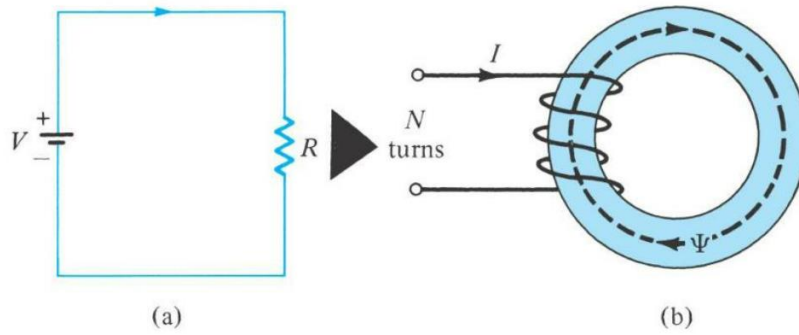


Fig. 4: Analogy between electric and magnetic circuits.

First, we notice from the table that two terms are new. We define the magneto-motive force (mmf) $\mathcal{F}$ (in ampere-turns) as:

$$\mathcal{F} = NI = \int \mathbf{H} \cdot d\mathbf{l}$$

The source of mmf in magnetic circuits is usually a coil carrying current, as shown in Fig. 4. We also define inductance $\mathcal{R}$ (in ampere-turns/Weber) as

$$\mathcal{R} = \frac{l}{\mu S}$$

where l and S are the mean length and cross-sectional area of the magnetic core, respectively, the reciprocal of reluctance is permeance μ . The basic relationship for circuit elements is Ohm's law:

$$V = IR$$

$$\mathcal{F} = \mathcal{R}\Psi$$

Based on this, Kirchhoff's current and voltage laws can be applied to nodes and loops of a given magnetic circuit just as in an electric circuit. The rules for adding voltages and combining series and parallel resistances also hold for mmfs and reluctance. Thus, for n magnetic circuit elements in series:

$$\Psi_1 = \Psi_2 = \Psi_3 \dots \dots = \Psi_n$$

and

$$\mathcal{F} = \mathcal{F}_1 + \mathcal{F}_2 + \mathcal{F}_3 \dots \dots \dots + \mathcal{F}_n$$

For n magnetic circuit elements in parallel:

$$\Psi = \Psi_1 + \Psi_2 + \Psi_3 \dots \dots + \Psi_n$$

and

$$\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}_3 \dots \dots \dots = \mathcal{F}_n$$

Some differences between electric and magnetic circuits should be pointed out. Unlike an electric circuit where current I flows, magnetic flux does not flow. In addition, conductivity σ is independent of current density $\mathbf{J}$ in an electric circuit, whereas permeability μ varies with flux density $\mathbf{B}$ in a magnetic circuit. Ferromagnetic (nonlinear) materials are usually used in most practical magnetic devices. These differences notwithstanding, the magnetic circuit concept is an approximate analysis of practical magnetic devices.

Force on Magnetic Materials

Determining the force a magnetic field exerts on a piece of magnetic material is of practical interest. This is useful in electromechanical systems such as electromagnets, relays, rotating machines, and magnetic levitation. Consider, for example, an electromagnet made of iron of constant relative permeability, as shown in Fig. 5. The coil has N turns and carries a current I .

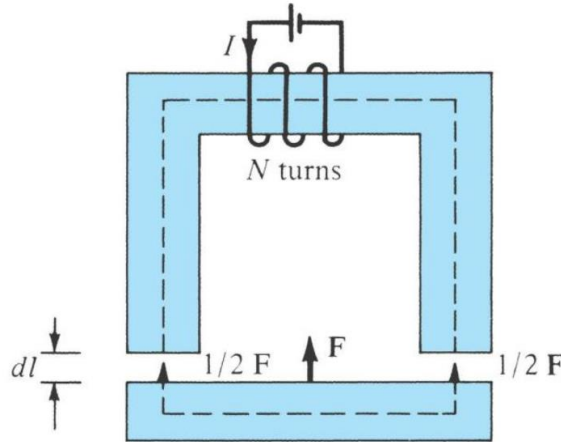


Fig. 5: An electromagnet.

If we ignore fringing, the magnetic field in the air gap is the same as that in iron ($B_{1n} = B_{2n}$). To find the force between the two pieces of iron, we calculate the change in the total energy resulting from the two pieces of the magnetic circuit separated by a differential displacement dl . The work required to effect the displacement is equal to the change in the stored energy in the air gap (assuming constant current), that is:

$$-Fdl = dW_m = 2 \left[\frac{1}{2} \frac{B^2}{\mu_0} S dl \right]$$

where S is the cross-sectional area of the gap, factor 2 accounts for the two air gaps, and the negative sign indicates that the force acts to reduce the air gap (or that the force is attractive). Thus,

$$F = -2 \frac{B^2 S}{2\mu_0}$$

Note that the force exerted on the lower piece, not the current-carrying upper piece, gives rise to the field. The attractive force across a single gap can be obtained as:

$$F = -\frac{B^2 S}{2\mu_0}$$

This equation can calculate the forces in many devices, including relays, rotating machines, and magnetic levitation. The attractive pressure (in N/m²) in the magnetized surface is:

$$p = \frac{F}{S} = \frac{B^2}{2\mu_0} = \frac{1}{2}BH$$

which is the same as the energy density w_m in the air gap.



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Electromagnetics I
Magnetic Force

The magnetic force exerted on charged particles, current elements, and loops is the principle that lies behind the operation of electrical devices such as ammeters, voltmeters, galvanometers, cyclotrons, plasmas, motors, and magneto-hydrodynamic generators. There are at least three ways in which force due to magnetic fields $\mathbf{B}$ can be experienced:

- a- Due to a moving charged particle in the field $\mathbf{B}$
- b- On a current element in an external field $\mathbf{B}$
- c- Between two current elements.

The electric force $\mathbf{F}_e$ on an electric charge Q is:

$$\mathbf{F}_e = Q \mathbf{E}$$

A magnetic field can also exert a force on a charge Q moving with velocity $\mathbf{u}$, thus:

$$\mathbf{F}_m = \mathbf{u} \times \mathbf{B}$$

where $\mathbf{F}_m$ is perpendicular to both $\mathbf{u}$ and $\mathbf{B}$. While $\mathbf{F}_e$ is independent of the velocity of the charge and can perform work on the charge and change its kinetic energy. $\mathbf{F}_m$ depends on the charge velocity and is normal to it. It cannot perform work because it is at right angles to the direction of motion of the charge and does not cause an increase in the kinetic energy of the charge. The magnitude of $\mathbf{F}_m$ is generally small compared to $\mathbf{F}_e$ except at high velocities. For a moving charge Q in the presence of both electric and magnetic fields, the total force on the charge is given by:

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m$$

or

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

This is known as the Lorentz force equation. It relates mechanical force to an electrical one. If the mass of the charged particle moving in $\mathbf{E}$ and $\mathbf{B}$ fields is m , then by Newton's second law of motion:

$$\mathbf{F} = m \frac{d\mathbf{u}}{dt} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

Force on a Current Element

A charge dQ moving with velocity $\mathbf{u}$ produces a convection current element $dQ \mathbf{u}$. For this convection current:

$$I = \frac{dQ}{dt} \frac{S dl}{dv} = \frac{dQ}{dv} \frac{dl}{dt} S$$

$$\mathbf{J} = \frac{I}{S} = \frac{dQ}{dv} \frac{dl}{dt} = \rho_v \mathbf{u}$$

$$\mathbf{J} = \rho_v \mathbf{u}$$

but the relationship between current elements gives:

$$I dl = K dS = J dv$$

combining the above equations yields:

$$\mathbf{J} dv = \rho_v \mathbf{u} dv = I dl = dQ \mathbf{u}$$

Thus, the force on a current element $I dl$ in a magnetic field $\mathbf{B}$ is found merely by replacing $dQ \mathbf{u}$ by $I dl$:

$$d\mathbf{F} = I dl \times \mathbf{B}$$

in a closed path L , the force on the circuit is given by:

$$\mathbf{F} = \oint I dl \times \mathbf{B}$$

The field that exerts force on $I dl$ must be due to another element, i.e., $\mathbf{B}$ is external to the current element $I dl$.

Force Between Two Current Elements

The force between two elements $I_1 d\mathbf{l}_1$ and $I_2 d\mathbf{l}_2$ can be found by applying Biot-Savart's law, according to which both current elements produce magnetic fields. Therefore, there is a force $d(d\mathbf{F}_1)$ on the element $I_1 d\mathbf{l}_1$ due to the field $d\mathbf{B}_2$ produced by the element:

$$d(d\mathbf{F}_1) = I_1 d\mathbf{l}_1 \times d\mathbf{B}_2$$

But from Biot-Savart's law:

$$d\mathbf{B}_2 = \frac{\mu_0 I_2 d\mathbf{l}_2 \times \mathbf{a}_{R_{21}}}{4\pi R_{21}^2}$$

hence

$$d(d\mathbf{F}_1) = \frac{\mu_0 I_1 d\mathbf{l}_1 \times (I_2 d\mathbf{l}_2 \times \mathbf{a}_{R_{21}})}{4\pi R_{21}^2}$$

This equation is essentially the law of force between two current elements and is analogous to Coulomb's law, which expresses the force between two stationary charges. The total force $\mathbf{F}_1$ on current loop 1 due to current loop 2 is shown in Fig. (1):

$$\mathbf{F}_1 = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{d\mathbf{l}_1 \times (I_2 d\mathbf{l}_2 \times \mathbf{a}_{R_{21}})}{R_{21}^2}$$

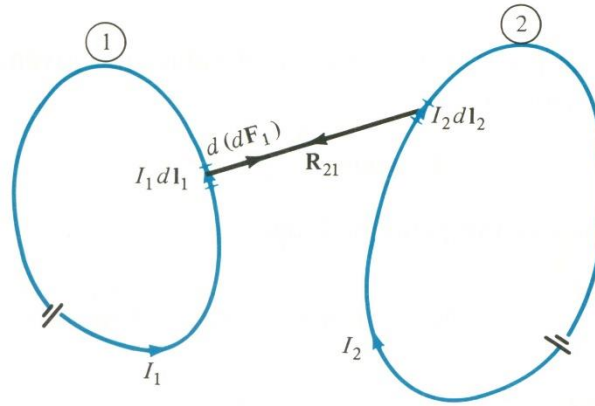


Fig. (1): Force between two current loops.

The force $\mathbf{F}_2$ on loop 2 due to the magnetic field $\mathbf{B}_1$ from loop 1 is obtained by interchanging subscripts 1 and 2. It can be shown that:

$$\mathbf{F}_2 = -\mathbf{F}_1$$

Thus, $\mathbf{F}_1$ and $\mathbf{F}_2$ obey Newton's third law, action and reaction must be equal and opposite.

Magnetic Torque and Moment

The concept of a current loop experiencing a torque in a magnetic field is paramount in understanding the behavior of orbiting charged particles, dc motors, and generators. If the loop is parallel to a magnetic field, it experiences a force that tends to rotate it. The torque $\mathbf{T}$ (or mechanical moment of force) on the loop is defined as the vector product of the force:

$$\mathbf{T} = \mathbf{r} \times \mathbf{F}$$

where $\mathbf{r}$ is the magnetic moment? The torque on a rectangular loop of length l and width is placed in a uniform magnetic field, as shown Fig. (2).

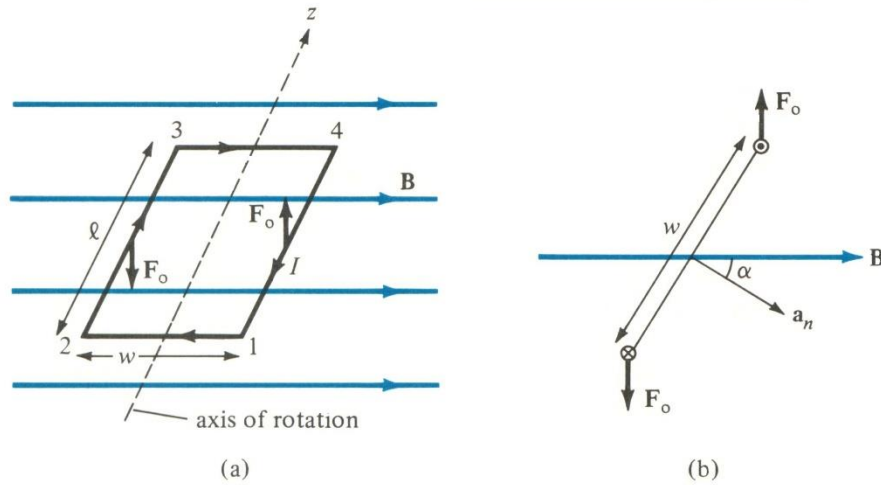


Fig. (2) Rectangular loop in a uniform $\mathbf{B}$.

Since $d\mathbf{l}$ is parallel to $\mathbf{B}$ along sides 12 and 34 of the loop no force is exerted on those sides. Thus:

or

$$\mathbf{F} = \mathbf{F}_0 - \mathbf{F}_0$$

where

$$|\mathbf{F}_0| = I B l$$

because $\mathbf{B}$ is uniform. Thus, no force is exerted on the loop as a whole. However, $\mathbf{F}_0$ and $-\mathbf{F}_0$ act at different points on the loop, thereby creating a couple. If the normal to the plane of the loop makes an angle α with $\mathbf{B}$, as shown in the cross-sectional view of Fig. (2b), the torque on the loop is;

$$|\mathbf{T}| = |\mathbf{F}_0| w \sin \alpha$$

or

$$\mathbf{T} = \frac{\mathbf{B} I S}{l w} \sin \alpha$$

but the area of the loop is:

$$S = l w$$

$$\mathbf{T} = \mathbf{B} I S \sin \alpha$$

we define the quantity

$$\mathbf{m} = I S \mathbf{a}_n$$

as the magnetic dipole moment of the loop. Here $\mathbf{a}_n$ is a unit vector normal to the plane of the loop, and its direction is determined by the right-hand rule. The magnetic dipole moment is essentially the product of the current and area of the loop; its direction is normal to the loop. Thus:

$$\mathbf{T} = \mathbf{m} \times \mathbf{B}$$

which is generally applicable in determining the torque on a planar loop of any arbitrary shape, although it was obtained using a rectangular loop. The only limitation is that the magnetic field must be uniform. It should be noted that the torque is in the direction of the axis of rotation (the z-axis in the case of the above figure). It is directed such as to reduce α so that $\mathbf{m}$ and $\mathbf{B}$ are in the same direction. In an equilibrium position (when $\mathbf{m}$ and $\mathbf{B}$ are in the same direction), the loop is perpendicular to the magnetic field and the torque will be zero and the sum of the forces on the loop.

Magnetic Dipole

A bar magnet or a small filamentary current loop is usually called a magnetic dipole. The reason for this and what we mean by "small" will soon be evident. Let us determine the magnetic field $\mathbf{B}$ at an observation point $P(r, \theta, \phi)$ due to a circular loop carrying current, as in Fig. (3).

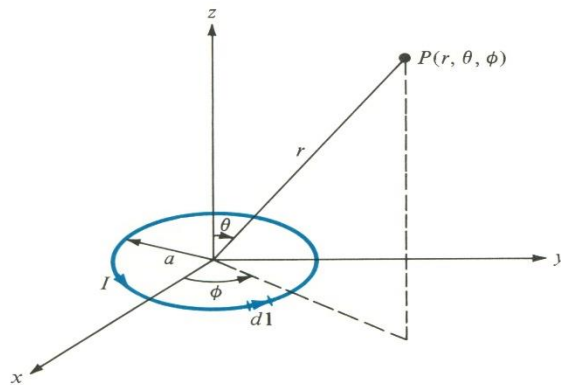


Fig. (3): The magnetic field at a point P due to a current loop.

The magnetic vector potential at P is:

$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}}{r}$$

It can be shown that at far-field $r \gg a$, so that the loop appears small at the observation point), $\mathbf{A}$ has only ϕ -component is given by:

$$\mathbf{A} = \frac{\mu_0 I \pi a^2 \sin\theta \mathbf{a}_\phi}{4\pi r^2}$$

$$\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_r}{4\pi r^2}$$

where the magnetic moment of the loop is:

$$\mathbf{m} = I \pi a^2 \mathbf{a}_z$$

and

$$\mathbf{a}_z \times \mathbf{a}_r = \sin\theta \mathbf{a}_\phi$$

The magnetic flux density is determined by:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

as

$$\mathbf{B} = \frac{\mu_0 \mathbf{m}}{4\pi r^3} (2 \cos\theta \mathbf{a}_r + \sin\theta \mathbf{a}_\theta)$$

These expressions are similar to those obtained for electrical potential V and electric field intensity $\mathbf{E}$ due to an electric dipole. Similarities between a far-field due to a small current loop and a far-field due to an electric dipole are seen clearly. It is, therefore, reasonable to regard a small current loop as a magnetic dipole. The $\mathbf{B}$ lines due to a magnetic dipole are similar to those due to an electric dipole. Fig. (4) illustrates the $\mathbf{B}$ lines around the magnetic dipole.

$$\mathbf{m} = I \mathbf{S}$$

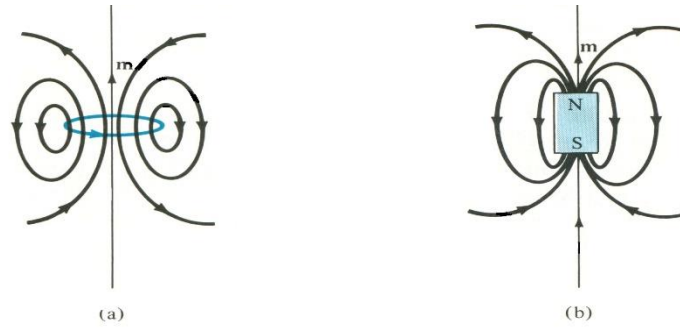


Fig. (4): The magnetic field lines due to magnetic dipoles: (a) a small current loop and (b) a bar magnet.

A short permanent magnetic bar may also be regarded as a magnetic dipole. Observe that the $\mathbf{B}$ lines due to the bar are similar to those due to a small current loop. Consider the bar magnet in Fig. (5).

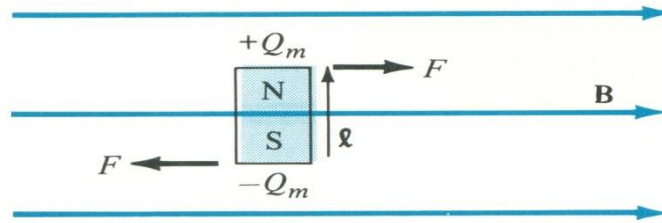


Fig. (5): A bar magnet in an external magnetic field.

If Q_m is an isolated magnetic charge (pole strength) and l is the length of the bar, the bar has a dipole moment:

$$\mathbf{m} = Q_m \mathbf{l}$$

When the bar is in a uniform magnetic field $\mathbf{B}$, it experiences a torque:

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} = Q_m \mathbf{l} \times \mathbf{B}$$

where $\mathbf{l}$ points in the direction south-to-north. The torque tends to align the bar with the external magnetic field. The force acting on the magnetic charge is given by:

$$\mathbf{F} = Q_m \mathbf{B}$$

since both a small current loop and a bar magnet produce magnetic dipoles, they are equivalent if they produce the same torque in a given $\mathbf{B}$ field; that is, when

$$Q_m \ell = I S$$

showing that they must have the same dipole moment.



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Electromagnetics I
Magnetization in Materials

In general, we may use the magnetic susceptibility χ_m or the relative permeability μ_r to classify materials in terms of their magnetic property or behavior. A material is said to be nonmagnetic if $\chi_m = 0$ or $\mu_r = 1$; it is magnetic otherwise. Free space and materials with $\chi_m = 0$ or $\mu_r = 1$ are regarded as non-magnetic. Magnetic materials may be grouped into three major classes: diamagnetic, paramagnetic, and ferromagnetic. This rough classification is depicted in Fig. 1.

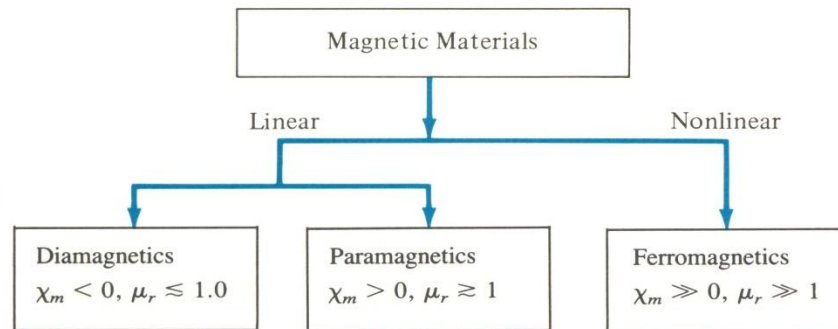


Fig. 1: Classification of magnetic materials.

A material is considered diamagnetic if it has $\mu_r \leq 1$, i.e., a very small negative χ_m . It is paramagnetic if $\mu_r \geq 1$, i.e., very small positive χ_m . If $\mu_r \gg 1$, i.e., very large positive χ_m , the material is ferromagnetic. For most practical purposes, it is assumed that $\mu_r \approx 1$ for diamagnetic and paramagnetic materials. Thus, diamagnetic and paramagnetic materials are regarded as linear and nonmagnetic. Ferromagnetic materials are always nonlinear and magnetic except when above the Curie temperature.

Diamagnetism occurs in materials where the magnetic fields due to the electronic motions of orbiting and spinning electrons completely cancel each other. Thus, the permanent magnetic moment of each atom is zero, and the materials are weakly affected by a magnetic field. Most diamagnetic materials (e.g., bismuth, lead, copper, silicon, diamond, sodium chloride) are ordered -10^{-5} . Diamagnetic materials are seldom used in practice. Materials whose atoms have non-zero permanent magnetic moments may be paramagnetic or

ferromagnetic. Paramagnetism occurs in materials where the magnetic fields produced by orbital and spinning electrons do not cancel completely. Unlike diamagnetism, paramagnetism is temperature-dependent. For most paramagnetic materials (e.g., air, platinum, tungsten, potassium), χ_m is of the order of 10^{-5} to 10^{-3} and is temperature-dependent. Such materials find application in Masers.

Ferromagnetism occurs in materials whose atoms have relatively large permanent magnetic moments. They are called ferromagnetic materials because the best-known member is iron. Other members are cobalt and nickel, as well as their alloys. Ferromagnetic materials are very useful in practice. As distinct from diamagnetic and paramagnetic materials, ferromagnetic materials have the following properties:

- They are capable of being magnetized very strongly by a magnetic field.
- They retain a considerable amount of their magnetization when removed from the field.
- They lose their ferromagnetic properties and become linear paramagnetic materials when the temperature is raised above a certain temperature known as the Curie temperature. Thus, if a permanent magnet is heated above its Curie temperature (770 °C for Iron), it loses its magnetization completely.
- They are nonlinear, that is, the constitutive relation:

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H}$$

does not hold for ferromagnetic materials because μ_r depends on $\mathbf{B}$ and cannot be represented by a single value. For example, for Nickel $\mu_r = 50$ under some conditions and 600 under other conditions.

Ferromagnetic materials, such as iron and steel, are used for screening or shielding to protect sensitive electrical devices from disturbances from strong magnetic fields. A typical example of an iron shield is shown in Fig. 2, where the compass is protected.

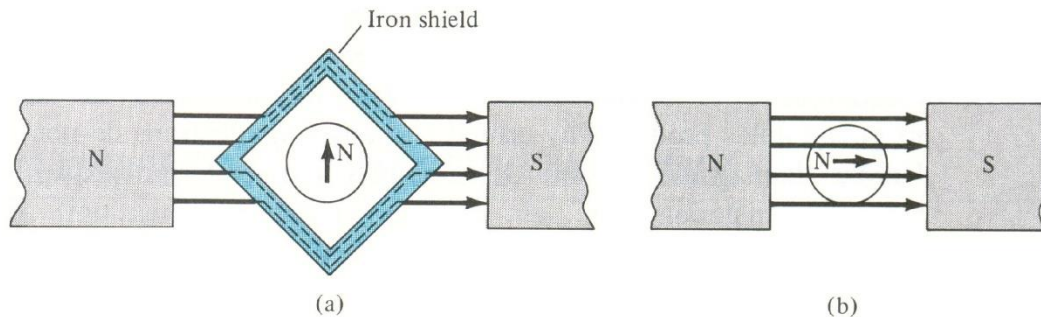


Fig. 2: Magnetic screening.

Without the iron shield, the compass gives an erroneous reading due to the effect of the external magnetic field. For perfect screening, it is required that the shield have infinite permeability. Even though:

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

holds for all materials, including ferromagnetism, the relationship between $\mathbf{B}$ and $\mathbf{H}$ depends on the previous magnetization of a ferromagnetic material or its “magnetic history.” Instead of having a linear relationship between $\mathbf{B}$ and $\mathbf{H}$ (i.e., $\mathbf{B} = \mu\mathbf{H}$), it is only possible to represent the relationship by a magnetization or $\mathbf{B}$ – $\mathbf{H}$ curve. A typical $\mathbf{H}$ curve is shown in Fig. 3. First, note the nonlinear relationship between $\mathbf{B}$ and $\mathbf{H}$.

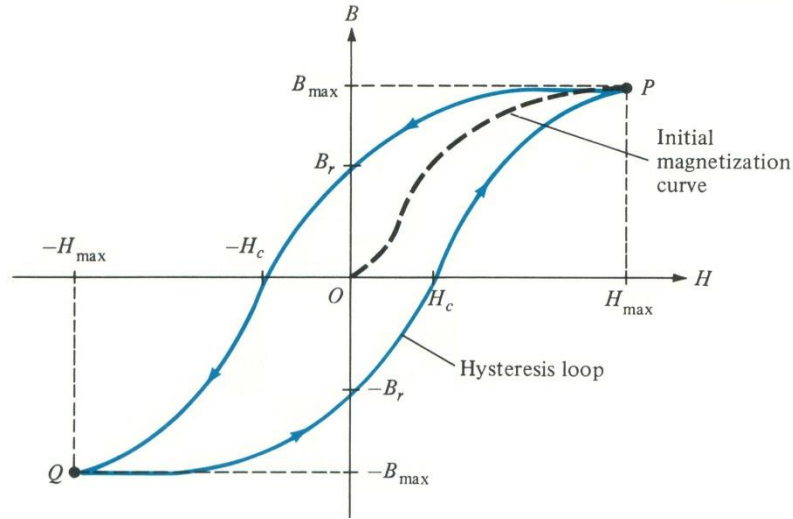


Fig. 3: A typical $\mathbf{B}$ – $\mathbf{H}$ curve.

If we assume that the ferromagnetic material is initially unmagnetized, as $\mathbf{H}$ increases (due to an increase in current) from 0 to maximum applied field intensity $\mathbf{H}_{\max}$, curve OP is produced. This curve is referred to as the version or initial magnetization curve. After reaching saturation at P, if $\mathbf{H}$ is decreased, $\mathbf{B}$ does not follow the initial curve but lags behind $\mathbf{H}$. This phenomenon of $\mathbf{B}$ lagging behind $\mathbf{H}$ is called hysteresis (which means “to lag” in Greek). If $\mathbf{H}$ is reduced to zero, $\mathbf{B}$ is not reduced to zero but to $\mathbf{B}_r$, referred to as the permanent flux density. The value of $\mathbf{B}_r$ depends on $\mathbf{H}_{\max}$, the maximum applied field intensity. The existence of $\mathbf{B}_r$ is the cause of having permanent magnets if $\mathbf{H}$ increases negatively (by reversing the direction of current), and $\mathbf{B}$ becomes zero when $\mathbf{H}$ becomes $\mathbf{H}_0$, which is known as the coercive field intensity. Materials for which $\mathbf{H}_c$ is small are said to be magnetically hard. The value of $\mathbf{H}_c$ also depends on $\mathbf{H}_{\max}$.

Further, an increase in $\mathbf{H}$ in the negative direction to reach Q and a reversal in its direction to reach P gives a closed curve called a hysteresis loop. The shape of hysteresis loops varies from one material to another. Some ferrites, for example, have an almost rectangular hysteresis loop and are used in digital computers as magnetic information storage devices. The area of a hysteresis loop gives the energy loss (hysteresis loss) per unit volume during one cycle of the periodic magnetization of the ferromagnetic material. The energy loss is in the form of heat. It is, therefore, desirable that materials used in electric generators, motors, and transformers should have tall but narrow hysteresis loops so that hysteresis losses are minimal.

Magnetic Boundary Conditions

We define magnetic boundary conditions as the conditions that the $\mathbf{H}$ (or $\mathbf{B}$) field must satisfy at the boundary between two different media. Applying:

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = I$$

Consider the boundary between two magnetic media 1 and 2, characterized respectively by μ_1 and μ_2 as in Fig. 4.

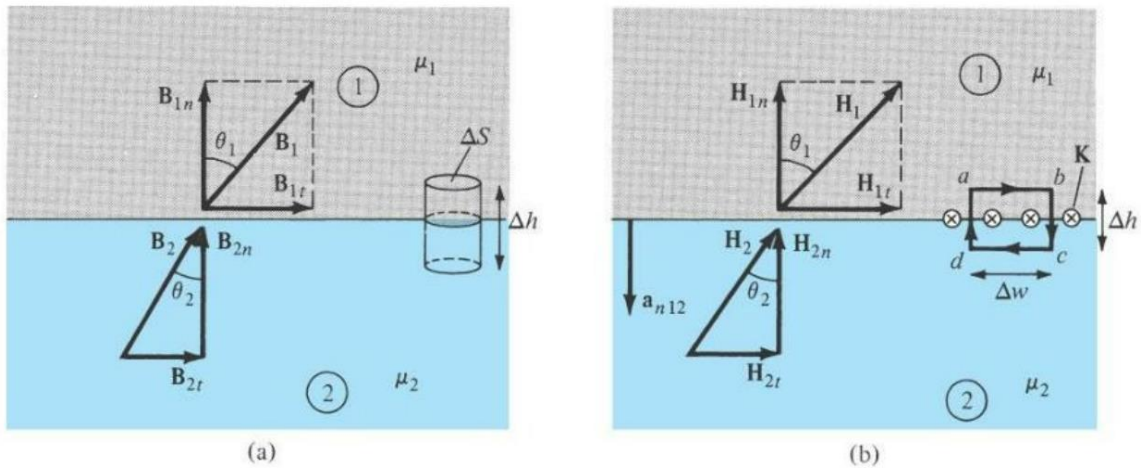


Fig. (4): Boundary conditions between two magnetic media.

As $\Delta h \rightarrow 0$

$$B_{1n}\Delta S - B_{2n}\Delta S = 0$$

thus

$$B_{1n} = B_{2n}$$

and

$$\mu_1 H_{1n} = \mu_2 H_{2n}$$

And in the closed path abcd:

$$K \cdot \Delta w = H_{1t}\Delta w + H_{1n}\frac{\Delta h}{2} + H_{2n}\Delta w - H_{2t}\Delta w - H_{2n}\frac{\Delta h}{2} - H_{1n}\frac{\Delta h}{2}$$

Where K is the sheet current, and

$$H_{1t} - H_{2t} = K$$

or

$$\frac{B_{1t}}{\mu_1} - \frac{B_{2t}}{\mu_2} = K$$

In general

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K}$$

If the fields make an angle θ to the surface, then

$$B_1 \cos \theta_1 = B_{1n} = B_{2n} = B_2 \cos \theta_2$$

$$\frac{B_1}{\mu_1} \sin \theta_1 = H_{1t} = H_{2t} = \frac{B_2}{\mu_2} \sin \theta_2$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$



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Electromagnetics I
Magnetostatics

There are two major laws governing magnetostatic fields: Biot-Savart's law and Ampere's circuit law. Like Coulomb's law, Biot-Savart's law is the general law of a magnetostatic. Just as Gauss's law is a special case of Coulomb's law, Ampere's law is a special case of Biot-Savart's law and is easily applied in problems involving symmetrical current distribution. The magnetic field strength $\mathbf{H}$ at a field point P due to a current I flowing in a wire as shown in Fig. (1) is given by:

$$d\mathbf{H} \propto \frac{I dl \sin \alpha}{R^2}$$

or

$$d\mathbf{H} = \frac{k I dl \sin \alpha}{R^2}$$

where k is the constant of proportionality. In SI units,

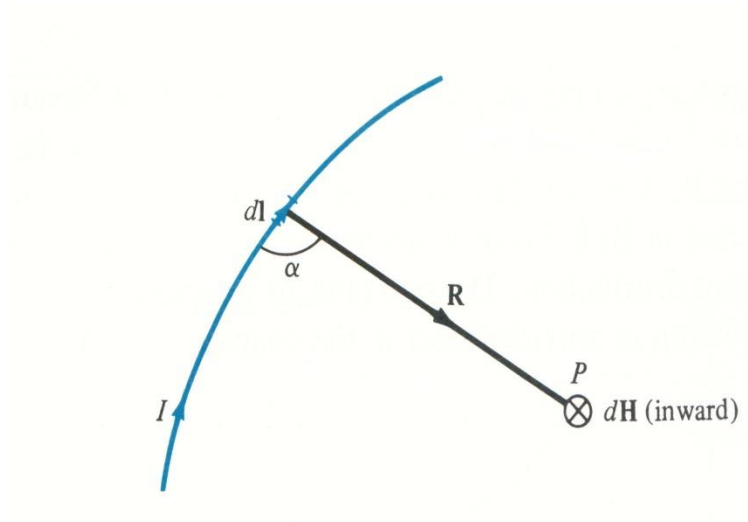


Fig. (1): Magnetic field $\mathbf{H}$ due to current I .

$$k = 1/4\pi$$

so:

$$d\mathbf{H} = \frac{I dl \sin \alpha}{4\pi R^2}$$

From the definition of cross product:

$$d\mathbf{H} = \frac{I dl \times \mathbf{a}_R}{4\pi R^3} = \frac{I dl \times \mathbf{R}}{4\pi R^3}$$

where

$$R = |\mathbf{R}|$$

and

$$\mathbf{a}_R = \mathbf{R}/R$$

Thus the direction of $d\mathbf{H}$ can be determined by the right-hand rule with the right-hand thumb pointing in the direction of the current, the Right-hand fingers encircling the wire in the direction of $d\mathbf{H}$ as shown in Fig. (2a).

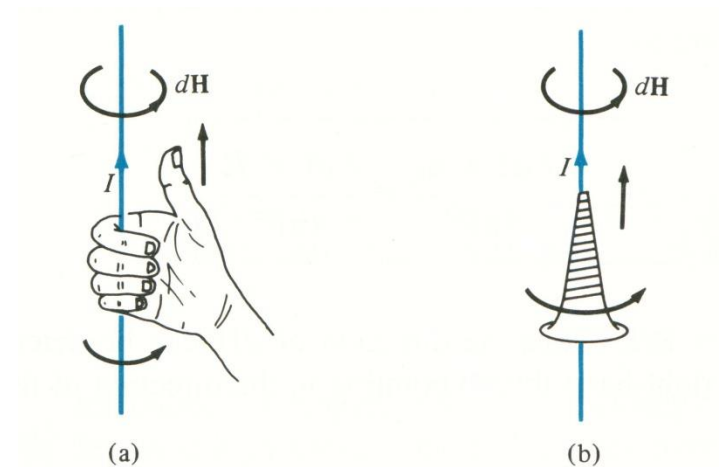


Fig. (2): Determining the direction of H using the right-hand or screw rules.

Alternatively, we can use the right-hand screw rule to determine the direction of $d\mathbf{H}$: with the screw placed along the wire and pointed in the direction of current flow, the direction of advance of the screw is the direction of $d\mathbf{H}$ as in Fig. (2b). It is customary to represent the direction of the magnetic field intensity $\mathbf{H}$ (or current I_0 by a small circle with a dot or cross sign depending on whether $\mathbf{H}$ (or I) is out of, or into, the page. Just as we can have different charge configurations, we can have different current distributions: line

current, surface current, and volume current as shown in Fig. (3). If we define K as the surface current density (in amperes/meter) and $\mathbf{J}$ as the volume current density (in amperes/meter square), the source element are related as:

$$I d\mathbf{l} \equiv K d\mathbf{S} \equiv \mathbf{J} dv$$

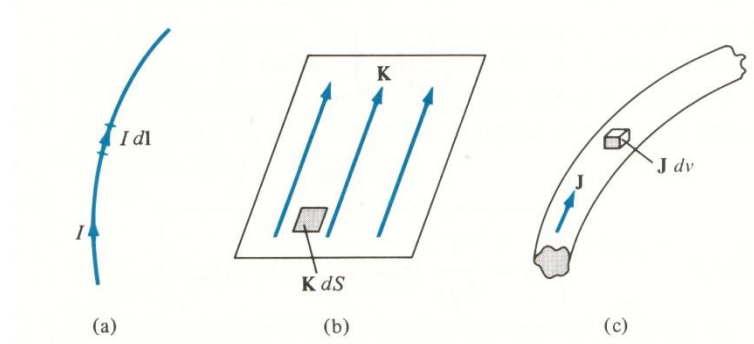


Fig. (3). Current distributions: line, surface, and volume.

Thus in terms of the distributed current sources, the Biot-Savart's law becomes:

$$\mathbf{H} = \int_L \frac{I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} \quad \text{line current}$$

$$\mathbf{H} = \int_S \frac{K d\mathbf{S} \times \mathbf{a}_R}{4\pi R^2} \quad \text{surface current}$$

$$\mathbf{H} = \int_V \frac{\mathbf{J} dv \times \mathbf{a}_R}{4\pi R^2} \quad \text{volume current}$$

As an example, let us apply Biot-Savart's law to determine the field due to a straight current carrying filamentary conductor of finite length AB as in Fig. (4). We assume that the conductor is along the z -axis with its upper and lower ends respectively subtending angles α_2 and α_1 at P , the point at which $\mathbf{H}$ is to be determined. If we consider the contribution $d\mathbf{H}$ at P due element $d\mathbf{l}$ at $(0, 0, z)$:

$$d\mathbf{H} = \frac{I d\mathbf{l} \times \mathbf{R}}{4\pi R^3}$$

but

$$d\mathbf{l} = dz \mathbf{a}_z$$

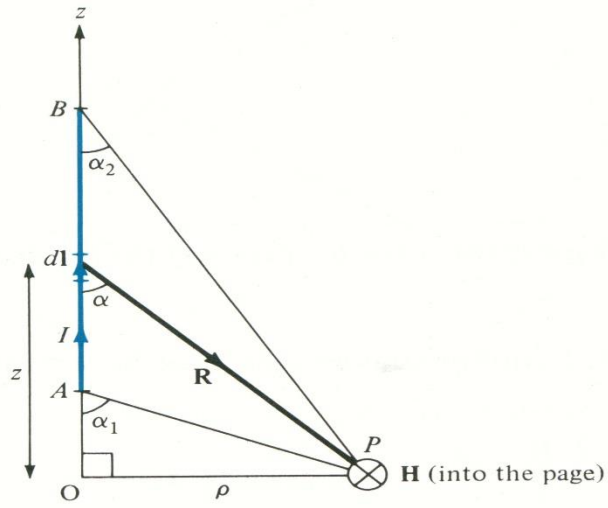


Fig. (4). Magnetic field at point A due to straight filament conductor.

and

$$\mathbf{R} = \rho \mathbf{a}_\rho - z \mathbf{a}_z$$

so

$$I d\mathbf{l} \times \mathbf{R} = \rho dz \mathbf{a}_\phi$$

hence,

$$\mathbf{H} = \int \frac{I \rho dz}{4\pi [\rho^2 + z^2]^{3/2}} \mathbf{a}_\phi$$

letting

$$z = \rho \cot \alpha$$

$$dz = -\rho \operatorname{cosec}^2 \alpha d\alpha$$

$$\mathbf{H} = -\frac{1}{4\pi} \int_{\alpha_1}^{\alpha_2} \frac{\rho^2 \operatorname{cosec}^2 \alpha d\alpha}{\rho^3 \operatorname{cosec}^3 \alpha} \mathbf{a}_\phi$$

$$\mathbf{H} = -\frac{1}{4\pi} \mathbf{a}_\phi \int_{\alpha_1}^{\alpha_2} \sin \alpha d\alpha$$

$$\mathbf{H} = \frac{1}{4\pi\rho} (\cos\alpha_2 - \cos\alpha_1) \mathbf{a}_\phi$$

This expression is generally applied for any straight filamentary conductor of finite length. Notice that $\mathbf{H}$ is always along the unit vector $\mathbf{a}_\phi$ (i.e., along concentric circular paths) irrespective of the length of the wire or the point of interest P . As a special case, when the conductor is semi-infinite (with respect to P) so that point A is now at $(0, 0, 0)$ while B is at $(0, 0, \infty)$; $\alpha_1 = 90^\circ$, $\alpha_2 = 0^\circ$, thus:

$$\mathbf{H} = \frac{1}{4\pi\rho} \mathbf{a}_\phi$$

Another special case is when the conductor is *infinite* in length. For this case, point A is at $(0, 0, -\infty)$ while B is at $(0, 0, \infty)$, $\alpha_1 = 180^\circ$, $\alpha_2 = 0^\circ$, so:

$$\mathbf{H} = \frac{1}{2\pi\rho} \mathbf{a}_\phi$$

To find unit vector $\mathbf{a}_\phi$ is not always easy. A simple approach is to determine $\mathbf{a}_\phi$ from:

$$\mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho$$

where $\mathbf{a}_l$ is a unit vector along the line current and $\mathbf{a}_\rho$ is a unit vector along the perpendicular line from the line current to the field point.



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Maxwell's Equation

The magnetic flux density $\mathbf{B}$ is similar to the electric flux density $\mathbf{D}$ as $\mathbf{D} = \epsilon_0 \mathbf{E}$ in free space, the magnetic flux density $\mathbf{B}$ is related to the magnetic field intensity $\mathbf{H}$ according to:

$$\mathbf{B} = \mu_0 \mathbf{H}$$

where μ_0 is a constant known as the permeability of free space. The constant is in henrys/meter (H/m) and has the value of:

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

The magnitude flux through a surface $\mathbf{S}$ is given by:

$$\psi = \int_S \mathbf{B} \cdot d\mathbf{S}$$

where the magnetic flux Ψ is in Webers (Wb) and the magnetic flux density is in Webers/square meter (Wb/m²) or tesla. The magnetic flux line is the path to which $\mathbf{B}$ is tangential at every point in a magnetic field. The needle of a magnetic compass will orient itself if placed in the magnetic field along the line. For example, the magnetic flux lines due to a straight long wire are shown in Fig. (1).

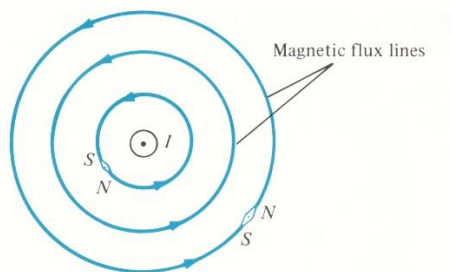


Fig. (1): Magnetic flux lines due to a straight with current out of the page.

The direction of $\mathbf{B}$ is taken as that indicated as "north" by the needle of the magnetic compass. Each flux line is closed and has no beginning or end. Though Fig. (1) is for a straight, current-carrying conductor, it is generally true that magnetic flux lines are closed and do not cross each other regardless of the circuit geometry. In an electrostatic field, the flux passing through a closed surface is the same as the charge enclosed:

$$\Psi = \oint \mathbf{D} \cdot d\mathbf{S} = Q$$

thus it is possible to have an isolated electric charge as shown in Fig. (2), which also reveals that electric flux lines are not necessarily closed. Unlike electric flux lines, magnetic flux lines always close upon themselves as in Fig. (2). This is due to the fact that it is not possible to have isolated magnetic poles by dividing a magnetic bar successively into two, we end up with magnetic flux lines due to a straight wire with current coming out of the page. We find it impossible to separate the north from the South Pole. In other words, an isolated magnetic charge does not exist.

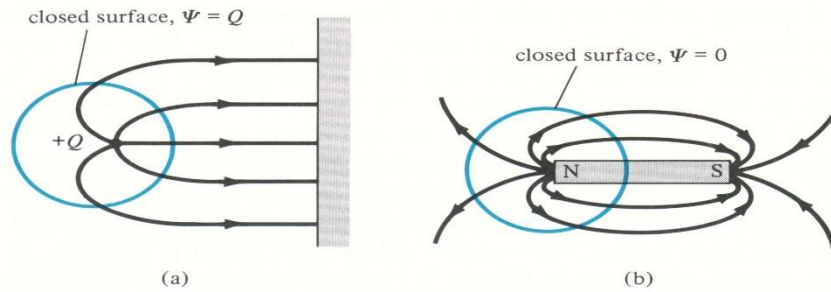


Fig. (2): Flux lines leaving a closed surface due to a) an isolated charge, b) magnet.

Thus, the total flux through a closed surface in a magnetic field must be zero, or the total flux through a closed surface in a magnetic field must be zero:

$$\int_S \mathbf{B} \cdot d\mathbf{S} = 0$$

This equation is referred to as the law of conservation of magnetic flux or Gauss's law for magnetostatic fields just that Gauss's law for electrostatic fields. Although the magnetostatic field is not conservative, magnetic flux is conserved. By applying the divergence theorem:

$$\int_S \mathbf{B} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{B} \, dv = 0$$

or

$$\nabla \cdot \mathbf{B} = 0$$

This equation is the fourth Maxwell's equation to be derived which shows that magnetostatic fields have no sources or sinks and suggest that magnetic field lines are always continuous.

Maxwell's Equations for Static EM Fields

Maxwell's four equations for static electromagnetic fields are now given as:

$\nabla \times \mathbf{E} = 0$	$\int_L \mathbf{E} \cdot d\mathbf{l} = 0$	<i>Conservativeness of electrostatic field</i>
$\nabla \times \mathbf{H} = \mathbf{J}$	$\int_L \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{S}$	<i>Ampere's Law</i>
$\nabla \cdot \mathbf{D} = \rho_v$	$\int_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	<i>Gauss's Law</i>
$\nabla \cdot \mathbf{B} = 0$	$\int_S \mathbf{B} \cdot d\mathbf{S} = 0$	<i>Nonexistence of magnetic monopole.</i>



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Electromagnetics I
Method of Images

The method of images, introduced by Lord Kelvin in 1848, is commonly used to determine V , $\mathbf{E}$, $\mathbf{D}$, and ρ_s due to charges in the presence of conductors. By this method, we avoid solving Poisson's or Laplace's equation but rather utilize the fact that a conducting surface is an equipotential. Although the method does not apply to all electrostatic problems, it can reduce a formidable problem to a simple one. The image theory states that a given charge configuration above an infinite grounded perfect conducting plane may be replaced by the charge configuration itself, its image, and an equipotential surface in place of the conducting plane. Typical examples of point line, and volume charge configurations are portrayed in the figure, and their corresponding image configurations are in the figure.

In applying the image method, two conditions must always be satisfied:

- The image charge(s) must be located in the conducting region.
- The image charge(s) must be located such that on the conducting surface(s) the potential is zero or constant.

The first condition is necessary to satisfy Poisson's equation, and the second condition ensures that the boundary conditions are satisfied. Let us now apply the image theory to two specific problems.

A Point Charge above a Grounded Conducting Plane

Consider a point charge Q placed at a distance h from a perfect conducting plane of infinite extent as in the figure. The image configuration is in Fig. (1). The electric field at point P (x, y, z) is given by:

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_-$$
$$\mathbf{E} = \frac{Q\mathbf{r}_1}{4\pi\epsilon_0 r_1^3} + \frac{-Q\mathbf{r}_2}{4\pi\epsilon_0 r_2^3}$$

The distance vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ are given by:

$$\mathbf{r}_1 = (x, y, z) - (0, 0, h) = (x, y, z - h)$$

$$\mathbf{r}_2 = (x, y, z) - (0, 0, -h) = (x, y, z + h)$$

so:

$$\mathbf{E} = \frac{Q}{4\pi \epsilon_0} \left[\frac{x\mathbf{a}_x + y\mathbf{a}_y + (z-h)\mathbf{a}_z}{[x^2 + y^2 + (z-h)^2]^{3/2}} - \frac{x\mathbf{a}_x + y\mathbf{a}_y + (z+h)\mathbf{a}_z}{[x^2 + y^2 + (z+h)^2]^{3/2}} \right]$$

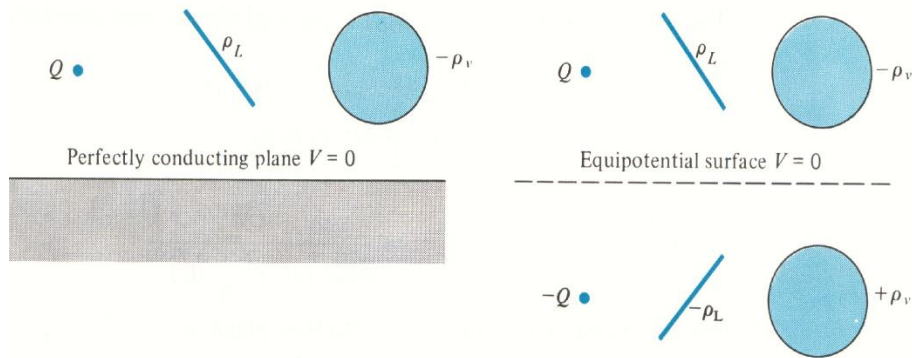


Fig. (1): Image System: (a) charge configuration above a perfectly conducting plane, (b) image configuration with the conducting plane replaced by equipotential surface.

It should be noted that when $z=0$, $\mathbf{E}$ has only the z -component, confirming that $\mathbf{E}$ is normal to the conducting surface. The potential at P is easily obtained using:

$$V = -\int \mathbf{E} \cdot d\mathbf{l}$$

thus

$$V = V_+ + V_-$$

$$V = \frac{Q}{4\pi \epsilon_0 r_1} + \frac{-Q}{4\pi \epsilon_0 r_2}$$

$$V = \frac{Q}{4\pi \epsilon_0} \left[\frac{1}{[x^2 + y^2 + (z-h)^2]^{1/2}} - \frac{1}{[x^2 + y^2 + (z+h)^2]^{1/2}} \right]$$

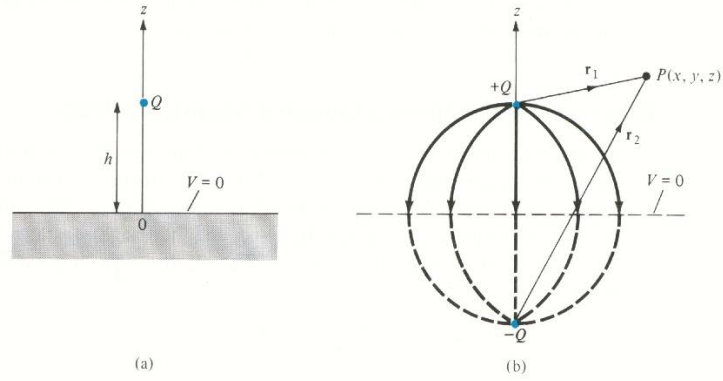


Fig. (2): (a) Point charge and grounded conducting plane (b) Image configuration and field lines.

For $z \geq 0$ and $V=0$ for $z \leq 0$. Note that $V(z=0)=0$. The surface charge density of the induced charge can also be obtained as:

$$\rho_s = D_n = \epsilon_0 E_{n_{z=0}}$$

$$\rho_s = \frac{-Qh}{2\pi [x^2 + y^2 + h^2]^{3/2}}$$

The total induced charge on the conducting plane is:

$$Q_i \int \rho_s dS = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{-Q h dx dy}{2\pi [x^2 + y^2 + h^2]^{3/2}}$$

By changing variables:

$$\rho^2 = x^2 + y^2$$

$$dx dy = \rho d\rho d\phi$$

$$Q_i = -\frac{Qh}{2\pi} \int_0^{2\pi} \int_0^{\infty} \frac{\rho d\rho d\phi}{[\rho^2 + h^2]^{3/2}}$$

Integrating over ϕ gives 2π , and letting:

$$\rho d\rho = \frac{1}{2} d(\rho^2)$$

thus:

$$Q_i = -\frac{Qh}{2\pi} \int_0^\infty [\rho^2 + h^2]^{-3/2} \frac{1}{2} d(\rho^2)$$

$$Q_i = \frac{Qh}{[\rho^2 + h^2]^{1/2}} \Big|_0^\infty$$

$$Q_i = -Q$$

as expected, because all flux lines terminating on the conductor would have terminated on the image charge if the conductor were absent.

A Line Charge above a Grounded Conducting Plane

Consider an infinite charge with density ρ_L C/m located at a distance h from the grounded conducting plane $z=0$. The same image system in the figure above applies to the line charge except that Q is replaced by ρ_L . The infinite line charge ρ_L may be assumed to be at $x=0, z=h$ and the image $-\rho_L$ at $x=0, z=-h$ so that the two are parallel to the y -axis. The electric field at point P is given by:

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_-$$

$$\mathbf{E} = \frac{\rho_L}{2\pi \epsilon_0 \rho_1} \mathbf{a}_{\rho_1} + \frac{-\rho_L}{2\pi \epsilon_0 \rho_2} \mathbf{a}_{\rho_2}$$

The distance vectors ρ_1 and ρ_2 are given by:

$$\rho_1 = (x, y, z) - (0, 0, h) = (x, y, z - h)$$

$$\rho_2 = (x, y, z) - (0, 0, -h) = (x, y, z + h)$$

so:

$$\mathbf{E} = \frac{\rho_L}{2\pi \epsilon_0} \left[\frac{x \mathbf{a}_x + (z - h) \mathbf{a}_z}{x^2 + (z - h)^2} - \frac{x \mathbf{a}_x + (z + h) \mathbf{a}_z}{x^2 + (z + h)^2} \right]$$

Again, notice that when $z=0$, $\mathbf{E}$ has only the z -component, confirming that $\mathbf{E}$ is normal to the conducting surface. The potential at P is obtained:

$$V = -\int \mathbf{E} \cdot d\mathbf{l}$$

Thus

$$V = V_+ + V_-$$

$$V = -\frac{\rho_L}{2\pi\epsilon_0} \ln \rho_1 - \frac{-\rho_L}{2\pi\epsilon_0} \ln \rho_2$$

$$V = -\frac{\rho_L}{2\pi\epsilon_0} \ln \frac{\rho_1}{\rho_2}$$

Substituting

$$\rho_1 = |\rho_1|$$

and

$$\rho_2 = |\rho_2|$$

gives:

$$V = -\frac{\rho_L}{2\pi\epsilon_0} \ln \left[\frac{x^2 + (z-h)^2}{x^2 + (z+h)^2} \right]^{1/2}$$

for $z \geq 0$ and $V=0$, and for $z \leq 0$. Note that $V(z=0) = 0$. The surface charge induced on the conducting plane is given by:

$$\rho_s = D_n = \epsilon_0 E_{z,z=0} = \frac{-\rho_L h}{\pi(x^2 + h^2)}$$

The induced charge per length on the conducting plane is:

$$\rho_i = \int \rho_s dx = -\frac{\rho_L h}{\pi} \int_{-\infty}^{\infty} \frac{dx}{x^2 + h^2}$$

By letting:

$$x = h \tan \alpha$$

Hence:

$$\rho_i = -\frac{\rho_L h}{\pi} \int_{-\pi/2}^{\pi/2} \frac{d\alpha}{h}$$

$$\rho_i = -\rho_L$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Poisson's and Laplace's Equations

The procedure for determining the electric field $\mathbf{E}$ has generally been using either Coulomb's law or Gauss's law when the charge distribution is known, or using:

$$\mathbf{E} = -\nabla V$$

when the potential V is known throughout the region. In most practical situations, however, neither the charge distribution nor the potential distribution is known. When only electrostatic conditions (charge and potential) at some boundaries are known and it is desired to find $\mathbf{E}$ and V throughout the region, then such problems are usually tackled using Poisson's or Laplace's equation or the method of images, and they are usually referred to as boundary-value problems.

Poisson's and Laplace's Equations

Poisson's and Laplace's equations are easily derived from Gauss's law:

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon \mathbf{E}) = \rho_v$$

and

$$\mathbf{E} = -\nabla V$$

Substituting gives:

$$\nabla \cdot (-\epsilon \nabla V) = \rho_v$$

for an inhomogeneous medium. For a homogeneous medium:

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}$$

This is known as Poisson's equation. A special case of this equation occurs when:

$$\rho_v = 0$$

i.e., for a charge-free region, which becomes:

$$\nabla^2 V = 0$$

which is known as Laplace's equation. Thus, Laplace's equation in Cartesian, cylindrical, or spherical coordinates respectively is given by:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

depending on whether the potential is $V(x, y, z)$, $V(\rho, \phi, z)$, $V(r, \theta, \phi)$. Poisson's equation in those coordinates systems may be obtained by simply replacing zero on the right-hand side of Laplace's equation with:

$$-\rho_v / \epsilon$$

Laplace's equation is of primary importance in solving electrostatic problems involving a set of conductors maintained at different potentials. Laplace's and Poisson's equations are not only useful in solving electrostatic field problem; they are used in various other field problems. For example, V would be interpreted as magnetic potential in magnetostatics, as temperature in heat conduction, as stress function in fluid flow, and as pressure head in seepage.

Uniqueness Theorem

Since there are several methods (analytical, graphical, numerical, experimental, etc.) of solving a given problem, we may wonder whether solving Laplace's equation in different ways gives different solutions. Therefore, before we begin to solve Laplace's equation, we should answer this question: If a solution of Laplace's equation satisfies a given set of boundary conditions, is this the only possible solution? The answer is yes: there is only one solution. We say that the solution is unique. Thus, any solution of Laplace's equation, which satisfies the same boundary conditions, must be the only solution regardless of the method used. This is known as the uniqueness theorem. The theorem applies to any solution of Poisson's or Laplace's equation in a given region or closed surface. The theorem is proved by contradiction. We assume that there are two solutions V_1 and V_2 of Laplace's equation both of which satisfy the prescribed boundary conditions. Thus:

$$\nabla^2 V_1 = 0$$

$$\nabla^2 V_2 = 0$$

$$V_1 = V_2$$

on the boundary. We consider their difference:

$$V_d = V_2 - V_1$$

which obeys:

$$\nabla^2 V_d = \nabla^2 V_2 - \nabla^2 V_1 = 0$$

$$V_d = 0$$

on the boundary. From the divergence theorem:

$$\int_V \nabla \cdot \mathbf{A} dv = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

Let

$$\mathbf{A} = V_d \nabla V_d$$

and use a vector identity:

$$\nabla \cdot \mathbf{A} = \nabla \cdot (V_d \nabla V_d) = V_d \nabla^2 V_d + \nabla V_d \cdot \nabla V_d$$

but

$$\nabla^2 V_d = 0$$

accordingly:

$$\nabla \cdot \mathbf{A} = \nabla V_d \cdot \nabla V_d$$

substituting gives:

$$\int_V \nabla V_d \cdot \nabla V_d dv = \oint_S V_d \nabla V_d \cdot d\mathbf{S}$$

The right-hand side of the above equation vanishes, hence:

$$\int_V |\nabla V_d|^2 dv = 0$$

since the integration is always positive:

$$\nabla V_d = 0$$

or

$$V_d = V_2 - V_1 = \text{cons}$$

everywhere in V which must be consistent, hence:

$$V_d = 0$$

or

$$V_1 = V_2$$

everywhere, showing that V_1 and V_2 cannot be different solutions of the same problem. This is the uniqueness theorem: if a solution to Laplace's equation can be found that satisfies the boundary conditions, then the solution is unique. Similar steps can be taken to show that the theorem applies to Poisson's equation and to prove the theorem for the case where the electric field (potential gradient) is specified on the boundary.

Three things that uniquely describe a problem:

- an appropriate differential equation, Laplace's or Poisson's equation,
- a solution region,
- a prescribed boundary conditions.

A problem does not have a unique solution and cannot be solved completely if any of the three items is missing.

Procedures for Solving Poisson's or Laplace's Equation

The following general procedure may be taken in solving a given boundary-value problem involving Poisson's or Laplace's equation:

- Solve Laplace's if $\rho_v = 0$ or Poisson's if $\rho_v \neq 0$ equation using either (a) direct integration when V is a function of one variable, or (b) separation of variables if V is a function of more than one variable. The solution at this point is not unique but expressed in terms of unknown integration constants to be determined.
- Apply the boundary conditions to determine a unique solution for V . Imposing the given boundary conditions makes the solution unique.
- Having obtained V , find E using:

$$\mathbf{E} = -\nabla V$$

and

$$\mathbf{D} = \varepsilon \mathbf{E}$$

if desired, find the charge Q induced on a conductor using:

$$Q = \int \rho_s \, dS$$

where

$$\rho_s = D_n$$

and D_n is the component of $\mathbf{D}$ normal to the conductor. If necessary, the capacitance between two conductors can be found using:

$$C = Q/V$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Polarization in Dielectrics

Consider an atom of the dielectric as consisting of a negative charge $-Q$ (electron cloud) and a positive charge $+Q$ (nucleus). Since we have equal amounts of positive and negative charge, the whole atom or molecule is electrically neutral. When an electric field $\mathbf{E}$ is applied, the positive charge is displaced from its equilibrium position in the direction of $\mathbf{E}$ by the force $F_+ = QE$ while the negative charge is displaced in the opposite direction by the force $F_- = QE$. A dipole results from the displacement of the charges and the dielectric is said to be polarized. In the polarized state, the electric cloud is distorted by the applied electric field $\mathbf{E}$. This distorted charge distribution is equivalent, by the superposition, to the original distribution plus a dipole whose moment is:

$$\mathbf{P} = Q\mathbf{d}$$

where $\mathbf{d}$ is the distance vector from $-Q$ to $+Q$ of the dipole. If there are N dipoles in a volume v of the dielectric, the total dipole moment due to the electric field is:

$$Q_1 \mathbf{d}_1 + Q_2 \mathbf{d}_2 + \dots + Q_N \mathbf{d}_N = \sum_{k=1}^N Q_k \mathbf{d}_k$$

As a measure of intensity of the polarization, a polarization vector $\mathbf{P}$ is defined as the dipole moment per unit volume of the dielectric; that is:

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^N Q_k \mathbf{d}_k}{\Delta v}$$

Thus, the major effect of the electric field $\mathbf{E}$ on a dielectric is the creation of dipole moments that align themselves in the direction of $\mathbf{E}$. The effect of the dielectric on the electric field $\mathbf{E}$ is thus to increase $\mathbf{D}$ inside it by amount $\mathbf{P}$. In other words, due to the polarization of $\mathbf{E}$ to the dielectric material, the flux density is greater than it would be in free space. Hence:

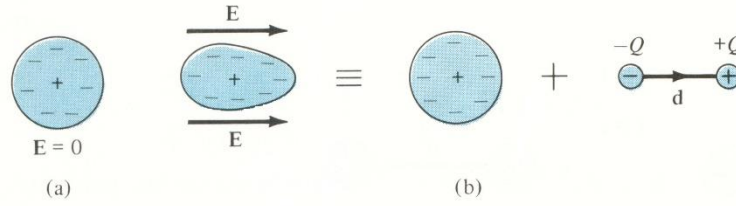


Fig. (1): Polarization of a non-polar atom or molecule.

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

The polarization $\mathbf{P}$ is expected to vary directly as the applied electric field $\mathbf{E}$. For some dielectrics, this is usually the case and we have

$$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$$

where χ_e is known as the electric susceptibility of the material, is more or less a measure of how sensitive a given dielectric is to electric fields.

Dielectric Constant and Strength

Since

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E}$$

or

$$\mathbf{D} = \epsilon \mathbf{E}$$

where

$$\epsilon = \epsilon_0 \epsilon_r$$

and

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$$

ϵ is called the permittivity of the dielectric, ϵ_0 is the permittivity of free space, defined as approximately $10^{-9}/36\pi$ F/m, and ϵ_r is called the dielectric constant or relative permittivity which is the ratio of the permittivity of the dielectric to that of free space. Practically speaking, no dielectric is ideal. When the electric field in a dielectric is sufficiently large, it begins to pull electrons completely out of the molecules, and the dielectric becomes conducting. Dielectric breakdown is said to have occurred when a dielectric becomes conducting. Dielectric breakdown occurs in all kinds of dielectric materials (gases, liquids, or solids) and depends on the nature of material, temperature, humidity, and the amount of time that field is applied. The minimum value of the electric

field at which dielectric breakdown occurs is called the dielectric strength of the dielectric material. The dielectric strength is the maximum electric field that a dielectric can tolerate or withstand without breakdown.

Linear, Isotropic, and Homogeneous Dielectrics

1. A material is said to be linear if $\mathbf{D}$ varies linearly with $\mathbf{E}$ and nonlinear otherwise.
2. Materials for which ϵ is the same at all points and does not depend on (x, y, z) are said to be homogeneous. They are said to be inhomogeneous (or non-homogeneous) when ϵ is dependent of the space coordinates. The atmosphere is a typical example of an inhomogeneous medium; its permittivity varies with altitude.
3. Materials for which $\mathbf{D}$ and $\mathbf{E}$ are in the same direction are said to be isotropic, they have the same properties in all directions. For anisotropic (or non-isotropic) materials, $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ are not parallel; ϵ (or χ_e) does has nine components that are collectively referred to as tensor.

A dielectric material in which $\mathbf{D} = \epsilon \mathbf{E}$ applies, is linear if ϵ does not change with the applied $\mathbf{E}$ field, homogeneous if ϵ does not change from point to point and isotropic if ϵ does not change with direction. The same idea holds for a conducting material in which:

$$\mathbf{J} = \sigma \mathbf{E}$$

applies. The material is linear if σ does not vary with $\mathbf{E}$, homogeneous if σ is the same at all points, and isotropic if σ does not vary with direction. Only linear, isotropic, and homogeneous media will be considered. For such media, all formulas for free space can be applied by merely replacing ϵ_0 with $\epsilon_0 \epsilon_r$. Thus, Coulomb's law becomes:

$$\mathbf{F} = \frac{Q_1 Q_2}{4\pi \epsilon_0 \epsilon_r R^2} \mathbf{a}_R$$

and

$$W = \frac{1}{2} \int \epsilon_0 \epsilon_r \mathbf{E}^2 dv$$

when applied to a dielectric medium.

Continuity Equation and Relaxation time:

Due to the principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the closed surface of the volume. Thus current I_{out} coming out of the closed surfaces is:

$$I_{out} = \oint \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_{in}}{dt}$$

where Q_{in} is the total charge enclosed by the closed surface. Invoking divergence theorem:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{J} dv$$

but

$$-\frac{dQ_{in}}{dt} = -\frac{d}{dt} \int_v \rho_v dv = -\int_v \frac{\partial \rho_v}{\partial t} dv$$

Hence:

$$\int_v \nabla \cdot \mathbf{J} dv = -\int_v \frac{\partial \rho_v}{\partial t} dv$$

or

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

which is called the continuity of current equation. It must be kept in mind that the continuity equation is derived from principle of conservation of charge and essentially states that there can be no accumulation of charge at any point. For steady currents:

$$\partial \rho_v / dt = 0$$

hence

$$\nabla \cdot \mathbf{J} = 0$$

showing that the total charge leaving a volume is the same as the total charge entering it. Kirchhoff's current law follows from this. Now from Ohm's law:

$$\mathbf{J} = \sigma \mathbf{E}$$

and Gauss's Law:

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}$$

Substituting yields:

$$\nabla \cdot \sigma \mathbf{E} = \frac{\sigma \rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

or

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\varepsilon} \rho_v = 0$$

By separating variables:

$$\frac{\partial \rho_v}{\rho_v} = -\frac{\sigma}{\varepsilon} \partial t$$

and integrating both sides gives:

$$\ln \rho_v = -\frac{\sigma t}{\varepsilon} + \ln \rho_{v0}$$

Thus:

$$\rho_v = \rho_{v0} e^{-t/T_r}$$

where

$$T_r = \frac{\varepsilon}{\sigma}$$

and ρ_{v0} is the initial charge density. The equation shows that as a result of introduction charge at some interior point of the material there is a decay of volume charge density ρ_v . Associated with the decay is charge movement from the interior point at which it was introduced to the surface of the material. The time constant T_r is known as the relaxation time or rearrangement time. Relaxation time is the time it takes a charge placed in the interior of a material to drop to $e^{-1} = 36.8$ percent of its initial value. It is short for good conductors and long for good dielectrics. For example, copper $\sigma = 5.8 \times 10^7$ mhos/m, $\varepsilon_r = 1$, and

$$T_r = \varepsilon_0 \varepsilon_r / \sigma = 1.53 \times 10^{-19} \text{ s}$$

showing a rapid decay of charge placed inside copper. This implies that for good conductors, the relaxation time is so short that most of charge will vanish from any interior point and appear at the surface as surface charge. On the other hand, for fused quartz, for instance, $\sigma = 10^{-7}$ mhos/m, $\varepsilon_r = 5$,

$$T_r = \varepsilon_0 \varepsilon_r / \sigma = 51.2 \text{ days}$$

showing a very large relaxation time. Thus for good dielectric, one may consider the introduced charge to remain wherever placed.



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Communication Engineering Department
Electromagnetics I
Properties of Materials

Materials may be classified in terms of their conductivity σ as conductors and nonconductors. The conductivity of a material usually depends on temperature and frequency. A material with high conductivity ($\sigma \gg 1$) is referred to as a metal whereas one with low conductivity ($\sigma \ll 1$) is referred to as an insulator. A material whose conductivity lies somewhere between those of metals and insulators is called a semiconductor. The major difference between a metal and an insulator lies in the amount of electrons available for conduction of current. Dielectric materials have few electrons available for conduction of current in contrast to metals, which have an abundance of free electrons.

Convection and Conduction Currents

Electric voltage (or potential difference) and current are two fundamental quantities in electrical engineering. Electric current is generally caused by the motion of electric charges. The current (in amperes) through a given area is defined as the electric charge flowing through the area per unit time:

$$I = \frac{dQ}{dt}$$

Thus in a current of one ampere, charge is being transferred at a rate of one Coulomb per second. If current ΔI flows through a surface ΔS , the current density is:

$$J_n = \frac{\Delta I}{\Delta S}$$

or

$$\Delta I = J_n \cdot \Delta S$$

assuming that the current density is perpendicular to the surface. If the current density is not normal to the surface:

$$\Delta I = \mathbf{J} \cdot \Delta \mathbf{S}$$

Thus, the total current flowing through a surface S is:

$$I = \int \mathbf{J} \cdot d\mathbf{S}$$

Depending on how I is produced, there are different kinds of currents densities: convection current density, conduction current density, and displacement current density. The current I through S is merely the flux of the current density $\mathbf{J}$. Convection current, as distinct from conduction current, does not involve conductors and consequently does not satisfy Ohm's law. It occurs when current flows through an insulating medium such as liquid, rarefied gas, or a vacuum. A beam of electrons in a vacuum tube, for example, is a convection current.

Consider a filament of Fig. (1). If there is a flow of charge, of density ρ_v , at velocity

$$\mathbf{u} = u_y \mathbf{u}_y$$

then, the current through the filament is:

$$\Delta I = \Delta Q / \Delta t = \rho_v \Delta S (\Delta l / \Delta t) = \rho_v \Delta S u_y$$

The current density at a given point as the current through a unit normal of area at that point.

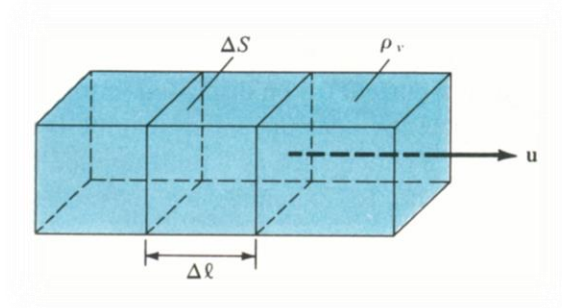


Fig. (1): Current in a filament.

The y-directed current density J_y is given by :

$$J_y = \frac{\Delta I}{\Delta S} = \rho_v u_y$$

hence, in general

$$\mathbf{J} = \rho_v \mathbf{u}$$

The current I is the convection current and J is the convection current density in amperes/square meter (A/m^2). Conduction current requires a conductor. A conductor is characterized by a large amount of free electrons that provide conduction current due to an impressed electric field. When an electric field E is applied, the force on electron with charge $-e$ is:

$$F = -e E$$

Since the electron is not in free space, it will not be accelerated under the influence of the electric field. Rather, it suffers constant collision with the atomic lattice and from drifts from one atom to another. If the electron with mass m is moving in an electric field E with an average drift velocity u , according to Newton's law, the average change in momentum of the free electron must match the applied force. Thus:

$$mu / \tau = -e E$$

or

$$u = -(e\tau / m) E$$

where τ is the average time interval between collisions. This indicates that the drift velocity of the electron is directly proportional to the applied field. If are n electrons per unit volume, the electronic charge density is given by:

$$\rho_v = -ne$$

Thus the conduction current density is:

$$J = \rho_v u = (ne^2\tau / m) E = \sigma E$$

or

$$J = \sigma E$$

where

$$\sigma = ne^2\tau / m$$

is the conductivity of the conductor.

Conductors

A conductor has abundance of charge that is free to move. Consider an isolated conductor, such as shown in Fig. (2). When an external electric field E_e is applied, the positive free charges are pushed along the same direction as the applied field, while the negative free charges move in the opposite direction. This charge migration takes place very quickly. The free charges do two things. First, they accumulate on the surface of the conductor and

form an induced surface charge. Second, the induced charges set up an internal induced field $\mathbf{E}_i$, which cancels the externally applied field $\mathbf{E}_e$. This leads to an important property of a conductor. A perfect conductor cannot contain an electrostatic field within it. A conductor is called an equipotential body, implying that the potential is the same every where in the conductor. This is based on the fact that:

$$\mathbf{E} = -\nabla V = 0$$

Another way of looking at this is to consider Ohm's law, $\mathbf{J} = \sigma \mathbf{E}$. To maintain a finite current density $\mathbf{J}$, in a perfect conductor (σ goes to infinity), requires that the electric field inside the conductor must vanish. In other words, $\mathbf{E}$ goes to zero because σ goes to infinity in a perfect conductor. If some charges are introduced in the interior of such a conductor, the charges will move to the surface and redistribute themselves quickly in such a manner that the field inside the conductor. According to Gauss's law, if $\mathbf{E} = 0$, the charge density ρ_v must be zero. Hence, a perfect conductor cannot contain an electrostatic field within it. Under static conditions:

$$\mathbf{E} = 0$$

$$\rho_v = 0$$

$$V_{ab} = 0$$

We now consider a conductor whose ends are maintained at potential difference V , as shown in Fig. (2). The conductor is not isolated but wired to a source of electromotive force, which compels the free charges to move and prevents eventual establishment of electrostatic equilibrium. Thus, an electric field must exist inside the conductor to sustain the flow of current. As the electrons move, they encounter some damping forces called resistance. Suppose the conductor has a uniform cross section S and is of length l . The direction of the electric field $\mathbf{E}$ produced is the same as the direction of the flow of positive charges or current I .

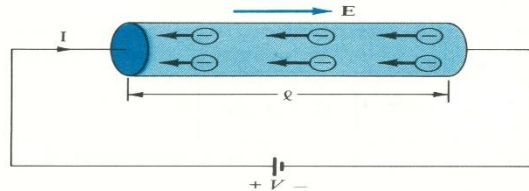


Fig. (2): A conductor of uniform cross section under an applied $\mathbf{E}$ field.

This direction is opposite to the direction of the flow of electrons. The electric field applied is uniform and its magnitude is given by:

$$E = \frac{V}{l}$$

Since the conductor has a uniform cross section:

$$J = \frac{I}{S}$$

And:

$$\frac{I}{S} = \sigma E = \frac{\sigma V}{\ell}$$

hence

$$R = \frac{V}{I} = \frac{\ell}{\sigma S}$$

or

$$R = \frac{\rho \ell}{S}$$

where

$$\rho = 1/\sigma$$

is the resistivity of the material. This is useful in determining the resistance of any conductor of uniform cross section but is not applicable to non-uniform cross section of the conductor. The basic definition of resistance R as the potential difference V between the two ends of the conductor to the current I through the conductor still applies. Therefore, for a non-uniform cross-section:

$$R = \frac{V}{I} = \frac{\int \mathbf{E} \cdot d\mathbf{l}}{\int \sigma \mathbf{E} \cdot d\mathbf{S}}$$

Power P (in watts) is defined as the rate of change of energy W (in Joules) or force times velocity. Hence:

$$\int \rho_v dv \mathbf{E} \cdot \mathbf{u} = \int \mathbf{E} \cdot \rho_v \mathbf{u} dv$$

or

$$P = \int \mathbf{E} \cdot \mathbf{J} dv$$

which is known as Joule's Law. The power density w_p is given:

$$w_p = \frac{dP}{dv} = \mathbf{E} \cdot \mathbf{J} = \sigma |\mathbf{E}|^2$$

For a conductor with uniform cross section,

$$dv = dS \, dl$$

and

$$P = \int_L \mathbf{E} \cdot d\mathbf{l} \int_S \mathbf{J} \cdot d\mathbf{S} = VI$$

or

$$P = I^2 R$$

which is the more common form of Joule's Law in electric circuit theory.



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Resistance and Capacitor

If the cross section of the conductor is not uniform, the resistance is obtained from:

$$R = \frac{V}{I} = \frac{\int \mathbf{E} \cdot d\mathbf{l}}{\oint \sigma \mathbf{E} \cdot d\mathbf{S}}$$

The problem of finding the resistance of a conductor of non-uniform cross section can be treated as a boundary-value problem. The resistance R (or conductance $G=1/R$) of a given conducting material can be found by following these steps:

- Choose a suitable coordinate system,
- Assume V_0 as the potential difference between conductor terminals.
- Solve Laplace's equation $\nabla^2 V$ to obtain V . Then determine $\mathbf{E}$ from:

$$\mathbf{E} = -\nabla V$$

and I from:

$$I = \int \sigma \mathbf{E} \cdot d\mathbf{S}$$

finally, obtain R as:

$$R = \frac{V_0}{I}$$

In essence, we assume V_0 , find I , and determine R . Alternatively, it is possible to assume current I_0 , find the corresponding potential difference V , and determine R from

$$R = \frac{V}{I_0}$$

Capacitance

The capacitance of a capacitor is obtained using a similar technique. In order to have a capacitor we must have two (or more) conductors carrying equal but opposite charges. This implies that all the flux lines leaving one conductor must necessarily terminate at the surface of the other conductor. The conductors are sometimes referred to as the plates of the capacitor. The plates may be separated by free space or a dielectric. Consider the two-conductor capacitor of Fig. (1). The conductors are maintained at a potential difference V given by:

$$V = V_1 - V_2 = -\int_2^1 \mathbf{E} \cdot d\mathbf{l}$$

Where $\mathbf{E}$ is the electric field existing between the conductors and conductor 1 is assumed to carry a positive charge. (Note that the $\mathbf{E}$ field is always normal to the conducting surfaces). We define the capacitance C of the capacitor as the ratio of the magnitude of the charge on one of the plates to the potential difference between them; that is,

$$C = \frac{Q}{V} = \frac{\epsilon \oint \mathbf{E} \cdot d\mathbf{S}}{\int \mathbf{E} \cdot d\mathbf{l}}$$

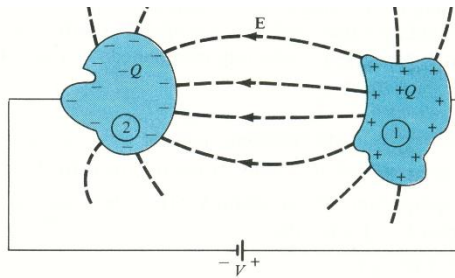


Fig. (1): Two dielectrics capacitor.

The negative sign before

$$V = -\int \mathbf{E} \cdot d\mathbf{l}$$

has been dropped because we are interested in the absolute value of V . The capacitance C is a physical property of the capacitor and is measured in Farads (F). C can be obtained for any given two-conductor capacitance by following either of these methods:

- Assuming Q and determining V in terms of Q (involving Gauss's law)
 - Assuming V and determining Q in terms of V (involving solving Laplace's equation).
- The former method is used here, which involves taking the following steps:
- Choose a suitable coordinate system,

- Let the two conducting plates carry charges $+Q$ and $-Q$,
- Determine $\mathbf{E}$ using Coulomb's or Gauss's law and find V from:

$$V = -\int \mathbf{E} \cdot d\mathbf{l}$$

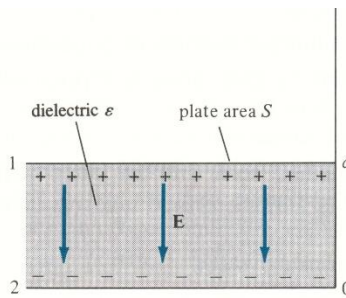
The negative sign may be ignored in this case because we are interested in the absolute value of V . Finally, obtain C from:

$$C = \frac{Q}{V}$$

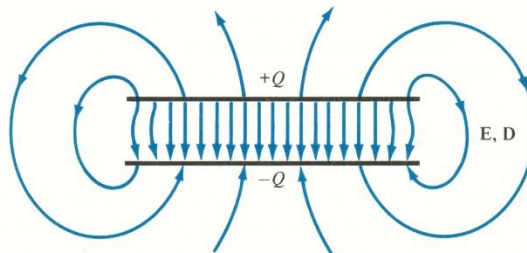
Parallel-Plate Capacitor

Consider the parallel-plate capacitor of Fig. (2). Suppose that each of the plates has an area S and they are separated by a distance d . We assume that plates 1 and 2 respectively carry charges $+Q$ and $-Q$ uniformly distributed on them so that:

$$\rho_s = \frac{Q}{S}$$



(a)



(b)

Fig. (2): Parallel-plates capacitor.

An ideal parallel-plate capacitor is one in which the plate separation d is very small compared with the dimensions of the plate. Assuming such an ideal case, the fringing field

at the edge of the plates, as illustrated in Fig. (2), can be ignored so that the field between them is considered uniform. If the space between the plates is filled with a homogeneous dielectric with permittivity ε and we ignore flux fringing at the edges of the plates, then:

$$\mathbf{D} = -\rho_S \mathbf{a}_x$$

or

$$\mathbf{E} = \frac{\rho_S}{\varepsilon} (-\mathbf{a}_x)$$

$$\mathbf{E} = -\frac{Q}{\varepsilon S} \mathbf{a}_x$$

hence

$$V = -\int_2^1 \mathbf{E} \cdot d\mathbf{l} = -\int_0^d \left[-\frac{Q}{\varepsilon S} \mathbf{a}_x \right] \cdot dx \mathbf{a}_x = \frac{Qd}{\varepsilon S}$$

and thus for a parallel-plate capacitor

$$C = \frac{Q}{V} = \frac{\varepsilon S}{d}$$

This formula offers a means of measuring the dielectric constant ε_r of a given dielectric. By measuring the capacitance C of a parallel-plate capacitor with the space between the plates filled with the dielectric and the capacitance C_0 with air between the plates, we find ε_r from:

$$\varepsilon_r = \frac{C}{C_0}$$

It can also be shown that the energy stored in a capacitor is given by:

$$W_E = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{Q^2}{2C}$$

To verify this for a parallel-plate capacitor, we substitute and obtain:

$$W_E = \frac{1}{2} \int_v \varepsilon \frac{Q^2}{\varepsilon^2 S^2} dv = \frac{\varepsilon Q^2 S d}{2 \varepsilon^2 S^2}$$

$$W_E = \frac{Q^2}{2} \left(\frac{d}{\epsilon S} \right) = \frac{Q^2}{2C} = \frac{1}{2} QV$$

Coaxial Capacitor

This is essentially a coaxial cable or coaxial cylindrical capacitor. Consider length L of two coaxial conductors of inner radius a and outer radius b ($b > a$) as shown in Fig. (3). Let the space between the conductors be filled with a homogeneous dielectric with permittivity ϵ .

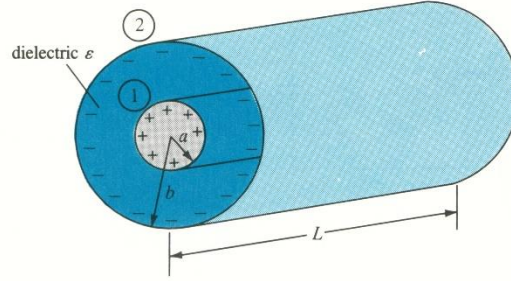


Fig. (3): Coaxial cable capacitor.

We assume that conductors 1 and 2 respectively carry $+Q$ and $-Q$ uniformly distributed on them. By applying Gauss's law to an arbitrary Gaussian cylindrical surface of radius ρ ($a < \rho < b$), we obtain:

$$Q = \epsilon \oint \mathbf{E} \cdot d\mathbf{S} = \epsilon E_\rho 2\pi \rho L$$

hence

$$\mathbf{E} = \frac{Q}{2\pi \epsilon \rho L} \mathbf{a}_\rho$$

Neglecting flux fringing at the cylinder ends,

$$V = -\int_2^1 \mathbf{E} \cdot d\mathbf{l} = -\int_b^a \left[\frac{Q}{2\pi \epsilon \rho L} \mathbf{a}_\rho \right] \cdot d\rho \mathbf{a}_\rho$$

$$V = \frac{Q}{2\pi \epsilon L} \ln \frac{b}{a}$$

Thus the capacitance of a coaxial cylinder is given by:

$$C = \frac{Q}{V} = \frac{2\pi \epsilon L}{\ln \frac{b}{a}}$$

Spherical Capacitor

This is the case of two concentric spherical conductors. Consider the inner sphere of radius a and outer sphere of radius b ($b > a$) separated by a dielectric medium with permittivity ϵ as shown in Fig. (4). We assume charges $+Q$ and $-Q$ on the inner and outer spheres respectively. By applying Gauss's law to an arbitrary Gaussian spherical surface of radius r ($a < r < b$):

$$Q = \epsilon \oint \mathbf{E} \cdot d\mathbf{S} = \epsilon E_r 4\pi r^2$$

that is:

$$\mathbf{E} = \frac{Q}{4\pi \epsilon r^2} \mathbf{a}_r$$

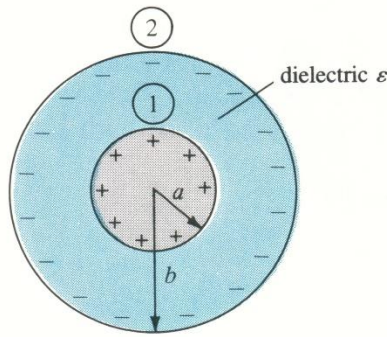


Fig. (4): Spherical capacitor.

The potential difference between the conductors is

$$V = -\int_2^1 \mathbf{E} \cdot d\mathbf{l} = -\int_b^a \left[\frac{Q}{4\pi \epsilon r^2} \mathbf{a}_r \right] \cdot dr \mathbf{a}_r$$

$$V = \frac{Q}{4\pi \epsilon} \left[\frac{1}{a} - \frac{1}{b} \right]$$

thus the capacitance of the spherical capacitor is:

$$C = \frac{Q}{V} = \frac{4\pi \epsilon}{\frac{1}{a} - \frac{1}{b}}$$

By letting $b \rightarrow \infty$,

$$C = 4\pi \epsilon a$$

which is the capacitance of a spherical capacitor whose outer plate is infinitely large. Such is the case of a spherical conductor at a large distance from other conducting bodies. Recall from network theory that if two capacitors with capacitance C_1 and C_2 are in series, the total capacitance is:

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

If the capacitors are in, the total capacitance is:

$$C = C_1 + C_2$$

Let us reconsider the expression for finding the resistance R and the capacitance C of an electrical system. The expressions were given:

$$R = \frac{V}{I} = \frac{\int \mathbf{E} \cdot d\mathbf{l}}{\oint \sigma \mathbf{E} \cdot d\mathbf{S}}$$

$$C = \frac{Q}{V} = \frac{\varepsilon \oint \mathbf{E} \cdot d\mathbf{S}}{\int \mathbf{E} \cdot d\mathbf{l}}$$

The product of these expressions yields:

$$RC = \frac{\varepsilon}{\sigma}$$

which is the relaxation time T_r of the homogeneous medium separating the conductors. In summary: for a parallel-plate capacitor:

$$C = \frac{\varepsilon S}{d}$$

$$R = \frac{d}{\sigma S}$$

for a cylindrical capacitor,

$$C = \frac{2\pi \varepsilon L}{\ln \frac{b}{a}}$$

$$R = \frac{\ln \frac{b}{a}}{2\pi \sigma L}$$

for a spherical capacitor,

$$C = \frac{4\pi \varepsilon}{\frac{1}{a} - \frac{1}{b}}$$

$$R = \frac{\frac{1}{a} - \frac{1}{b}}{4\pi \sigma}$$

and finally for an isolated spherical conductor,

$$C = 4\pi \varepsilon a$$

$$R = \frac{1}{4\pi \sigma a}$$

where the resistance R in each equation is not the resistance of the capacitor plate but the leakage resistance between the plates; therefore, σ in those equations is the conductivity of the dielectric medium separating the plates?



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Solved Example on E and V

For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_0(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

- Find $\mathbf{E}$ and V for $r \geq a$.
- Find $\mathbf{E}$ and V for $r \leq a$.
- Find the total charge.
- Show that $\mathbf{E}$ is maximum when $r = 0.745 a$.

Solution

For $r \geq a$

$$Q_{enc} = \int \rho_v dv = \iiint \rho_0(a^2 - r^2)r^2 \sin \theta d\theta d\phi dr$$

$$Q_{enc} = \rho_0 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^a (a^2 r^2 - r^4) dr$$

$$Q_{enc} = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^a = \frac{8\pi}{15} \rho_0$$

$$\Psi = \int \mathbf{D} \cdot d\mathbf{S} = \epsilon_0 E_r (4\pi r^2)$$

$$\Psi = Q_{enc}$$

$$\epsilon_0 E_r (4\pi r^2) = \frac{8\pi}{15} \rho_0$$

$$E_r = \frac{2\rho_0}{15\epsilon_0 r^2}$$

$$\mathbf{E} = \frac{2\rho_0}{15\epsilon_0 r^2} \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{2\rho_0}{15\varepsilon_o} \int r^{-2} \cdot dr = \frac{2\rho_0}{15\varepsilon_o r} + C_1$$

Since

$$V(r^- > 0) = 0$$

$$C_1 = 0$$

$$V = \frac{2\rho_0}{15\varepsilon_o r}$$

b. For $r \leq a$

$$Q_{enc} = \rho_0(4\pi) \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^r = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right)$$

$$E_r = \frac{Q_{enc}}{4\pi\varepsilon_o r^2} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\mathbf{E} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right) \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r^2}{6} - \frac{r^4}{20} \right) + C_2$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + C_2$$

Since $V(r = a^+) = V(r = a^-)$:

$$\frac{2\rho_0}{15\varepsilon_o a} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^4}{20} - \frac{a^4}{6} \right) + C_2$$

$$C_2 = \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

c. The total charge is found in part (a) as:

$$Q = \frac{8\pi}{15} \rho_0$$

d. For $r \geq a$, E decays to zero with no maxima. For $r \leq a$:

$$E_r = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\frac{\partial E_r}{\partial r} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^2}{3} - \frac{3r^2}{5} \right) = 0$$

$$r = \frac{a\sqrt{5}}{3}$$

hence

$$r = 0.7453 a$$



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Electromagnetics I
Vector Algebra

A quantity can be either a scalar or a vector. A scalar has only magnitude whereas a vector has both magnitude and direction. Thus:

$$\mathbf{A} = A\mathbf{a}_A = |\mathbf{A}|\mathbf{a}_A$$

where $\mathbf{a}_A$ is a unit vector along $\mathbf{A}$. The magnitude of $\mathbf{A}$ in Cartesian coordinates is:

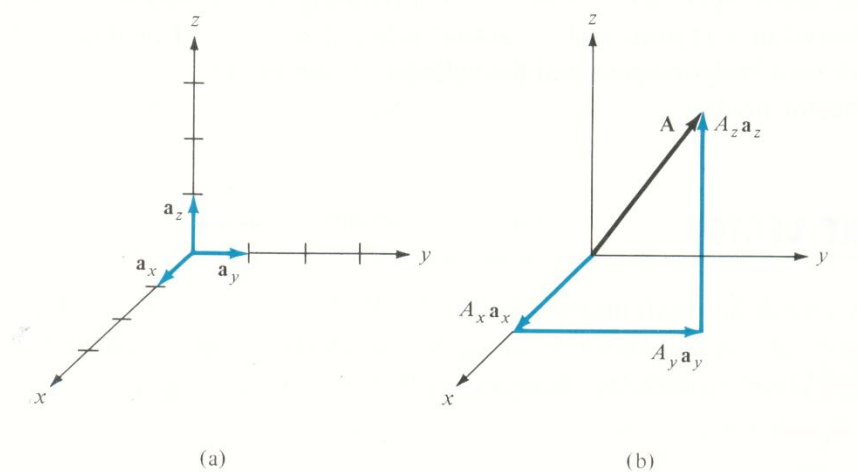


Fig. (1): (a) Unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$ and $\mathbf{a}_z$, (b) components of $\mathbf{A}$ along $\mathbf{a}_x$, $\mathbf{a}_y$ and $\mathbf{a}_z$.

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

and the unit vector is:

$$\mathbf{a}_A = \frac{A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

Vectors can be added or subtracted, thus:

$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$\mathbf{D} = \mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

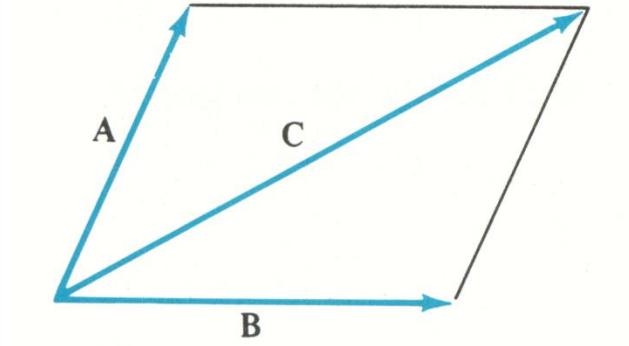


Fig. (2) Vector addition $\mathbf{C} = \mathbf{A} + \mathbf{B}$.

Furthermore, a point P is represented by (x, y, z) . Its position $\mathbf{r}_P$ is defined as the directed distance from the origin O to P ,

$$\mathbf{r}_P = \overrightarrow{OP} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z$$

Point (3, 4, 5) for example and its position vector $3\mathbf{a}_x + 4\mathbf{a}_y + 5\mathbf{a}_z$ are shown in Fig. (3).

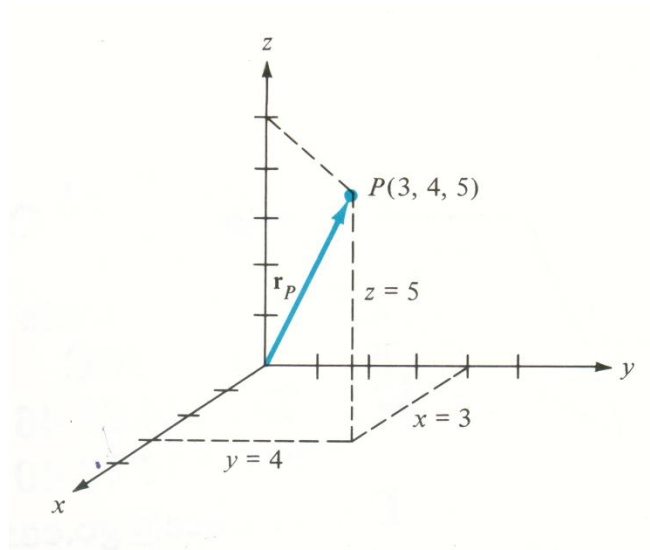


Fig. (3): $\mathbf{r}_P = 3\mathbf{a}_x + 4\mathbf{a}_y + 5\mathbf{a}_z$.

The distance vector is defined as the displacement from one point to another. Thus:

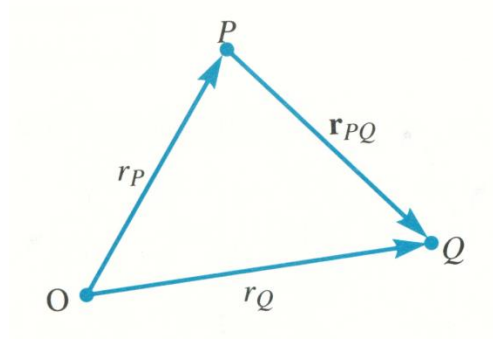


Fig. (4): Distance vector $\mathbf{r}_{PQ}$.

$$\mathbf{r}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P = (x_Q - x_P)\mathbf{a}_x + (y_Q - y_P)\mathbf{a}_y + (z_Q - z_P)\mathbf{a}_z$$

Now when two vector $\mathbf{A}$ and $\mathbf{B}$ are multiplied, the results is either a scalar or a vector depending on how they are multiplied. Thus, the dot product $\mathbf{A} \cdot \mathbf{B}$ gives a scalar and the cross product $\mathbf{A} \times \mathbf{B}$ results in a vector. The dot product is defined as:

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

with

$$\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = 0$$

$$\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

The cross product is defined as:

$$\mathbf{A} \times \mathbf{B} = AB \sin \theta_{AB} \mathbf{a}_n$$

where $\mathbf{a}_n$ is a unit vector normal to the plane containing $\mathbf{A}$ and $\mathbf{B}$. Also:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\mathbf{a}_x - (A_x B_z - A_z B_x)\mathbf{a}_y + (A_x B_y - A_y B_x)\mathbf{a}_z$$

thus, the cross product is a vector with a magnitude equal to the area of the parallelogram and in the direction normal to it.

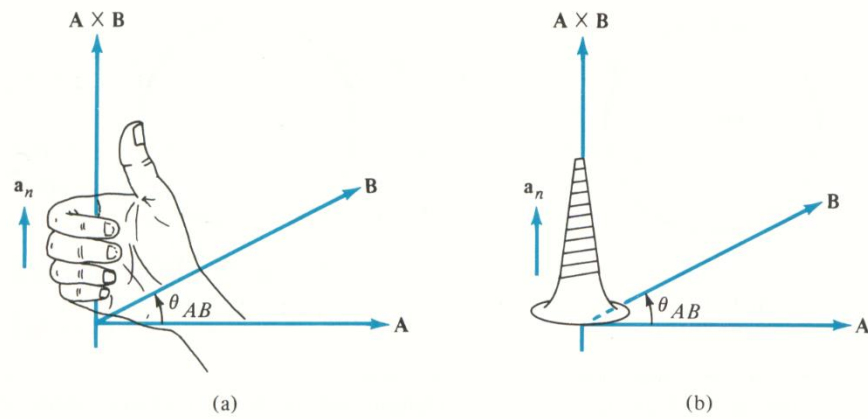


Fig. (5): Direction of $\mathbf{A} \times \mathbf{B}$ and $\mathbf{a}_n$.

Cross product has the following basic properties:

$$\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}$$

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$

$$\mathbf{A} \times \mathbf{A} = \mathbf{0}$$

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z$$

$$\mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x$$

$$\mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

which can be obtained in cyclic permutation as shown below for a right-hand rule.

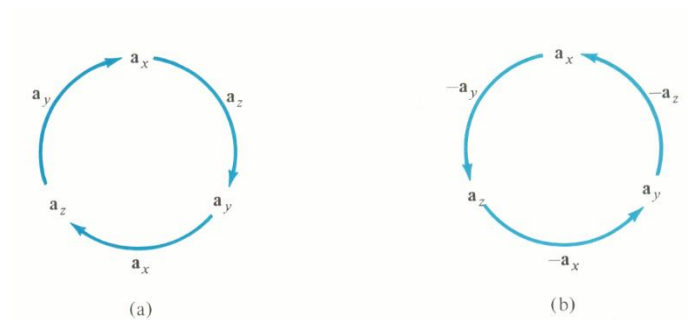


Fig. (6): Cross product using cyclic permutation.

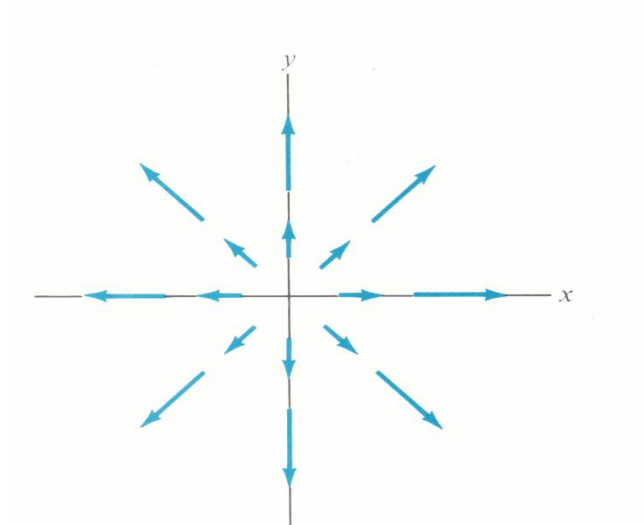


Fig. (7): Coulomb vector force on the point charges Q_1 and Q_2 .

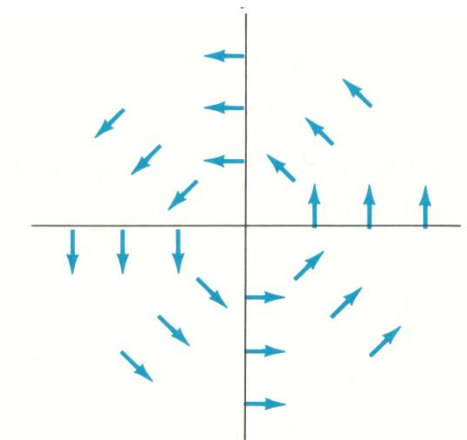


Fig. (8): Coulomb vector force on the point charges Q_1 and Q_2 .

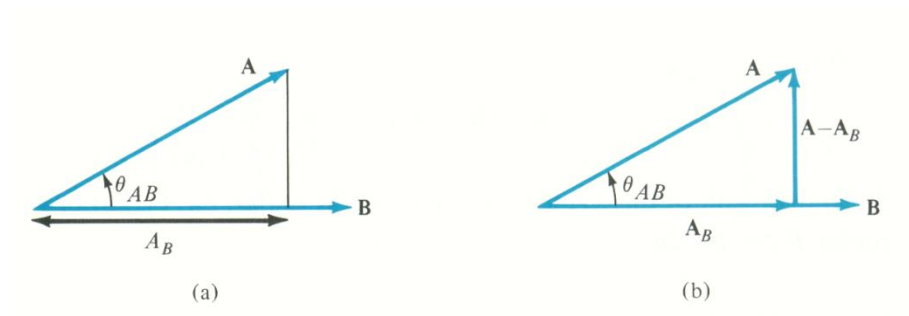


Fig. (9): Coulomb vector force on the point charges Q_1 and Q_2 .

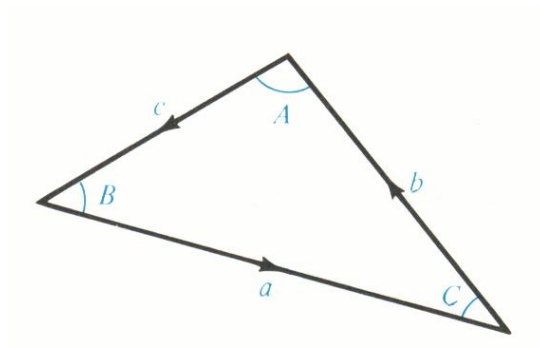


Fig. (10): Coulomb vector force on the point charges Q_1 and Q_2 .



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Vector Calculus

Differential Length, Area, and Volume

Differential elements in length, area, and volume are defined in the Cartesian, cylindrical, and spherical coordinate systems. In Cartesian coordinates, the differential length is:

$$dl = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

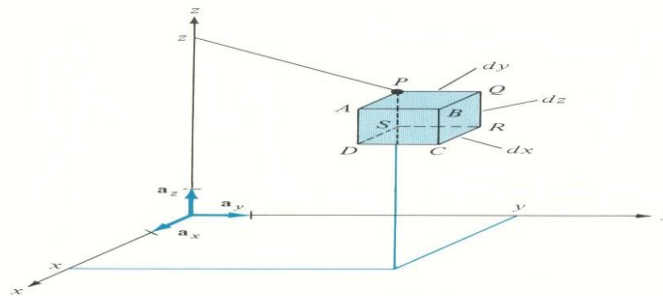


Fig. (1) Differential element in Cartesian coordinate system.

Differential normal area is given by:

$$dS = dy \, dz \mathbf{a}_x = dx \, dz \mathbf{a}_y = dx \, dy \mathbf{a}_z$$

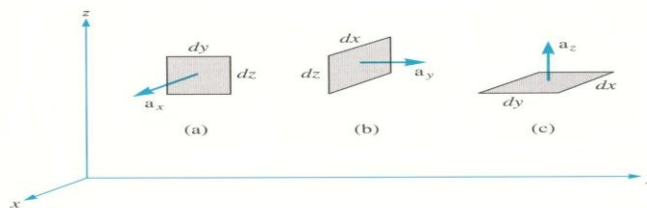


Fig. (2) Differential normal areas in Cartesian coordinates.

Differential volume is given by:

$$dv = dx \, dy \, dz$$

These differential elements are very important, as they will be referred to repeatedly throughout the book. If we move from Point P to Q or Q to P), for example:

$$dl = dy \mathbf{a}_y$$

because movement is in the y -direction and if:

$$dl = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

The way $d\mathbf{S}$ is defined is important. The differential surface element $d\mathbf{S}$ may generally be defined as:

$$d\mathbf{S} = dS \mathbf{a}_n$$

where dS is the area of the surface element and $\mathbf{a}_n$ is a unit vector normal to the surface dS (and directed away from the volume if dS is part of the surface describing a volume). If surface ABCD is considered as in Fig. (3), for example:

$$d\mathbf{S} = dy \, dz \, \mathbf{a}_x$$

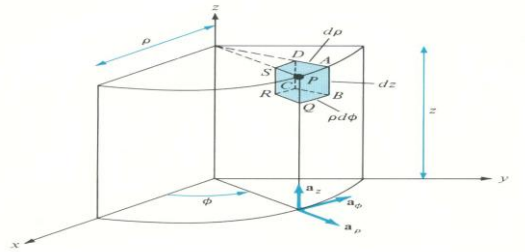


Fig. (3) Differential elements in cylindrical coordinates.

whereas for surface PQRS,

$$d\mathbf{S} = -dy \, dz \, \mathbf{a}_x$$

because

$$\mathbf{a}_n = -\mathbf{a}_x$$

is normal to PQRS.

Cylindrical Coordinates

Differential displacement is given by:

$$dl = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$

Differential normal area is given by:

$$d\mathbf{S} = \rho \, d\phi \, dz \, \mathbf{a}_\rho = d\rho \, dz \, \mathbf{a}_\phi = \rho \, d\phi \, d\rho \, \mathbf{a}_z$$

Differential volume is given by:

$$dv = \rho \, d\rho \, d\phi \, dz$$

$d\mathbf{S}$ and dv can easily be obtained from $d\mathbf{l}$. For example, $d\mathbf{S}$ along $\mathbf{a}_z$ is the product of the components of $d\mathbf{l}$ along $\mathbf{a}_\rho$ and $\mathbf{a}_\phi$ that is $\rho \, d\rho \, d\phi \, \mathbf{a}_z$. Also dv is the product of the three components of $d\mathbf{l}$, $\rho \, d\rho \, d\phi \, dz$.

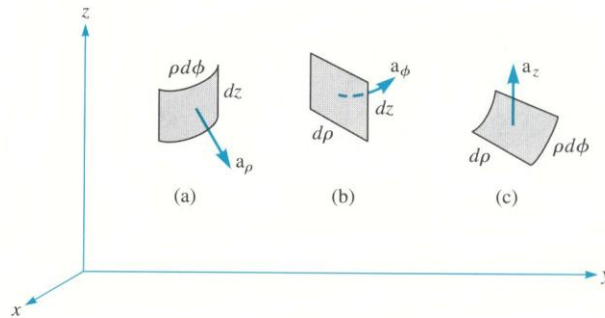


Fig. (4) Differential areas in cylindrical coordinates.

Spherical coordinates

In spherical coordinates, the differential displacement is:

$$d\mathbf{l} = dr \, \mathbf{a}_r + r \, d\theta \, \mathbf{a}_\theta + r \, \sin \theta \, d\phi \, \mathbf{a}_\phi$$

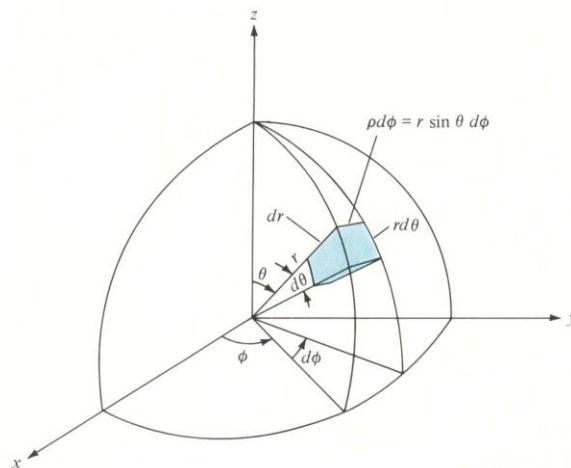


Fig. (5) Differential elements in the spherical coordinate system.

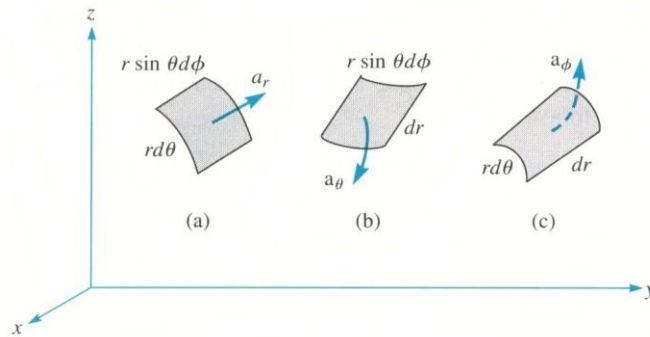


Fig. (6) Differential areas in spherical coordinates.

The differential normal area is:

$$d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r = r \sin \theta dr d\phi \mathbf{a}_\theta = r dr d\theta \mathbf{a}_\phi$$

and the differential volume is:

$$dv = r^2 \sin \theta dr d\theta d\phi$$

$d\mathbf{S}$ and dv can easily be obtained from $d\mathbf{l}$. For example, $d\mathbf{S}$ along $\mathbf{a}_\theta$ is the product of the components of $d\mathbf{l}$ along $\mathbf{a}_r$ and $\mathbf{a}_\phi$ that is $r dr \sin \theta d\phi$. Also dv is the product of the three components of $d\mathbf{l}$, $r dr \sin \theta d\phi$.

DEL Operator

The del operator in Cartesian coordinates is:

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

also known as the gradient operator, is not a vector in itself, but when it operates on a scalar function, for example, a vector ensue The operator is useful in defining.

Gradient of a Scalar

$$\rho = \sqrt{x^2 + y^2}$$

$$\tan = \frac{y}{x}$$

hence

$$\frac{\partial}{\partial x} = \cos \phi \frac{\partial}{\partial \rho} - \frac{\sin \phi}{\rho} \frac{\partial}{\partial \phi}$$

$$\frac{\partial}{\partial y} = \sin \phi \frac{\partial}{\partial \rho} + \frac{\cos \phi}{\rho} \frac{\partial}{\partial \phi}$$

$$\nabla = \mathbf{a}_\rho \frac{\partial}{\partial \rho} + \mathbf{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{a}_z \frac{\partial}{\partial z}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\tan \theta = \frac{\sqrt{x^2 + y^2}}{z}$$

$$\tan \phi = \frac{y}{x}$$

$$\frac{\partial}{\partial x} = \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{\cos \theta \cos \phi}{r} \frac{\partial}{\partial \theta} - \frac{\sin \phi}{\rho} \frac{\partial}{\partial \phi}$$

$$\frac{\partial}{\partial y} = \sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{\cos \theta \sin \phi}{r} \frac{\partial}{\partial \theta} + \frac{\cos \phi}{\rho} \frac{\partial}{\partial \phi}$$

$$\frac{\partial}{\partial z} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}$$

Similarly

$$\nabla = \mathbf{a}_r \frac{\partial}{\partial r} + \mathbf{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

Gradient of a Scalar

By definition, the gradient of a scalar field V is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V . A mathematical expression for the can be obtained by evaluating the difference in the field dV between points P_1 and P_2 of Fig. (7), where V_1 , V_2 , and V_3 are contours on which V is constant.

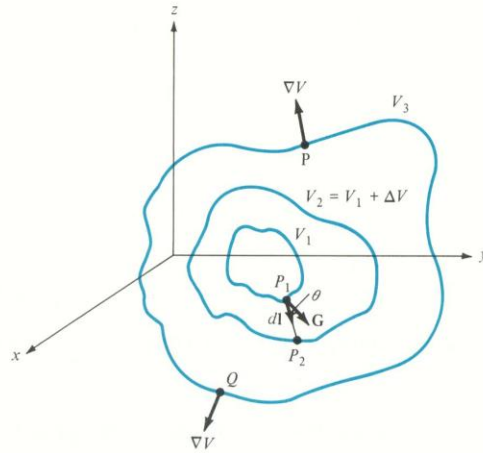


Fig. (7) Gradient of a scalar.

Form calculus:

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz$$

$$dV = \left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right) \cdot (dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z)$$

and by definition the gradient of V is:

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

the gradient of V can be expressed in Cartesian, cylindrical, and spherical coordinates. For Cartesian coordinates:

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

for cylindrical coordinates:

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

and for spherical coordinates,

$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

Divergence Theorem

The divergence of $\mathbf{A}$ at a given point P is the outward flux per unit volume as the volume shrinks about P . Hence:

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

where Δv is the volume enclosed by the closed surface S in which P is located. Physically, the divergence of the vector field $\mathbf{A}$ may regard at a given point as a measure of how much the field diverges or emanates from that point. Fig. (8) shows when the divergence of a vector field at point P is positive as the vector diverges or spreads out at P (a), a vector field has negative divergence or convergence at P (b), and when a vector field has zero divergence at P (c).

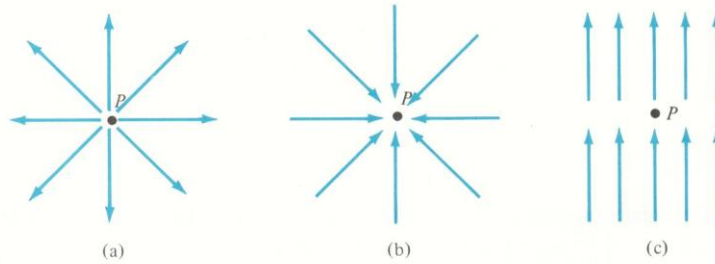


Fig. (8) Illustration of the divergence of a vector field at P , (a) positive divergence, (b) negative divergence (c) zero divergence.

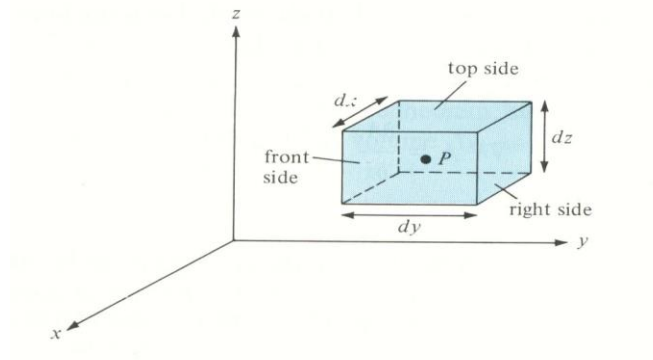


Fig. (9) Evaluation of $\nabla \cdot \mathbf{A}$ at point P .

Mathematically:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

In cylindrical coordinates, substituting yields:

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

the divergence of $\mathbf{A}$ in spherical coordinates is thus obtained as:

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

It is expected that:

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dv$$

This is called the divergence theorem, otherwise known as the *Gauss-Ostrogradsky theorem*. The theorem states that the total outward flux of a vector field $\mathbf{A}$ through the dosed surface S is the same as the volume integral of the divergence of $\mathbf{A}$.

Stokes's Theorem

The circulation of a vector field $\mathbf{A}$ around a closed path L is defined as:

$$\oint_L \mathbf{A} \cdot d\mathbf{l}$$

the curl of $\mathbf{A}$ is defines as an axial or rotational vector whose magnitude is the maximum circulation of $\mathbf{A}$ per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented so as to make the circulation maximum:

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right) \mathbf{a}_n$$

where the area ΔS is bounded by the curve L and $\mathbf{a}_n$ is the unit vector normal to the surface ΔS and is determined using the right-hand rule. In Cartesian coordinates the curl of $\mathbf{A}$ is easily found using:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] \mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z$$

and in cylindrical coordinates :

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z$$

and in spherical coordinates as:

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

or

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{r \sin \theta} \left[\frac{\partial(A_\phi \sin \theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r \\ & + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial(r A_\phi)}{\partial r} \right] \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial(r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right] \mathbf{a}_\phi \end{aligned}$$

The physical significance of the curl of a vector field is to provide the maximum value of the circulation of the field per unit area and indicates the direction along which this maximum value occurs. The curl of a vector field $\mathbf{A}$ at a point P may be regarded as a measure of the circulation or how much the field curls around P . For example, Fig.(10 a) shows that the curl of a vector field around P is directed out of the page.

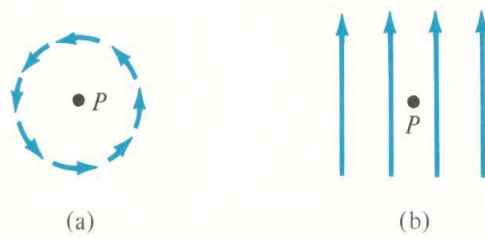


Fig. (10): Illustration of a curl: (a) curl at P points out of the page; (b) curl at P is zero.

Fig. (10 b) shows a vector field with zero curl. From the definition of the curl of $\mathbf{A}$ it is expected that:

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

This is called Stokes's theorem. It states that the circulation of $\mathbf{A}$ around a (closed) path L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S bounded by L as shown in Fig. (11), provided that $\mathbf{A}$ and $\nabla \times \mathbf{A}$ are continuous on S .

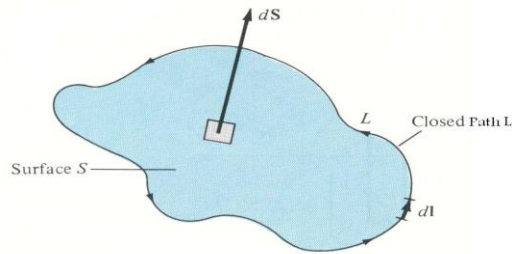


Fig. (11): Determining the sense of $d\mathbf{l}$ and $d\mathbf{S}$ involved in Stokes's theorem.

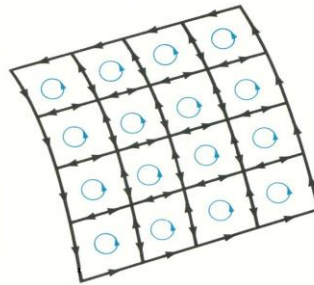


Fig. (12): Illustration of Stokes's theorem.

The direction of $d\mathbf{l}$ and $d\mathbf{S}$ must be chosen using the right-hand rule or right-handed screw rule. If we let the fingers point in the direction of $d\mathbf{l}$, the thumb will indicate the direction of $d\mathbf{S}$ as shown in Fig. (12). It is worth noting that whereas the divergence theorem relates a surface integral to a volume integral, Stokes's theorem relates a line integral (circulation) to a surface integral.

Laplacian of a Scalar

For practical reasons, it is expedient to introduce a single operator, which is the composite of gradient and divergence operators. This operator is known as the Laplacian.

$$\nabla^2 V = \nabla \cdot \nabla V = \nabla^2 V$$

$$\nabla^2 V = \left[\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right] \cdot \left[\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right]$$

that is:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Notice that the Laplacian of a scalar field is another scalar field. The Laplacian of V in other coordinate systems can be obtained from the above equation by transformation. In cylindrical coordinates:

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial r} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

and in spherical coordinates:

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

A scalar field V is said to be *harmonic* in a given region if its Laplacian vanishes in that region. In other words, if:

$$\nabla^2 V = 0$$

is satisfied in the region, the solution for V is harmonic. This equation is called *Laplace's equation*.

Classifications of Vector Fields

A vector field is uniquely characterized by its divergence and curl. Thus, all vector fields can be classified in terms of their vanishing or no vanishing divergence or curl as follows:

$$\nabla \cdot \mathbf{A} = 0 \quad \nabla \times \mathbf{A} = 0$$

$$\nabla \cdot \mathbf{A} \neq 0 \quad \nabla \times \mathbf{A} = 0$$

$$\nabla \cdot \mathbf{A} = 0 \quad \nabla \times \mathbf{A} \neq 0$$

$$\nabla \cdot \mathbf{A} \neq 0 \quad \nabla \times \mathbf{A} \neq 0$$

Fig. (13) illustrates typical fields in these four categories for a given vector field $\mathbf{A}$, if

$$\nabla \cdot \mathbf{A} = 0$$

then $\mathbf{A}$ is said to be *solenoidal* or *divergenceless*. Such a field has neither source nor sink of flux- From the divergence theorem.

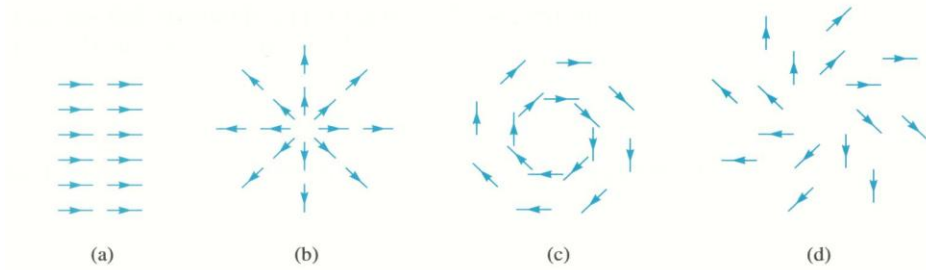


Fig. (13) Typical fields with vanishing and non-vanishing divergence or curl.

$$\nabla \cdot \mathbf{A} = 0$$

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = 0$$

$$\mathbf{F} = \nabla \times \mathbf{A}$$

A vector field $\mathbf{A}$ is said to be *irrotational* (or *potential*) if:

$$\nabla \times \mathbf{A} = 0$$

That is, a *curl-free* vector is irrotational. From Stokes's theorem

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_L \mathbf{A} \cdot d\mathbf{l} = 0$$

Thus in an irrotational field $\mathbf{A}$, the circulation of $\mathbf{A}$ around a closed path is identically zero. This implies that the line integral of $\mathbf{A}$ is independent of the chosen path. Therefore, an irrotational field is also known as a conservative field. Examples of irrotational fields include the electrostatic field and the gravitational field. In general, the field of gradient V (for any scalar V) is purely irrotational since:

$$\nabla \times (\nabla V) = 0$$

Thus, an irrotational field $\mathbf{A}$ can always be expressed in terms of a scalar field V , if

$$\nabla \times \mathbf{A} = 0$$

And

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = 0$$

then

$$\mathbf{A} = -\nabla V$$

For this reason, $\mathbf{A}$ may be called a potential field and V the scalar potential of $\mathbf{A}$. The negative sign in the above equation has been inserted for physical reasons. A vector $\mathbf{A}$ is uniquely prescribed within a region by its divergence and its curl. If we let:

$$\nabla \cdot \mathbf{A} = \rho_v$$

and

$$\nabla \times \mathbf{A} = \rho_s$$

ρ_v can be regarded as the source density of $\mathbf{A}$ and ρ_s its circulation density. Any vector $\mathbf{A}$ satisfying the above equations with both ρ_v and ρ_s vanishing at infinity can be written as the sum of two vectors: one irrotational (zero curl), and the other solenoidal (zero divergence). This is called Helmholtz's theorem.



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Vectors-Examples

Example 1

If $\mathbf{A} = 10\mathbf{a}_x - 4\mathbf{a}_y + 6\mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_x + \mathbf{a}_y$, find (a) the component of $\mathbf{A}$ along $\mathbf{a}_y$, (b) the magnitude of $3\mathbf{A} - \mathbf{B}$, (c) a unit vector along $\mathbf{A} + 2\mathbf{B}$.

Solution

(a) The component of $\mathbf{A}$ along $\mathbf{a}_y$ is $A_y = -4$.

$$\begin{aligned}\text{(b) } 3\mathbf{A} - \mathbf{B} &= 3(10, -4, 6) - (2, 1, 0) \\ &= (30, -12, 18) - (2, 1, 0) \\ &= (28, -13, 18)\end{aligned}$$

Hence,

$$\begin{aligned} |3\mathbf{A} - \mathbf{B}| &= \sqrt{28^2 + (-13)^2 + (18)^2} = \sqrt{1277} \\ &= 35.74 \end{aligned}$$

(c) Let $\mathbf{C} = \mathbf{A} + 2\mathbf{B} = (10, -4, 6) + (4, 2, 0) = (14, -2, 6)$.

A unit vector along $\mathbf{C}$ is

$$\mathbf{a}_c = \frac{\mathbf{C}}{|\mathbf{C}|} = \frac{(14, -2, 6)}{\sqrt{14^2 + (-2)^2 + 6^2}}$$

or

$$\mathbf{a}_c = 0.9113\mathbf{a}_x - 0.1302\mathbf{a}_y + 0.3906\mathbf{a}_z$$

Note that $|\mathbf{a}_c| = 1$ as expected.

Example 2

Points P and Q are located at $(0, 2, 4)$ and $(-3, 1, 5)$. Calculate

- (a) The position of vector $\mathbf{r}_P$
- (b) The distance vector from P to Q

- (c) The distance between P and Q
 (d) A vector parallel to PQ with magnitude of 10

Solution

- (a) $\mathbf{r}_P = 0\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z = 2\mathbf{a}_y + 4\mathbf{a}_z$
 (b) $\mathbf{r}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P = (-3, 1, 5) - (0, 2, 4) = (-3, -1, 1)$
 or $\mathbf{r}_{PQ} = -3\mathbf{a}_x - \mathbf{a}_y + \mathbf{a}_z$
 (c) Since $\mathbf{r}_{PQ}$ is the distance vector from P to Q , the distance between P and Q is the magnitude of this vector; that is,

$$d = |\mathbf{r}_{PQ}| = \sqrt{9 + 1 + 1} = 3.317$$

Alternatively:

$$\begin{aligned} d &= \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2 + (z_Q - z_P)^2} \\ &= \sqrt{9 + 1 + 1} = 3.317 \end{aligned}$$

- (d) Let the required vector be $\mathbf{A}$, then

$$\mathbf{A} = A\mathbf{a}_A$$

where $A = 10$ is the magnitude of $\mathbf{A}$. Since $\mathbf{A}$ is parallel to PQ , it must have the same unit vector as $\mathbf{r}_{PQ}$ or $\mathbf{r}_{QP}$. Hence,

$$\mathbf{a}_A = \pm \frac{\mathbf{r}_{PQ}}{|\mathbf{r}_{PQ}|} = \pm \frac{(-3, -1, 1)}{3.317}$$

and

$$\mathbf{A} = \pm \frac{10(-3, -1, 1)}{3.317} = \pm(-9.045\mathbf{a}_x - 3.015\mathbf{a}_y + 3.015\mathbf{a}_z)$$

Example 3

Given vectors $\mathbf{A} = 3\mathbf{a}_x + 4\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_y - 5\mathbf{a}_z$, find the angle between $\mathbf{A}$ and $\mathbf{B}$.

Solution

The angle θ_{AB} can be found by using either dot product or cross product.

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (3, 4, 1) \cdot (0, 2, -5) \\ &= 0 + 8 - 5 = 3 \\ |\mathbf{A}| &= \sqrt{3^2 + 4^2 + 1^2} = \sqrt{26} \\ |\mathbf{B}| &= \sqrt{0^2 + 2^2 + (-5)^2} = \sqrt{29} \\ \cos \theta_{AB} &= \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}| |\mathbf{B}|} = \frac{3}{\sqrt{(26)(29)}} = 0.1092 \\ \theta_{AB} &= \cos^{-1} 0.1092 = 83.73^\circ \end{aligned}$$

Alternatively:

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 3 & 4 & 1 \\ 0 & 2 & -5 \end{vmatrix} \\ &= (-20 - 2)\mathbf{a}_x + (0 + 15)\mathbf{a}_y + (6 - 0)\mathbf{a}_z \\ &= (-22, 15, 6) \\ |\mathbf{A} \times \mathbf{B}| &= \sqrt{(-22)^2 + 15^2 + 6^2} = \sqrt{745} \\ \sin \theta_{AB} &= \frac{|\mathbf{A} \times \mathbf{B}|}{|\mathbf{A}||\mathbf{B}|} = \frac{\sqrt{745}}{\sqrt{(26)(29)}} = 0.994 \\ \theta_{AB} &= \sin^{-1} 0.994 = 83.73^\circ\end{aligned}$$



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 1

1. Three field quantities are given by:

$$\mathbf{P} = 2 \mathbf{a}_x - \mathbf{a}_z$$

$$\mathbf{Q} = 2 \mathbf{a}_x - \mathbf{a}_y + 2 \mathbf{a}_z$$

$$\mathbf{R} = 2 \mathbf{a}_x - 3 \mathbf{a}_y + \mathbf{a}_z$$

Determine the following:

- $(\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q})$.
- $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$
- $\mathbf{P} \cdot \mathbf{Q} \times \mathbf{R}$
- $\sin \theta_{QR}$
- $\mathbf{P} \times (\mathbf{Q} \times \mathbf{R})$
- A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$
- The component of $\mathbf{P}$ along $\mathbf{Q}$

2. $\mathbf{E}$ and $\mathbf{F}$ are vector fields given by $\mathbf{E} = 2x \mathbf{a}_x + \mathbf{a}_y + yz \mathbf{a}_z$ and $\mathbf{F} = xy \mathbf{a}_x - y^2 \mathbf{a}_y + xyz \mathbf{a}_z$.

Determine:

- $|\mathbf{E}|$ at (1, 2, 3).
- The component of $\mathbf{E}$ along $\mathbf{F}$ at (1, 2, 3).
- A vector perpendicular to both $\mathbf{E}$ and $\mathbf{F}$ at (0, 1, -3) whose magnitude is unity.

3. If $\mathbf{A} = -\mathbf{a}_x + 6 \mathbf{a}_y + 5 \mathbf{a}_z$ and $\mathbf{B} = \mathbf{a}_x + 2 \mathbf{a}_y + 3 \mathbf{a}_z$. Then, find:

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$.
- The vector projection of $\mathbf{B}$ on $\mathbf{A}$.
- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 1-Solution

1. Three field quantities are given by:

$$\mathbf{P} = 2 \mathbf{a}_x - \mathbf{a}_z$$

$$\mathbf{Q} = 2 \mathbf{a}_x - \mathbf{a}_y + 2 \mathbf{a}_z$$

$$\mathbf{R} = 2 \mathbf{a}_x - 3 \mathbf{a}_y + \mathbf{a}_z$$

Determine the followings:

- a. $(\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q})$.
- b. $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$
- c. $\mathbf{P} \cdot \mathbf{Q} \times \mathbf{R}$
- d. $\sin \theta_{QR}$
- e. $\mathbf{P} \times (\mathbf{Q} \times \mathbf{R})$
- f. A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$
- g. The component of $\mathbf{P}$ along $\mathbf{Q}$

Solution

a.

$$\begin{aligned} (\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q}) &= \mathbf{P} \times (\mathbf{P} - \mathbf{Q}) + \mathbf{Q} \times (\mathbf{P} - \mathbf{Q}) = \mathbf{P} \times \mathbf{P} - \mathbf{P} \times \mathbf{Q} + \mathbf{Q} \times \mathbf{P} - \mathbf{Q} \times \mathbf{Q} \\ &= \mathbf{0} + \mathbf{Q} \times \mathbf{P} + \mathbf{Q} \times \mathbf{P} - \mathbf{0} = 2\mathbf{Q} \times \mathbf{P} \end{aligned}$$

$$= 2 \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & -1 & 2 \\ 2 & 0 & -1 \end{vmatrix}$$

$$= 2(1 - 0)\mathbf{a}_x + 2(4 + 2)\mathbf{a}_y + 2(0 + 2)\mathbf{a}_z = 2 \mathbf{a}_x + 12 \mathbf{a}_y + 4 \mathbf{a}_z$$

b. The only way $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$ makes sense is:

$$\mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = (2, -1, 2) \cdot \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & -1 & 2 \\ 2 & 0 & -1 \end{vmatrix}$$

$$= (2, -1, 2) \cdot (3, 4, 6) = 6 - 4 + 12 = 14$$

Alternatively:

$$\mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = \begin{vmatrix} 2 & -1 & 2 \\ 2 & -3 & 1 \\ 2 & 0 & -1 \end{vmatrix}$$

To find the determinant of a 3×3 matrix, we repeat the first two rows and cross multiply; when the cross multiplication is from right to left, the result should be negated as shown below. This technique of finding a determinant applies only to a 3×3 matrix. Hence:

$$\begin{aligned} \mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) &= \begin{vmatrix} 2 & -1 & 2 \\ 2 & -3 & 1 \\ 2 & 0 & -1 \\ 2 & -1 & 2 \\ 2 & -3 & 1 \end{vmatrix} \\ &= +6 + 0 - 2 + 12 - 0 - 2 = 14 \end{aligned}$$

c. From the equation:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

Then:

$$\mathbf{P} \cdot (\mathbf{Q} \times \mathbf{R}) = \mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = 14$$

or

$$\begin{aligned} \mathbf{P} \cdot (\mathbf{Q} \times \mathbf{R}) &= (2, 0, -1) \cdot (5, 2, -4) \\ &= 10 + 0 + 4 = 14 \end{aligned}$$

d.

$$\begin{aligned} \sin \theta_{QR} &= \frac{|\mathbf{Q} \times \mathbf{R}|}{|\mathbf{Q}||\mathbf{R}|} = \frac{|(5, 2, -4)|}{|(2, -1, 2)||(-2, -3, 1)|} \\ &= \frac{\sqrt{45}}{3\sqrt{14}} = \frac{\sqrt{5}}{\sqrt{14}} = 0.5976 \end{aligned}$$

e.

$$\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) = (2, 0, -1) \times (5, 2, -4) = (2, 3, 4)$$

Alternatively, using the bac-cab rule:

$$\begin{aligned} \mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) &= \mathbf{Q}(\mathbf{P} \cdot \mathbf{R}) - \mathbf{R}(\mathbf{P} \cdot \mathbf{Q}) \\ &= (2, -4, 2)(4 + 0 - 1) - (2, -3, 1)(4 + 0 - 2) = (2, 3, 4) \end{aligned}$$

f. A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$ is given by:

g.

$$\mathbf{a} = \frac{\pm \mathbf{Q} \times \mathbf{R}}{|\mathbf{Q} \times \mathbf{R}|} = \frac{\pm(5, 2, -4)}{\sqrt{45}} = \pm(0.745, 0.298, -0.596)$$

Note that $|\mathbf{a}| = 1$, $\mathbf{a} \cdot \mathbf{Q} = 0 = \mathbf{a} \cdot \mathbf{R}$. Any of these can be used to check $\mathbf{a}$.

h. The component of $\mathbf{P}$ along $\mathbf{Q}$ is:

$$\begin{aligned} \mathbf{P}_Q &= |\mathbf{P}| \cos \theta_{PQ} \mathbf{a}_Q \\ &= (\mathbf{P} \cdot \mathbf{a}_Q) \mathbf{a}_Q = \frac{(\mathbf{P} \cdot \mathbf{Q}) \mathbf{Q}}{|\mathbf{Q}|^2} \\ &= \frac{(4 + 0 - 2)(2, -1, 2)}{(4 + 1 + 4)} = \frac{2}{9} (2, -1, 2) \\ &= 0.4444 \mathbf{a}_x - 0.2222 \mathbf{a}_y + 0.4444 \mathbf{a}_z \end{aligned}$$

2. $\mathbf{E}$ and $\mathbf{F}$ are vector fields given by $\mathbf{E} = 2x \mathbf{a}_x + \mathbf{a}_y + yz \mathbf{a}_z$ and $\mathbf{F} = xy \mathbf{a}_x - y^2 \mathbf{a}_y + xyz \mathbf{a}_z$. Determine:

- $|\mathbf{E}|$ at (1, 2, 3).
- The component of $\mathbf{E}$ along $\mathbf{F}$ at (1, 2, 3).
- A vector perpendicular to both $\mathbf{E}$ and $\mathbf{F}$ at (0, 1, -3) whose magnitude is unity.

Solution

- At (1,2,3), $\mathbf{E} = (2, 1, 6)$

$$|\mathbf{E}| = \sqrt{4 + 1 + 36} = \sqrt{41} = 6.403$$

- At (1,2,3), $\mathbf{F} = (2, -4, 6)$

$$\begin{aligned}\mathbf{E}_F &= (\mathbf{E} \cdot \mathbf{a}_F) \mathbf{a}_F = \frac{(\mathbf{E} \cdot \mathbf{F}) \mathbf{F}}{|\mathbf{F}|^2} = \frac{36}{56} (2, -4, 6) \\ &= 1.286 \mathbf{a}_x - 2.571 \mathbf{a}_y + 3.857 \mathbf{a}_z\end{aligned}$$

- At (0,1, -3), $\mathbf{E} = (0, 1, -3)$, $\mathbf{F} = (0, -1, 0)$

$$\mathbf{E} \times \mathbf{F} = \begin{vmatrix} 0 & 1 & -3 \\ 0 & -1 & 0 \end{vmatrix} = (-3, 0, 0)$$

$$\mathbf{a}_{E \times F} = \frac{\pm \mathbf{E} \times \mathbf{F}}{|\mathbf{E} \times \mathbf{F}|} = \pm \mathbf{a}_x$$

- If $\mathbf{A} = -\mathbf{a}_x + 6 \mathbf{a}_y + 5 \mathbf{a}_z$ and $\mathbf{B} = \mathbf{a}_x + 2 \mathbf{a}_y + 3 \mathbf{a}_z$. Then, find:

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$.
- The vector projection of $\mathbf{B}$ on $\mathbf{A}$.
- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.

Solution

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$:

$$\mathbf{A}_B = \mathbf{A} \mathbf{a}_B = \frac{\mathbf{A} \mathbf{B}}{|\mathbf{B}|} = \frac{-1 + 12 + 15}{\sqrt{1 + 4 + 9}} = \frac{26}{\sqrt{14}} = 6.95$$

- The vector projection of $\mathbf{B}$ on $\mathbf{A}$:

$$\begin{aligned}\mathbf{B}_A &= (\mathbf{B} \cdot \mathbf{a}_A) \mathbf{a}_A = \frac{(\mathbf{B} \cdot \mathbf{A}) \mathbf{A}}{|\mathbf{A}|^2} = \frac{26}{(1 + 36 + 25)} (-1, 6, 5) \\ &= -0.4193 \mathbf{a}_x + 2.516 \mathbf{a}_y + 2.097 \mathbf{a}_z\end{aligned}$$

- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} -1 & 6 & 5 \\ 1 & 2 & 3 \end{vmatrix} = 8 \mathbf{a}_x + 8 \mathbf{a}_y - 8 \mathbf{a}_z$$

$$\begin{aligned}\mathbf{a}_{A \times B} &= \frac{8 \mathbf{a}_x + 8 \mathbf{a}_y - 8 \mathbf{a}_z}{8\sqrt{1 + 1 + 1}} = \frac{\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z}{\sqrt{3}} \\ &= 0.577 \mathbf{a}_x + 0.577 \mathbf{a}_y - 0.577 \mathbf{a}_z\end{aligned}$$



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 2

1. Given the following vector:

$$\mathbf{B} = \frac{10}{r} \mathbf{a}_r + r \cos \theta \mathbf{a}_\theta + \mathbf{a}_\phi$$

- Express $\mathbf{B}$ in Cartesian and Cylindrical coordinates.
- Find $\mathbf{B}(-3, 4, 0)$ and $\mathbf{B}(5, \frac{\pi}{2}, -2)$.

2. Two uniform vector fields are given by:

$$\mathbf{E} = -5 \mathbf{a}_\rho + 10 \mathbf{a}_\phi + 3 \mathbf{a}_z$$

$$\mathbf{F} = \mathbf{a}_\rho + 2 \mathbf{a}_\phi - 6 \mathbf{a}_z$$

Calculate:

- $|\mathbf{E} \times \mathbf{F}|$.
 - The vector component of $\mathbf{E}$ at $\mathbf{P}(5, \frac{\pi}{2}, 3)$ parallel to the line $x = 2, z = 3$.
 - The angle $\mathbf{E}$ makes with the surface $z = 3$ at $\mathbf{P}$.
3. Let $\mathbf{H} = -5\rho \sin \phi \mathbf{a}_\rho - \rho z \cos \phi \mathbf{a}_\phi + 3\rho \mathbf{a}_z$. At point $\mathbf{P}(2, 30^\circ, -1)$, find:
- A unit vector along $\mathbf{H}$.
 - The component of $\mathbf{H}$ parallel to $\mathbf{a}_x$.
 - The component of $\mathbf{H}$ normal to $\rho = 2$.
 - The component of $\mathbf{H}$ tangential to $\phi = 30^\circ$.



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Worksheet 2-Solution

1.a. Expressing $\mathbf{B}$ in Cartesian and Cylindrical coordinates. Using the equation:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Then

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{r} \\ r \cos \theta \\ 1 \end{bmatrix}$$

or

$$B_x = \frac{10}{r} \sin \theta \cos \phi + r \cos^2 \theta \cos \phi - \sin \phi$$

$$B_y = \frac{10}{r} \sin \theta \sin \phi + r \cos^2 \theta \sin \phi + \cos \phi$$

$$B_z = \frac{10}{r} \cos \theta - r \cos \theta \sin \theta$$

$$\text{but } r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}$$

$$\text{and } \phi = \tan^{-1} \left(\frac{y}{x} \right)$$

Hence

$$\sin \theta = \frac{\rho}{r} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\cos \theta = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\sin \phi = \frac{y}{\rho} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\cos \phi = \frac{x}{\rho} = \frac{x}{\sqrt{x^2 + y^2}}$$

Substituting all these gives:

$$B_x = \frac{10\sqrt{x^2 + y^2}}{(x^2 + y^2 + z^2)} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\sqrt{x^2 + y^2 + z^2}}{(x^2 + y^2 + z^2)} \cdot \frac{z^2 x}{\sqrt{x^2 + y^2}} - \frac{y}{\sqrt{x^2 + y^2}}$$

$$B_x = \frac{10x}{x^2 + y^2 + z^2} + \frac{xz^2}{\sqrt{(x^2 + y^2)(x^2 + y^2 + z^2)}} - \frac{y}{\sqrt{x^2 + y^2}}$$

$$B_y = \frac{10\sqrt{x^2 + y^2}}{(x^2 + y^2 + z^2)} \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\sqrt{x^2 + y^2 + z^2}}{(x^2 + y^2 + z^2)} \cdot \frac{z^2 y}{\sqrt{x^2 + y^2}} - \frac{y}{\sqrt{x^2 + y^2}}$$

$$B_y = \frac{10y}{x^2 + y^2 + z^2} + \frac{yz^2}{\sqrt{(x^2 + y^2)(x^2 + y^2 + z^2)}} + \frac{y}{\sqrt{x^2 + y^2}}$$

$$B_z = \frac{10z}{(x^2 + y^2 + z^2)} - \frac{z\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

Where B_x , B_y , and B_z are as found before.

b. Finding $\mathbf{B}$ $(-3, 4, 0)$ and $(5, \frac{\pi}{2}, -2)$:

At $(-3, 4, 0)$, $x = -3$, $y = 4$, and $z = 0$, so:

$$B_x = -\frac{30}{25} + 0 - \frac{4}{5} = -2$$

$$B_y = \frac{40}{25} + 0 - \frac{3}{5} = 1$$

$$B_z = 0 - 0 = 0$$

Thus:

$$\mathbf{B} = -2 \mathbf{a}_x + \mathbf{a}_y$$

For spherical to cylindrical vector transformation:

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{r} \\ r \\ \cos \theta \end{bmatrix}$$

or

$$B_\rho = \frac{10}{r} \sin \theta + r \cos^2 \theta$$

$$B_\phi = 1$$

$$B_z = \frac{10}{r} \cos \theta - r \sin \theta \cos \theta$$

But

$$r = \sqrt{\rho^2 + z^2}$$

And

$$\theta = \tan^{-1} \frac{\rho}{z}.$$

Thus

$$\sin \theta = \frac{\rho}{\sqrt{\rho^2 + z^2}}$$

$$\cos \theta = \frac{z}{\sqrt{\rho^2 + z^2}}$$

$$B_\rho = \frac{10\rho}{\rho^2 + z^2} + \sqrt{\rho^2 + z^2} \cdot \frac{z^2}{\rho^2 + z^2}$$

$$B_z = \frac{10z}{\rho^2 + z^2} - \sqrt{\rho^2 + z^2} \cdot \frac{\rho z}{\rho^2 + z^2}$$

Hence

$$\mathbf{B} = \left(\frac{10\rho}{\rho^2 + z^2} + \frac{z^2}{\sqrt{\rho^2 + z^2}} \right) \mathbf{a}_\rho + \mathbf{a}_\phi + \left(\frac{10z}{\rho^2 + z^2} - \frac{\rho z}{\sqrt{\rho^2 + z^2}} \right) \mathbf{a}_z$$

At $(5, \frac{\pi}{2}, -2)$, $\rho = 5$, $\phi = \frac{\pi}{2}$, and $z = -2$

So

$$\begin{aligned}\mathbf{B} &= \left(\frac{50}{29} + \frac{4}{\sqrt{29}}\right) \mathbf{a}_\rho + \mathbf{a}_\phi + \left(\frac{-20}{29} + \frac{10}{\sqrt{29}}\right) \mathbf{a}_z \\ &= 2.467 \mathbf{a}_\rho + \mathbf{a}_\phi + 1.167 \mathbf{a}_z\end{aligned}$$

Note that at $(-3, 4, 0)$

$$|\mathbf{B}(x, y, z)| = |\mathbf{B}(\rho, \phi, z)| = |\mathbf{B}(\rho, \theta, \phi)| = 2.907$$

2. a.

$$\mathbf{E} \times \mathbf{F} = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ -5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix}$$

$$\mathbf{E} \times \mathbf{F} = (-60 - 6) \mathbf{a}_\rho + (3 - 30) \mathbf{a}_\phi + (-10 - 10) \mathbf{a}_z = (-66, -27, -20)$$

$$|\mathbf{E} \times \mathbf{F}| = \sqrt{66^2 + 27^2 + 20^2} = 74.06$$

b. The vector component of $\mathbf{E}$ at $P(5, \frac{\pi}{2}, 3)$ parallel to the line $x = 2, z = 3$:

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y$$

But at $P(5, \frac{\pi}{2}, 3)$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \frac{\pi}{2} \mathbf{a}_\rho + \cos \frac{\pi}{2} \mathbf{a}_\phi = \mathbf{a}_\rho$$

Therefore

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y = (\mathbf{E} \cdot \mathbf{a}_\rho) \mathbf{a}_\rho = -5 \mathbf{a}_\rho \text{ (or } -5 \mathbf{a}_y)$$

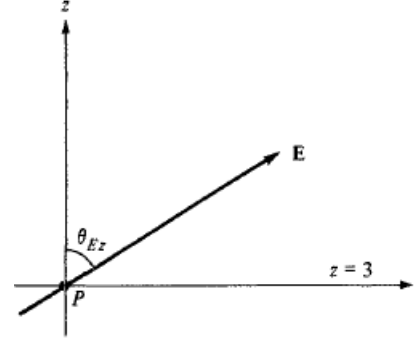
c. The angle $\mathbf{E}$ makes with the surface $z = 3$ at P . Utilizing the fact that the z -axis is normal to the surface $z = 3$, the angle between the z -axis and $\mathbf{E}$, as shown in the following figure, can be found using the dot product:

$$\mathbf{E} \cdot \mathbf{a}_z = |\mathbf{E}|(1) \cos \theta_{Ez} \rightarrow 3 = \sqrt{134} \cos \theta_{Ez}$$

$$\cos \theta_{Ez} = \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98^\circ$$

Hence, the angle between $z = 3$ and $\mathbf{E}$ is:

$$90^\circ - \theta_{Ez} = 15.02^\circ$$



3. a. A unit vector along $\mathbf{H}$:

At P , $\rho = 2$, $\phi = 30^\circ$, $z = -1$

$$\mathbf{H} = -5 \mathbf{a}_\rho + 2 \cos 30^\circ \mathbf{a}_\phi - 6 \mathbf{a}_z$$

$$\mathbf{H} = -5 \mathbf{a}_\rho + 1.732 \mathbf{a}_\phi - 6 \mathbf{a}_z$$

$$\mathbf{a}_n = \frac{(-5, 1.732, -6)}{\sqrt{5^2 + 1.732^2 + 6^2}} = -0.625 \mathbf{a}_\rho + 0.2165 \mathbf{a}_\phi - 0.75 \mathbf{a}_z$$

The component of $\mathbf{H}$ parallel to $\mathbf{a}_x$:

$$H_x = H_\rho \cos \phi - H_\phi \sin \phi = -5 \rho \sin \phi \cos \phi - \rho z \cos \phi \sin \phi$$

Or

P at $\rho = 2$, $\phi = 30^\circ$, $z = -1$:]

$$\mathbf{H} = H_x \mathbf{a}_x = (-10 \sin 30^\circ \cos 30^\circ + 2 \sin 30^\circ \cos 30^\circ) \mathbf{a}_x$$

$$\mathbf{H} = -3.4641 \mathbf{a}_x$$

The component of $\mathbf{H}$ normal to $\rho = 2$:

$$\mathbf{H}_n = H_\rho \mathbf{a}_\rho = -0.625 \mathbf{a}_\rho$$

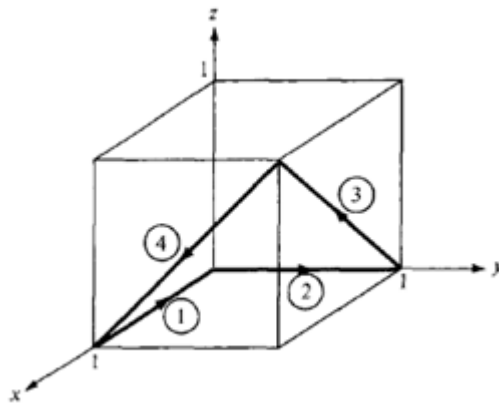
The component of $\mathbf{H}$ tangential to $\phi = 30^\circ$:

$$\mathbf{H}_t = H_\rho \mathbf{a}_\rho - H_z \mathbf{a}_z = -0.625 \mathbf{a}_\rho - 0.75 \mathbf{a}_z$$



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Applied Electromagnetics
Worksheet 3

1. Given that $\mathbf{F} = x^2 \mathbf{a}_x - xz \mathbf{a}_y - y^2 \mathbf{a}_z$, calculate the circulation of $\mathbf{F}$ around the (closed) path shown in the figure below:



2. Find the gradient of these scalar fields:
- $U = 4xz^2 + 3yz$
 - $W = 2\rho(z^2 + 1) \cos \phi$
 - $H = r^2 \cos \theta \cos \phi$
3. Find the divergence and curl of the following vectors:
- $\mathbf{A} = e^{xy} \mathbf{a}_x + \sin xy \mathbf{a}_y + \cos^2 xz \mathbf{a}_z$
 - $\mathbf{B} = \rho z^2 \cos \phi \mathbf{a}_\rho + z \sin^2 \phi \mathbf{a}_z$
 - $\mathbf{C} = r \cos \theta \mathbf{a}_r - \frac{1}{r} \sin \theta \mathbf{a}_\theta + 2r^2 \sin \theta \mathbf{a}_\phi$

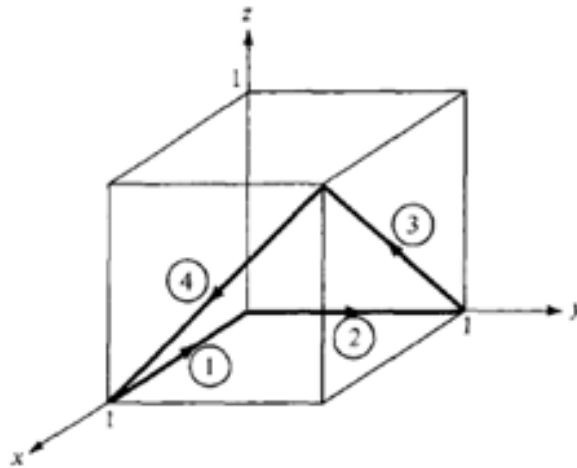


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Worksheet 3-Solutions

1. The circulation of $\mathbf{F}$ around path L is given by:

$$\oint_L \mathbf{F} \cdot d\mathbf{l} = \left(\int_1 + \int_2 + \int_3 + \int_4 \right) \mathbf{F} \cdot d\mathbf{l}$$

where the path is broken into segments numbered 1 to 4 as shown in the figure below.



For segment 1: $y = 0 = z$, $\mathbf{F} = x^2 \mathbf{a}_x$

and

$$d\mathbf{l} = dx \mathbf{a}_x$$

Notice that $d\mathbf{l}$ is always taken as along $+\mathbf{a}_x$ so that the direction on segment 1 is taken care of by the limits of integration. Thus,

$$\int_1 \mathbf{F} \cdot d\mathbf{l} = \int_1^0 x^2 dx = \left. \frac{x^3}{3} \right|_1^0 = -\frac{1}{3}$$

For segment 2, $x = 0 = z$:

$$\mathbf{F} = -y^2 \mathbf{a}_z$$

and

$$d\mathbf{l} = dy \mathbf{a}_y$$

$$\mathbf{F} \cdot d\mathbf{l} = 0$$

hence:

$$\int_2 \mathbf{F} \cdot d\mathbf{l} = 0$$

For segment 3, $y = 1$,

$$\mathbf{F} = x^2 \mathbf{a}_x - xz \mathbf{a}_y - \mathbf{a}_z$$

and

$$d\mathbf{l} = dx \mathbf{a}_x + dz \mathbf{a}_z$$

$$\int_3 \mathbf{F} \cdot d\mathbf{l} = \int (x^2 dx - dz)$$

But on 3, $z = x$,

$$dx = dz$$

hence:

$$\int_3 \mathbf{F} \cdot d\mathbf{l} = \int_0^1 (x^2 - 1) dx = \left. \frac{x^3}{3} - x \right|_0^1 = -\frac{2}{3}$$

For segment 4, $x = 1$,

$$\mathbf{F} = \mathbf{a}_x - z \mathbf{a}_y - y^2 \mathbf{a}_z$$

and

$$d\mathbf{l} = dy \mathbf{a}_y + dz \mathbf{a}_z$$

hence:

$$\int_4 \mathbf{F} \cdot d\mathbf{l} = \int (-z dy - y^2 dz)$$

But on 4, $z = y$,

$$dz = dy$$

so:

$$\int_3 \mathbf{F} \cdot d\mathbf{l} = \int_1^0 (-y - y^2) dy = -\frac{y^2}{2} - \frac{y^3}{3} \Big|_1^0 = \frac{5}{6}$$

By putting all these together, we obtain:

$$\oint_L \mathbf{F} \cdot d\mathbf{l} = -\frac{1}{3} + 0 - \frac{2}{3} + \frac{5}{6} = -\frac{1}{6}$$

2. a.

$$\bar{\nabla} U = \frac{\partial U}{\partial x} \mathbf{a}_x + \frac{\partial U}{\partial y} \mathbf{a}_y + \frac{\partial U}{\partial z} \mathbf{a}_z$$

$$\bar{\nabla} U = 4z^2 \mathbf{a}_x + 3z \mathbf{a}_y + (8xz + 3y) \mathbf{a}_z$$

b.

$$\bar{\nabla} W = \frac{\partial W}{\partial \rho} \mathbf{a}_\rho + \frac{\partial W}{\partial \phi} \mathbf{a}_\phi + \frac{\partial W}{\partial z} \mathbf{a}_z$$

$$\bar{\nabla} W = 2(z^2 + 1) \cos \phi \mathbf{a}_\rho - 2(z^2 + 1) \sin \phi \mathbf{a}_\phi + 4\rho z \cos \phi \mathbf{a}_z$$

c.

$$\bar{\nabla} H = \frac{\partial H}{\partial r} \mathbf{a}_r + \frac{\partial H}{\partial \theta} \mathbf{a}_\theta + \frac{\partial H}{\partial \phi} \mathbf{a}_\phi$$

$$\bar{\nabla} \mathbf{H} = 2r \cos \theta \cos \phi \mathbf{a}_r - r \sin \theta \cos \phi \mathbf{a}_\theta - r \cot \theta \sin \phi \mathbf{a}_\phi$$

3. a.

$$\nabla \cdot \mathbf{A} = ye^{xy} + x \cos xy - 2x \cos xz \sin xz$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xy} & \sin xy & \cos^2 xz \end{vmatrix}$$

$$\nabla \times \mathbf{A} = (0 - 0) \mathbf{a}_x + (0 + 2xz \cos xz \sin xz) \mathbf{a}_y + (y \cos xy - xe^{xy}) \mathbf{a}_z$$

$$\nabla \times \mathbf{A} = z \sin 2xz \mathbf{a}_y + (y \cos xy - xe^{xy}) \mathbf{a}_z$$

b.

$$\nabla \cdot \mathbf{B} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho^2 z^2 \cos \phi) + 0 + \sin^2 \phi$$

$$\nabla \cdot \mathbf{B} = 2z^2 \cos \phi + \sin^2 \phi$$

$$\nabla \times \mathbf{B} = \left(\frac{1}{\rho} \frac{\partial B_z}{\partial \phi} - 0 \right) \mathbf{a}_\rho + \left(\frac{\partial B_\rho}{\partial z} - \frac{\partial B_z}{\partial \rho} \right) \mathbf{a}_\phi + \frac{1}{\rho} \left(0 - \frac{\partial B_\rho}{\partial \phi} \right) \mathbf{a}_z$$

$$\nabla \times \mathbf{B} = \frac{z \sin 2\phi}{\rho} \mathbf{a}_\rho + 2\rho z \cos \phi \mathbf{a}_\phi + z^2 \sin \phi \mathbf{a}_z$$

c.

$$\nabla \cdot \mathbf{C} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^3 \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(-\frac{1}{r} \sin^2 \phi \right) + 0$$

$$\nabla \cdot \mathbf{C} = 3 \cos \theta - \frac{2 \cos \theta}{r^2}$$

$$\begin{aligned} \nabla \times \mathbf{C} &= \frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (2r^2 \sin^2 \theta) - 0 \right) \mathbf{a}_r \\ &+ \frac{1}{r} \left(0 - \frac{\partial}{\partial r} (2r^3 \sin \theta) \right) \mathbf{a}_\theta + \frac{1}{r} \left(\frac{\partial}{\partial r} (-\sin \theta) + r \sin \theta \right) \mathbf{a}_\phi \end{aligned}$$

$$\nabla \times \mathbf{C} = 4r \cos \theta \mathbf{a}_r - 6r \sin \theta \mathbf{a}_\theta + \sin \theta \mathbf{a}_\phi$$



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Worksheet 4

1. A circular ring of radius a carries a uniform charge ρ_L C/m and is placed on the xy -plane with axis the same as the z -axis. Show that:

$$\mathbf{E}(0, 0, h) = \frac{\rho_L a h}{2\epsilon_0 [h^2 + a^2]^{3/2}} \mathbf{a}_z$$

- What values of h gives the maximum value of $\mathbf{E}$?
- If the total charge on the ring is Q , find $\mathbf{E}$ as $a \rightarrow 0$.

2. For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_0(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

- Find $\mathbf{E}$ and V for $r \geq a$.
- Find $\mathbf{E}$ and V for $r \leq a$.
- Find the total charge.
- Show that $\mathbf{E}$ is maximum when $r = 0.145 a$.



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Applied Electromagnetics
Worksheet 4-Solutions

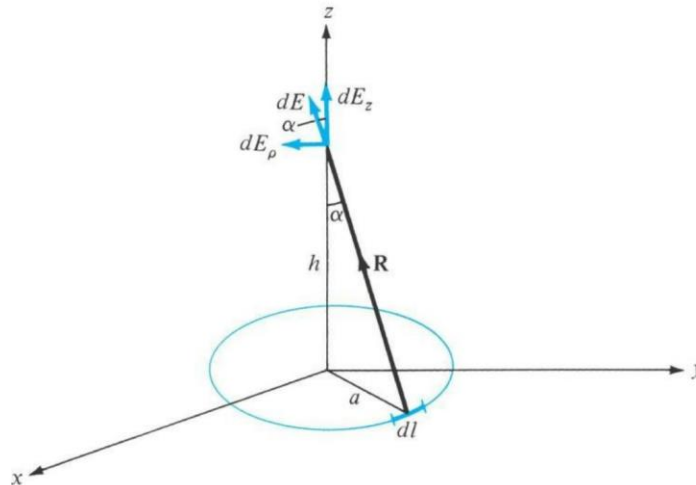
1. A circular ring of radius a carries a uniform charge ρ_L C/m and is placed on the xy -plane with axis the same as the z -axis. Show that:

$$\mathbf{E}(0, 0, h) = \frac{\rho_L a h}{2\epsilon_o [h^2 + a^2]^{3/2}} \mathbf{a}_z$$

- a. What values of h give the maximum value of $\mathbf{E}$?
- b. If the total charge on the ring is Q , find $\mathbf{E}$ as $a \rightarrow 0$.

Solution

Consider the system as shown in the following figure.



Using the equation:

$$\mathbf{E} = \int \frac{\rho_L dl}{4\pi\epsilon_o R^2} \mathbf{a}_R$$

and deriving each term in the equation. In this case:

$$d\mathbf{l} = a d\phi \mathbf{a}_\phi$$

$$\mathbf{R} = a(-\mathbf{a}_\rho) + h \mathbf{a}_z$$

$$R = |\mathbf{R}| = [a^2 + h^2]^{1/2}$$

$$\mathbf{a}_R = \frac{\mathbf{R}}{R}$$

$$\frac{\mathbf{a}_R}{R^2} = \frac{\mathbf{R}}{|\mathbf{R}|^3} = \frac{-a \mathbf{a}_\rho + h \mathbf{a}_z}{[a^2 + h^2]^{3/2}}$$

Hence:

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_o} \int_{\phi=0}^{2\pi} \frac{(-a\mathbf{a}_\rho + h\mathbf{a}_z)}{[a^2 + h^2]^{3/2}} a d\phi$$

By symmetry, the contributions along $\mathbf{a}_\rho$ add up to zero. This is evident from the fact that for every element $d\mathbf{l}$ there is a corresponding element diametrically opposite to it that gives an equal but opposite dE_ρ so that the two contributions cancel each other. Thus, we are left with the z-component. That is:

$$\mathbf{E} = \frac{\rho_L a h \mathbf{a}_z}{4\pi\epsilon_o [a^2 + h^2]^{3/2}} \int_{\phi=0}^{2\pi} d\phi = \frac{\rho_L a h \mathbf{a}_z}{2\epsilon_o [a^2 + h^2]^{3/2}}$$

As required.

1.

$$\frac{d|E|}{dh} = \frac{\rho_L a}{2\epsilon_o} \left\{ \frac{[a^2 + h^2]^{\frac{3}{2}}(1) - \frac{3}{2}(h)2h[a^2 + h^2]^{1/2}}{[a^2 + h^2]^{3/2}} \right\}$$

For maximum $\mathbf{E}$,

$$\frac{d|E|}{dh} = 0$$

which implies that:

$$[a^2 + h^2]^{1/2}[h^2 + a^2 - 3h^2] = 0$$

$$a^2 - 2h^2 = 0$$

$$h = \pm \frac{a}{\sqrt{2}}$$

2. Since the charge is uniformly distributed, the line charge density:

$$\rho_L = \frac{Q}{2\pi a}$$

so that:

$$\mathbf{E} = \frac{Qh}{4\pi\epsilon_0[a^2 + h^2]^{3/2}} \mathbf{a}_z$$

As $a \rightarrow 0$:

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 h^2} \mathbf{a}_z$$

or in general:

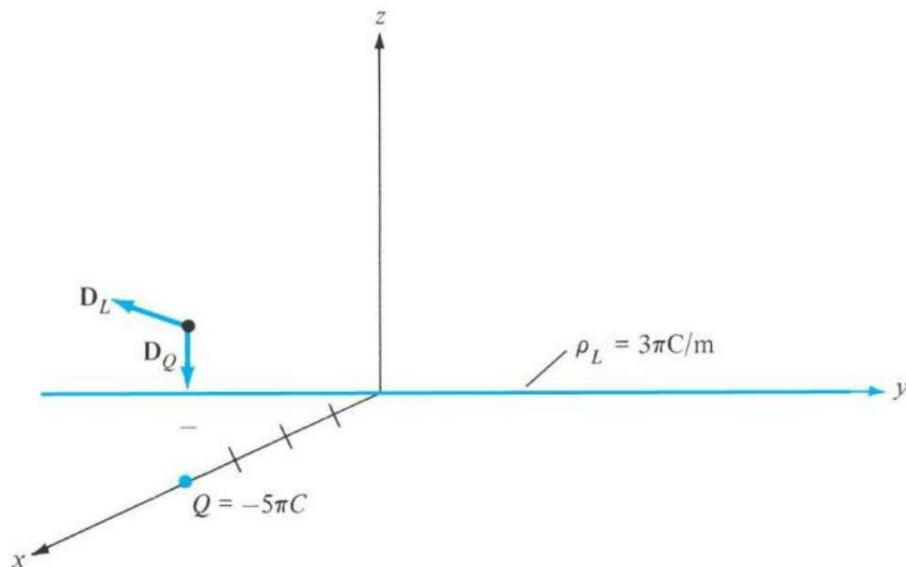
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_R$$

which is the same as that of a point charge, as one would expect.

2. Determine $\mathbf{D}$ at $(4, 0, 3)$ if there is a point charge -5π mC at $(4, 0, 0)$ and a line charge 3π mC/m along the y -axis.

Solution

Let $\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L$, where $\mathbf{D}_Q$ and $\mathbf{D}_L$ are flux densities due to the point charge and line charge, respectively, as shown in the figure:



$$\mathbf{D}_Q = \varepsilon_o \mathbf{E} = \frac{Q}{4\pi R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi |\mathbf{r} - \mathbf{r}'|^3}$$

where $\mathbf{r} - \mathbf{r}' = (4, 0, 3) - (4, 0, 0) = (0, 0, 3)$.

Hence:

$$\mathbf{D}_Q = \frac{-5\pi \cdot 10^{-3}(0, 0, 3)}{4\pi |(0, 0, 3)|^3} = -0.139 \mathbf{a}_z \text{ mC/m}^2$$

also:

$$\mathbf{D}_L = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho$$

In this case:

$$\mathbf{a}_\rho = \frac{(4, 0, 3) - (0, 0, 0)}{|(4, 0, 3) - (0, 0, 0)|} = \frac{(4, 0, 3)}{5}$$

$$\rho = |(4, 0, 3) - (0, 0, 0)| = 5$$

hence:

$$\mathbf{D}_L = \frac{3\pi}{2\pi(25)}(4 \mathbf{a}_x + 3 \mathbf{a}_z) = 0.24 \mathbf{a}_x + 0.18 \mathbf{a}_z \text{ mC/m}^2$$

Thus:

$$\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L = 240 \mathbf{a}_x + 41.1 \mathbf{a}_z \text{ } \mu\text{C/m}^2$$

3. For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_o(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

2. Find E and V for $r \geq a$.

3. Find E and V for $r \leq a$.

4. Find the total charge.

5. Show that E is maximum when $r = 0.745 a$.

Solution

For $r \geq a$:

$$Q_{enc} = \int \rho_v dv = \iiint \rho_o(a^2 - r^2)r^2 \sin \theta \, d\theta \, d\phi \, dr$$

$$Q_{enc} = \rho_o \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi \int_0^a (a^2 r^2 - r^4) \, dr$$

$$Q_{enc} = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^a = \frac{8\pi}{15} \rho_0$$

$$\Psi = \int \mathbf{D} \cdot d\mathbf{S} = \varepsilon_o E_r (4\pi r^2)$$

$$\Psi = Q_{enc}$$

$$\varepsilon_o E_r (4\pi r^2) = \frac{8\pi}{15} \rho_0$$

$$E_r = \frac{2\rho_0}{15\varepsilon_o r^2}$$

$$\mathbf{E} = \frac{2\rho_0}{15\varepsilon_o r^2} \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{2\rho_0}{15\varepsilon_o} \int r^{-2} \cdot dr = \frac{2\rho_0}{15\varepsilon_o r} + C_1$$

Since

$$V(r^- > 0) = 0$$

$$C_1 = 0$$

$$V = \frac{2\rho_0}{15\varepsilon_o r}$$

b. For $r \leq a$:

$$Q_{enc} = \rho_0 (4\pi) \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^r = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right)$$

$$E_r = \frac{Q_{enc}}{4\pi\varepsilon_o r^2} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\mathbf{E} = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right) \mathbf{a}_r$$

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r^2}{6} - \frac{r^4}{20} \right) + C_2$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + C_2$$

Since $V(r = a^+) = V(r = a^-)$:

$$\frac{2\rho_0}{15\varepsilon_o a} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^4}{20} - \frac{a^4}{6} \right) + C_2$$

$$C_2 = \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

$$V = \frac{\rho_0}{\varepsilon_o} \left(\frac{r^4}{20} - a^2 \frac{r^2}{6} \right) + \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

c. The total charge is found in part (a) as:

$$Q = \frac{8\pi}{15} \rho_0$$

d. For $r \geq a$, E decays to zero with no maxima. For $r \leq a$:

$$E_r = \frac{\rho_0}{\varepsilon_o} \left(a^2 \frac{r}{3} - \frac{r^3}{5} \right)$$

$$\frac{\partial E_r}{\partial r} = \frac{\rho_0}{\varepsilon_o} \left(\frac{a^2}{3} - \frac{3r^2}{5} \right) = 0$$

$$r = \frac{a\sqrt{5}}{3}$$

hence:

$$r = 0.7453 a$$



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Electromagnetics I
Worksheet 5

1. Given the potential:

$$V = \frac{10}{r^2} \sin \theta \cos \phi$$

- Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.
 - Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point A($1, 30^\circ, 120^\circ$) to B($4, 90^\circ, 60^\circ$).
2. Two dipoles with dipole moments $-5\mathbf{a}_z \text{ nC/m}$ and $9\mathbf{a}_z \text{ nC/m}$ are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.
3. Three point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, respectively. Find the energy in the system.



Princess Sumaya University for Technology
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Applied Electromagnetics
Worksheet 5-Solutions

1. Given the potential:

$$V = \frac{10}{r^2} \sin \theta \cos \phi$$

a. Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.

b. Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.

Solution

a. $\mathbf{D} = \epsilon_0 \mathbf{E}$

But

$$\mathbf{E} = -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \right]$$

$$\mathbf{E} = \frac{20}{r^3} \sin \theta \cos \phi \mathbf{a}_r - \frac{10}{r^3} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{10}{r^3} \sin \phi \mathbf{a}_\phi$$

At $(2, \pi/2, 0)$:

$$\mathbf{D} = \epsilon_0 \mathbf{E} \left(r = 2, \theta = \frac{\pi}{2}, \phi = 0 \right) = \epsilon_0 \left(\frac{20}{8} \mathbf{a}_r - 0 \mathbf{a}_\theta + 0 \mathbf{a}_\phi \right)$$

$$\mathbf{D} = 2.5 \epsilon_0 \mathbf{a}_r \text{ C/m}^2 = 22.1 \mathbf{a}_r \text{ pC/m}^2$$

b. The work done can be found in two ways, using either $\mathbf{E}$ or V .

Method 1:

$$W = -Q \int \mathbf{E} \cdot d\mathbf{l}$$

or

$$-\frac{W}{Q} = \int \mathbf{E} \cdot d\mathbf{l}$$

and because the electrostatic field is conservative, the integration path is immaterial. Hence, the work done in moving Q from $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$ is the same as that in moving Q from A to $\hat{A}$, from $\hat{A}$ to $\hat{B}$, and from $\hat{B}$ to B , where:

$$A(1, 30^\circ, 120^\circ) \xrightarrow{dl=dr\mathbf{a}_r} \hat{A}(4, 30^\circ, 120^\circ) \xrightarrow{dl=r d\theta\mathbf{a}_\theta} \hat{B}(4, 90^\circ, 60^\circ) \xrightarrow{dl=r \sin\theta d\phi\mathbf{a}_\phi} B(4, 90^\circ, 60^\circ)$$

That is, instead of moving Q directly from A and B , it is moved from $A \rightarrow \hat{A}$, $\hat{A} \rightarrow \hat{B}$, $\hat{B} \rightarrow B$ so that only one variable is changed at a time. This makes the line integral a lot easier to evaluate. Thus:

$$\begin{aligned} -\frac{W}{Q} &= -\frac{1}{Q} (W_{A\hat{A}} + W_{\hat{A}\hat{B}} + W_{\hat{B}B}) \\ -\frac{W}{Q} &= \left(\int_{A\hat{A}} + \int_{\hat{A}\hat{B}} + \int_{\hat{B}B} \right) \mathbf{E} \cdot d\mathbf{l} \\ -\frac{W}{Q} &= \int_{r=1}^4 \frac{20 \sin\theta \cos\phi}{r^3} dr \Big|_{\theta=30^\circ, \phi=120^\circ} + \int_{\theta=30^\circ}^{90^\circ} \frac{-10 \cos\theta \cos\phi}{r^3} r d\theta \Big|_{r=4, \phi=120^\circ} \\ &\quad + \int_{\phi=120^\circ}^{60^\circ} \frac{10 \sin\phi}{r^3} r \sin\theta d\phi \Big|_{r=4, \theta=90^\circ} \\ -\frac{W}{Q} &= 20 \left(\frac{1}{2} \right) \left(\frac{-1}{2} \right) \left[-\frac{1}{2r^2} \right]_{r=1}^4 - \frac{10(-1)}{16} \frac{1}{2} \sin\theta \Big|_{30^\circ}^{90^\circ} + \frac{10}{16} (1) [-\cos\phi]_{120^\circ}^{60^\circ} \\ -\frac{W}{Q} &= -\frac{75}{32} + \frac{5}{32} - \frac{10}{16} \end{aligned}$$

or

$$W = \frac{45}{16} Q = 28.125 \mu J$$

Method 2:

Since V is known, this method is a lot easier.

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} = QV_{AB}$$

$$W = Q(V_B - V_A) = 10 \left(\frac{10}{16} \sin 90^\circ \cos 60^\circ - \frac{10}{1} \sin 30^\circ \cos 120^\circ \right) \cdot 10^{-6}$$

$$W = 10 \left(\frac{10}{32} - \frac{-5}{2} \right) \cdot 10^{-6} = 28.125 \mu J$$

As obtained before.

2. Two dipoles with dipole moments $-5\mathbf{a}_z \text{ nC/m}$ and $9\mathbf{a}_z \text{ nC/m}$ are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.

Solution

$$V = \sum_{k=1}^2 \frac{p_k \cdot r_k}{4\pi\epsilon_o r_k^3} = \frac{1}{4\pi\epsilon_o} \left[\frac{p_1 \cdot r_1}{r_1^3} + \frac{p_2 \cdot r_2}{r_2^3} \right]$$

where:

$$p_1 = -5\mathbf{a}_z \quad r_1 = (0, 0, 0) - (0, 0, -2) = 2\mathbf{a}_z \quad r_1 = |r_1| = 2$$

$$p_2 = 9\mathbf{a}_z \quad r_2 = (0, 0, 0) - (0, 0, 3) = -3\mathbf{a}_z \quad r_2 = |r_2| = 3$$

hence:

$$V = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left[\frac{-10}{2^3} - \frac{27}{3^3} \right] \cdot 10^{-9} = -20.25 \text{ V}$$

3. Three point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, respectively. Find the energy in the system.

Solution

$$W = W_1 + W_2 + W_3 = 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32})$$

$$W = Q_2 \cdot \frac{Q_1}{4\pi\epsilon_o |(0,0,1) - (0,0,0)|} + \frac{Q_3}{4\pi\epsilon_o} \left[\frac{Q_1}{|(1,0,0) - (0,0,0)|} + \frac{Q_2}{|(1,0,0) - (0,0,1)|} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right)$$

$$W = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left(-4 - 3 + \frac{12}{\sqrt{2}} \right) \cdot 10^{-18}$$

$$W = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ}$$

Alternatively:

$$W = \frac{1}{2} \sum_{k=1}^3 Q_k V_k = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3)$$

$$W = \frac{Q_1}{2} \left[\frac{Q_2}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(1)} \right] + \frac{Q_2}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(\sqrt{2})} \right] + \frac{Q_3}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_2}{4\pi\epsilon_o(\sqrt{2})} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right) = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ}$$

as obtained previously.



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Worksheet 5-Solutions

1.

a. $\mathbf{D} = \varepsilon_o \mathbf{E}$

but:

$$\mathbf{E} = -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \right]$$

$$\mathbf{E} = \frac{20}{r^3} \sin \theta \cos \phi \mathbf{a}_r - \frac{10}{r^3} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{10}{r^3} \sin \phi \mathbf{a}_\phi$$

At $(2, \pi/2, 0)$:

$$\mathbf{D} = \varepsilon_o \mathbf{E} \left(r = 2, \theta = \frac{\pi}{2}, \phi = 0 \right) = \varepsilon_o \left(\frac{20}{8} \mathbf{a}_r - 0 \mathbf{a}_\theta + 0 \mathbf{a}_\phi \right)$$

$$\mathbf{D} = 2.5 \varepsilon_o \mathbf{a}_r \text{ C/m}^2 = 22.1 \mathbf{a}_r \text{ pC/m}^2$$

b. The work done can be found in two ways, using either $\mathbf{E}$ or V .

Method 1:

$$W = -Q \int \mathbf{E} \cdot d\mathbf{l}$$

or

$$-\frac{W}{Q} = \int \mathbf{E} \cdot d\mathbf{l}$$

and because the electrostatic field is conservative, the integration path is immaterial. Hence, the work done in moving Q from $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$ is the same as that in moving Q from A to $\hat{A}$, from $\hat{A}$ to $\hat{B}$, and from $\hat{B}$ to B where:

$$A(1, 30^\circ, 120^\circ) \xrightarrow{dl=dr\mathbf{a}_r} \hat{A}(4, 30^\circ, 120^\circ) \xrightarrow{dl=r d\theta\mathbf{a}_\theta} \hat{B}(4, 90^\circ, 60^\circ) \xrightarrow{dl=r \sin\theta d\phi\mathbf{a}_\phi} B(4, 90^\circ, 60^\circ)$$

That is, instead of moving Q directly from A and B , it is moved from $A \rightarrow \hat{A}$, $\hat{A} \rightarrow \hat{B}$, $\hat{B} \rightarrow B$ so that only one variable is changed at a time. This makes the line integral a lot easier to evaluate. Thus:

$$-\frac{W}{Q} = -\frac{1}{Q} (W_{A\hat{A}} + W_{\hat{A}\hat{B}} + W_{\hat{B}B})$$

$$-\frac{W}{Q} = \left(\int_{A\hat{A}} + \int_{\hat{A}\hat{B}} + \int_{\hat{B}B} \right) \mathbf{E} \cdot d\mathbf{l}$$

$$-\frac{W}{Q} = \int_{r=1}^4 \frac{20 \sin\theta \cos\phi}{r^3} dr \Big|_{\theta=30^\circ, \phi=120^\circ} + \int_{\theta=30^\circ}^{90^\circ} \frac{-10 \cos\theta \cos\phi}{r^3} r d\theta \Big|_{r=4, \phi=120^\circ} + \int_{\phi=120^\circ}^{60^\circ} \frac{10 \sin\phi}{r^3} r \sin\theta d\phi \Big|_{r=4, \theta=90^\circ}$$

$$-\frac{W}{Q} = 20 \left(\frac{1}{2} \right) \left(\frac{-1}{2} \right) \left[-\frac{1}{2r^2} \Big|_{r=1}^4 \right] - \frac{10(-1)}{16} \frac{1}{2} \sin\theta \Big|_{30^\circ}^{90^\circ} + \frac{10}{16} (1) [-\cos\phi]_{120^\circ}^{60^\circ}$$

$$-\frac{W}{Q} = -\frac{75}{32} + \frac{5}{32} - \frac{10}{16}$$

or

$$W = \frac{45}{16} Q = 28.125 \mu J$$

Method 2:

Since V is known, this method is a lot easier.

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} = QV_{AB}$$

$$W = Q(V_B - V_A) = 10 \left(\frac{10}{16} \sin 90^\circ \cos 60^\circ - \frac{10}{1} \sin 30^\circ \cos 120^\circ \right) \cdot 10^{-6}$$

$$W = 10 \left(\frac{10}{32} - \frac{-5}{2} \right) \cdot 10^{-6} = 28.125 \mu J$$

As obtained before.

2.

$$V = \sum_{k=1}^2 \frac{p_k \cdot r_k}{4\pi\epsilon_o r_k^3} = \frac{1}{4\pi\epsilon_o} \left[\frac{p_1 \cdot r_1}{r_1^3} + \frac{p_2 \cdot r_2}{r_2^3} \right]$$

where:

$$\begin{array}{lll} p_1 = -5 \mathbf{a}_z & r_1 = (0,0,0) - (0,0,-2) = 2 \mathbf{a}_z & r_1 = |r_1| = 2 \\ p_2 = 9 \mathbf{a}_z & r_2 = (0,0,0) - (0,0,3) = -3 \mathbf{a}_z & r_2 = |r_2| = 3 \end{array}$$

hence:

$$V = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left[\frac{-10}{2^3} - \frac{27}{3^3} \right] \cdot 10^{-9} = -20.25 V$$

3.

$$W = W_1 + W_2 + W_3 = 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32})$$

$$W = Q_2 \cdot \frac{Q_1}{4\pi\epsilon_o |(0,0,1) - (0,0,0)|} + \frac{Q_3}{4\pi\epsilon_o} \left[\frac{Q_1}{|(1,0,0) - (0,0,0)|} + \frac{Q_2}{|(1,0,0) - (0,0,1)|} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right)$$

$$W = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left(-4 - 3 + \frac{12}{\sqrt{2}} \right) \cdot 10^{-18}$$

$$W = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) nJ = 13.37 nJ$$

Alternatively:

$$W = \frac{1}{2} \sum_{k=1}^3 Q_k V_k = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3)$$

$$W = \frac{Q_1}{2} \left[\frac{Q_2}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(1)} \right] + \frac{Q_2}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(\sqrt{2})} \right] + \frac{Q_3}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_2}{4\pi\epsilon_o(\sqrt{2})} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right) = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) nJ = 13.37 \, nJ$$

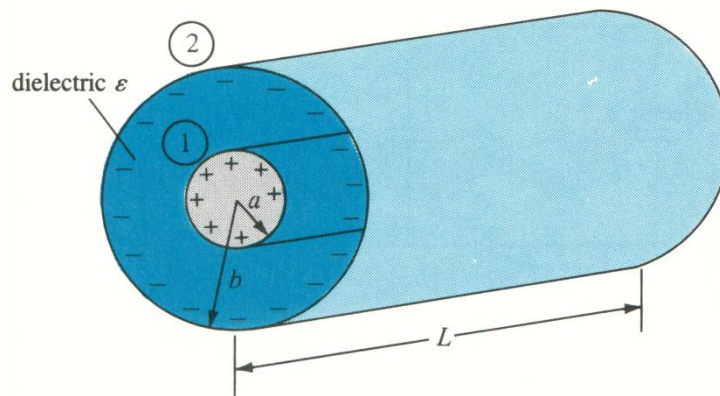
as obtained previously.



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 6

1. Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\epsilon_{r1} = 4$ and for $z \leq 0$, $\epsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$. Find:
 - a. $\mathbf{E}_2$ for $z \leq 0$.
 - b. The angles E_1 and E_2 make with the interface.
 - c. The energy densities in J/m³ in both dielectrics.
 - d. The energy within a cube of side 2 m centered at (3, 4, -5).

2. Consider the length L of two coaxial conductors of inner radius a and outer radius b ($b > a$) as shown in the following figure. Let the space between the conductors be filled with a homogeneous dielectric with permittivity ϵ . Find the capacitance of a coaxial cylinder.

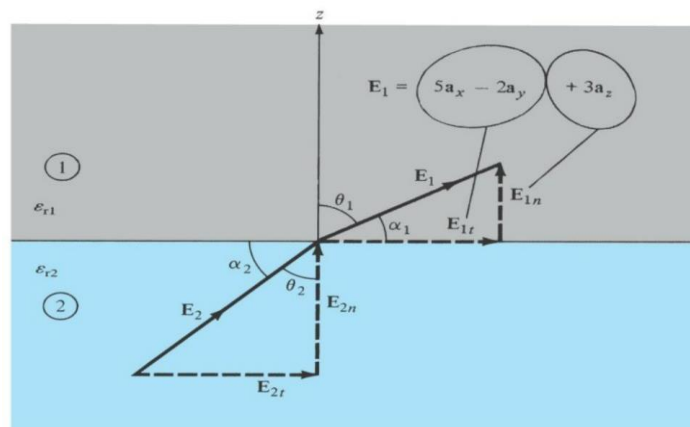


3. Conducting spherical shells with radii $a = 10$ cm and $b = 30$ cm are maintained at a potential difference of 100 V such that $V(r = b) = 0$ and $V(r = a) = 100$ V. Determine V and $\mathbf{E}$ in the region between the shells. If $\epsilon_r = 2.5$ in the region, determine the total charge induced on the shells and the capacitance of the system?



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Worksheet 6-Solutions

1. Let the problem be as illustrated in the following figure:



a. Since $\mathbf{a}_z$ is normal to the boundary plane, we obtain the normal components as:

$$E_{1n} = \mathbf{E}_1 \cdot \mathbf{a}_n = \mathbf{E}_1 \cdot \mathbf{a}_z = 3$$

$$\mathbf{E}_{1n} = 3\mathbf{a}_z$$

$$E_{2n} = (\mathbf{E}_2 \cdot \mathbf{a}_z) \mathbf{a}_z$$

also:

$$\mathbf{E} = \mathbf{E}_n + \mathbf{E}_t$$

hence:

$$\mathbf{E}_{1t} = \mathbf{E}_1 - \mathbf{E}_{1n} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

thus:

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

Similarly:

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \rightarrow \epsilon_{r2}\mathbf{E}_{2n} = \epsilon_{r1}\mathbf{E}_{1n}$$

or:

$$\mathbf{E}_{2n} = \frac{\epsilon_{r1}}{\epsilon_{r2}}\mathbf{E}_{1n} = \frac{4}{3}(3\mathbf{a}_z) = 4\mathbf{a}_z$$

thus:

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}$$

$$\mathbf{E}_2 = 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m}$$

b. Let α_1 and α_2 be the angle of $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface while θ_1 and θ_2 are the angles they make with the normal to the interface as shown in the previous figure; that is:

$$\alpha_1 = 90 - \theta_1$$

$$\alpha_2 = 90 - \theta_2$$

since:

$$E_{1n} = 3$$

and

$$E_{1t} = \sqrt{25 + 4} = \sqrt{29}$$

$$\tan \theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} = 1.795 \rightarrow \theta_1 = 60.9^\circ$$

hence:

$$\alpha_1 = 29.1^\circ$$

Alternatively:

$$\mathbf{E}_1 \cdot \mathbf{a}_n = |\mathbf{E}_1| \cdot 1 \cdot \cos \theta_1$$

or:

$$\cos \theta_1 = \frac{3}{\sqrt{38}} = 0.4867 \rightarrow \theta_1 = 60.9^\circ$$

Similarly:

$$E_{2n} = 4 \text{ and } E_{2t} = E_{1t} = \sqrt{29}$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \rightarrow \theta_2 = 53.4^\circ$$

hence:

$$\alpha_2 = 36.6^\circ$$

Note that:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$$

is satisfied.

c. The energy densities are given by:

$$w_{E1} = \frac{1}{2} \varepsilon_1 |\mathbf{E}_1|^2 = \frac{1}{2} \cdot 4 \cdot \frac{10^{-9}}{36\pi} \cdot (25 + 4 + 9) \cdot 10^6 = 672 \mu J/m^3$$

$$w_{E2} = \frac{1}{2} \varepsilon_2 |\mathbf{E}_2|^2 = \frac{1}{2} \cdot 3 \cdot \frac{10^{-9}}{36\pi} \cdot (25 + 4 + 16) \cdot 10^6 = 597 \mu J/m^3$$

d. At the center (3,4,-5) of the cube of side 2 m, $z = -5 < 0$; that is, the cube is in region 2 with $2 \leq x \leq 4$, $3 \leq y \leq 5$, $-6 \leq z \leq -4$. Hence:

$$W_E = \int w_{E2} dv = \int_{x=2}^4 \int_{y=3}^5 \int_{z=-6}^{-4} w_{E2} dx dy dz = w_{E2}(2)(2)(2) = 597 * 8 \mu J = 4.776 mJ$$

2. Assuming that conductors 1 and 2, respectively, carry $+Q$ and $-Q$ uniformly distributed on them. By applying Gauss's law to an arbitrary Gaussian cylindrical surface of radius ρ ($a < \rho < b$). Thus:

$$Q = \varepsilon \int \mathbf{E} \cdot d\mathbf{S} = \varepsilon E_{\rho} 2\pi\rho L$$

hence:

$$\mathbf{E} = \frac{Q}{2\pi\varepsilon\rho L} \mathbf{a}_{\rho}$$

Neglecting flux fringing at the cylinder ends:

$$V = - \int_a^b \mathbf{E} d\mathbf{l} = - \int_a^b \left[\frac{Q}{2\pi\varepsilon\rho L} \mathbf{a}_{\rho} \right] \cdot d\rho \mathbf{a}_{\rho}$$

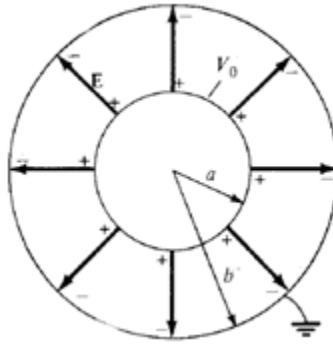
$$V = \frac{Q}{2\pi\varepsilon L} \ln \frac{b}{a}$$

Thus, the capacitance of a coaxial cylinder is given by:

$$C = \frac{Q}{V} = \frac{2\pi\varepsilon L}{\ln \frac{b}{a}}$$

3. Consider the spherical shells shown in the following figure. V depends only r and hence Laplace's equation becomes:

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dV}{dr} \right] = 0$$



Since $r \neq 0$ in the region of interest, we multiply through r^2 to obtain:

$$\frac{d}{dr} \left[r^2 \frac{dV}{dr} \right] = 0$$

integrating once gives:

$$r^2 \frac{dV}{dr} = A$$

or:

$$\frac{dV}{dr} = \frac{A}{r^2}$$

Integrating again gives:

$$V = -\frac{A}{r} + B$$

As usual, constants A and B are determined from the boundary conditions.

When:

$$r = b$$

$$V = 0 \rightarrow 0 = -\frac{A}{r} + B$$

or:

$$B = \frac{A}{b}$$

hence:

$$V = A \left[\frac{1}{b} - \frac{1}{r} \right]$$

also, when

$$r = a$$

$$V = V_o \rightarrow V_o = A \left[\frac{1}{b} - \frac{1}{a} \right]$$

or:

$$A = \frac{V_o}{\left[\frac{1}{b} - \frac{1}{a} \right]}$$

thus:

$$V = V_o \frac{\left[\frac{1}{r} - \frac{1}{b} \right]}{\frac{1}{a} - \frac{1}{b}}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dr} \mathbf{a}_r$$

$$\mathbf{E} = -\frac{A}{r^2} \mathbf{a}_r = \frac{V_o}{r^2 \left[\frac{1}{a} - \frac{1}{b} \right]} \mathbf{a}_r$$

$$Q = \int \varepsilon \mathbf{E} \cdot d\mathbf{S} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{\varepsilon_o \varepsilon_r V_o}{r^2 \left[\frac{1}{a} - \frac{1}{b} \right]} r^2 \sin \theta \, d\phi d\theta$$

$$Q = \frac{4\pi\epsilon_o\epsilon_r V_o}{\frac{1}{a} - \frac{1}{b}}$$

The capacitance is easily determined as:

$$C = \frac{Q}{V_o} = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$$

Substituting $a = 0.1 \text{ m}$, $b = 0.3 \text{ m}$, $V_o = 100 \text{ V}$ yields:

$$V = 100 \frac{\left[\frac{1}{r} - \frac{10}{3}\right]}{10 - \frac{10}{3}} = 15 \left[\frac{1}{r} - \frac{10}{3}\right] \text{ V}$$

Check:

$$\nabla^2 V = 0$$

$$V(r = 0.3\text{m}) = 0$$

$$V(r = 0.1\text{m}) = 100$$

$$\mathbf{E} = \frac{100}{r^2 \left[10 - \frac{10}{3}\right]} \mathbf{a}_r = \frac{15}{r^2} \mathbf{a}_r \text{ V/m}$$

$$Q = \pm 4\pi \cdot \frac{10^{-9}}{36\pi} \cdot \frac{(2.5) \cdot (100)}{10 - \frac{10}{3}} = \pm 4.167 \text{ nC}$$

The positive charge is induced on the inner shell; the negative charge is induced on the outer shell. Also:

$$C = \frac{|Q|}{V_o} = \frac{4.167 \times 10^{-9}}{100} = 41.67 \text{ pF}$$



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 7

1. State the Biot-Savart's law and then derive an expression for the magnetic field intensity $\mathbf{H}$ at the field point P due to a straight current-carrying filamentary conductor of finite length AB?
2. Find the magnetic field intensity $\mathbf{H}$ at the field point P (-3, 4, 0) due to current-carrying filamentary conductor as shown in Fig. (1).

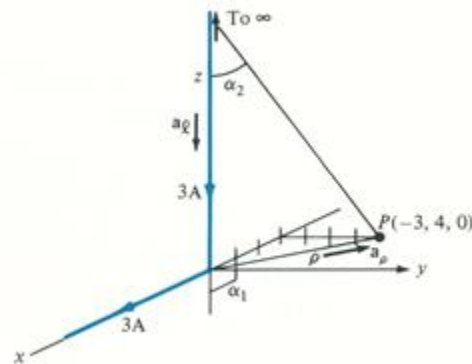


Fig. (1). A current filament along the semi-infinite x- and z- axes.

3. Derive an expression for the magnetic field $\mathbf{H}$ at a point P (0, 0, h) due to a circular loop of radius a and carrying a current I. Then find $\mathbf{H}$ at (0, 0, 1) cm if $a=5\text{ cm}$ and $I=50\text{ mA}$ if the ring is placed at $z=1$ plane?



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 8

1. Find the force $\mathbf{F}$ on a rectangular loop $a \times b$ carrying a current I_2 placed parallel to an infinitely long wire carrying a current I_1 at a distant ρ_0 from the near side? Find $\mathbf{F}$ if $I_1 = 10 \text{ A}$, $I_2 = 5 \text{ A}$, $a = 10 \text{ cm}$, $b = 30 \text{ cm}$ and $\rho_0 = 20 \text{ cm}$?
2. Determine the self-inductance of a coaxial cable of inner radius a and radius b ?
3. A toroid of radius $\rho_0 = 10 \text{ cm}$ and a circular cross section $a = 1 \text{ cm}$ is made of steel ($\mu_r = 1000$), and a coil of 200 turns. Calculate the current I required to produce a flux of 0.5 Wb in the core.

MATHEMATICAL FORMULAS

A.1 TRIGONOMETRIC IDENTITIES

$$\tan A = \frac{\sin A}{\cos A}, \quad \cot A = \frac{1}{\tan A}$$

$$\sec A = \frac{1}{\cos A}, \quad \csc A = \frac{1}{\sin A}$$

$$\sin^2 A + \cos^2 A = 1, \quad 1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \csc^2 A$$

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$2 \sin A \sin B = \cos (A - B) - \cos (A + B)$$

$$2 \sin A \cos B = \sin (A + B) + \sin (A - B)$$

$$2 \cos A \cos B = \cos (A + B) + \cos (A - B)$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos (A \pm 90^\circ) = \mp \sin A$$

$$\sin (A \pm 90^\circ) = \pm \cos A$$

$$\tan (A \pm 90^\circ) = -\cot A$$

$$\cos (A \pm 180^\circ) = -\cos A$$

$$\sin (A \pm 180^\circ) = -\sin A$$

$$\tan(A \pm 180^\circ) = \tan A$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan(A \pm B) = \frac{\tan A \pm B}{1 \mp \tan A \tan B}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin A = \frac{e^{jA} - e^{-jA}}{2j}, \quad \cos A = \frac{e^{jA} + e^{-jA}}{2}$$

$$e^{jA} = \cos A + j \sin A \quad (\text{Euler's identity})$$

$$\pi = 3.1416$$

$$1 \text{ rad} = 57.296^\circ$$

A.2 COMPLEX VARIABLES

A complex number may be represented as

$$z = x + jy = r \angle \theta = re^{j\theta} = r(\cos \theta + j \sin \theta)$$

$$\text{where } x = \operatorname{Re} z = r \cos \theta, \quad y = \operatorname{Im} z = r \sin \theta$$

$$r = |z| = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1} \frac{y}{x}$$

$$j = \sqrt{-1}, \quad \frac{1}{j} = -j, \quad j^2 = -1$$

The complex conjugate of $z = z^* = x - jy = r \angle -\theta = re^{-j\theta}$

$$= r(\cos \theta - j \sin \theta)$$

$$(e^{j\theta})^n = e^{jn\theta} = \cos n\theta + j \sin n\theta \quad (\text{de Moivre's theorem})$$

If $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$, then $z_1 = z_2$ only if $x_1 = x_2$ and $y_1 = y_2$.

$$z_1 \pm z_2 = (x_1 + x_2) \pm j(y_1 + y_2)$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1)$$

or

$$z_1 z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)} = r_1 r_2 \angle \theta_1 + \theta_2$$

$$\frac{z_1}{z_2} = \frac{(x_1 + jy_1)}{(x_2 + jy_2)} \cdot \frac{(x_2 - jy_2)}{(x_2 - jy_2)} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + j \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}$$

or

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)} = \frac{r_1}{r_2} \angle \theta_1 - \theta_2$$

$$\sqrt{z} = \sqrt{x + jy} = \sqrt{r} e^{j\theta/2} = \sqrt{r} \angle \theta/2$$

$$z^n = (x + jy)^n = r^n e^{jn\theta} = r^n \angle n\theta \quad (n = \text{integer})$$

$$z^{1/n} = (x + jy)^{1/n} = r^{1/n} e^{j\theta/n} = r^{1/n} \angle \theta/n + 2\pi k/n \quad (k = 0, 1, 2, \dots, n-1)$$

$$\ln(re^{j\theta}) = \ln r + \ln e^{j\theta} = \ln r + j\theta + j2k\pi \quad (k = \text{integer})$$

A.3 HYPERBOLIC FUNCTIONS

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}, \quad \coth x = \frac{1}{\tanh x}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}, \quad \operatorname{sech} x = \frac{1}{\cosh x}$$

$$\sin jx = j \sinh x, \quad \cos jx = \cosh x$$

$$\sinh jx = j \sin x, \quad \cosh jx = \cos x$$

$$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$$

$$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$$

$$\sinh(x \pm jy) = \sinh x \cos y \pm j \cosh x \sin y$$

$$\cosh(x \pm jy) = \cosh x \cos y \pm j \sinh x \sin y$$

$$\tanh(x \pm jy) = \frac{\sinh 2x}{\cosh 2x + \cos 2y} \pm j \frac{\sin 2y}{\cosh 2x + \cos 2y}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\operatorname{sech}^2 x + \tanh^2 x = 1$$

$$\sin(x \pm jy) = \sin x \cosh y \pm j \cos x \sinh y$$

$$\cos(x \pm jy) = \cos x \cosh y \mp j \sin x \sinh y$$

A.4 LOGARITHMIC IDENTITIES

$$\log xy = \log x + \log y$$

$$\log \frac{x}{y} = \log x - \log y$$

$$\log x^n = n \log x$$

$$\log_{10} x = \log x \text{ (common logarithm)}$$

$$\log_e x = \ln x \text{ (natural logarithm)}$$

$$\text{If } |x| \ll 1, \ln(1 + x) \approx x$$

A.5 EXPONENTIAL IDENTITIES

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\text{where } e \approx 2.7182$$

$$e^x e^y = e^{x+y}$$

$$[e^x]^n = e^{nx}$$

$$\ln e^x = x$$

A.6 APPROXIMATIONS FOR SMALL QUANTITIES

$$\text{If } |x| \ll 1,$$

$$(1 \pm x)^n \approx 1 \pm nx$$

$$e^x \approx 1 + x$$

$$\ln(1 + x) \approx x$$

$$\sin x \approx x \quad \text{or} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\cos x \approx 1$$

$$\tan x \approx x$$

A.7 DERIVATIVES

If $U = U(x)$, $V = V(x)$, and $a = \text{constant}$,

$$\frac{d}{dx}(aU) = a \frac{dU}{dx}$$

$$\frac{d}{dx}(UV) = U \frac{dV}{dx} + V \frac{dU}{dx}$$

$$\frac{d}{dx} \left[\frac{U}{V} \right] = \frac{V \frac{dU}{dx} - U \frac{dV}{dx}}{V^2}$$

$$\frac{d}{dx}(aU^n) = naU^{n-1} \frac{dU}{dx}$$

$$\frac{d}{dx} \log_a U = \frac{\log_a e}{U} \frac{dU}{dx}$$

$$\frac{d}{dx} \ln U = \frac{1}{U} \frac{dU}{dx}$$

$$\frac{d}{dx} a^U = a^U \ln a \frac{dU}{dx}$$

$$\frac{d}{dx} e^U = e^U \frac{dU}{dx}$$

$$\frac{d}{dx} U^V = VU^{V-1} \frac{dU}{dx} + U^V \ln U \frac{dV}{dx}$$

$$\frac{d}{dx} \sin U = \cos U \frac{dU}{dx}$$

$$\frac{d}{dx} \cos U = -\sin U \frac{dU}{dx}$$

$$\frac{d}{dx} \tan U = \sec^2 U \frac{dU}{dx}$$

$$\frac{d}{dx} \sinh U = \cosh U \frac{dU}{dx}$$

$$\frac{d}{dx} \cosh U = \sinh U \frac{dU}{dx}$$

$$\frac{d}{dx} \tanh U = \text{sech}^2 U \frac{dU}{dx}$$

A.8 INDEFINITE INTEGRALS

If $U = U(x)$, $V = V(x)$, and $a = \text{constant}$,

$$\int a \, dx = ax + C$$

$$\int U \, dV = UV - \int V \, dU \quad (\text{integration by parts})$$

$$\int U^n \, dU = \frac{U^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{dU}{U} = \ln U + C$$

$$\int a^U \, dU = \frac{a^U}{\ln a} + C, \quad a > 0, a \neq 1$$

$$\int e^U \, dU = e^U + C$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$$

$$\int x e^{ax} \, dx = \frac{e^{ax}}{a^2} (ax - 1) + C$$

$$\int x^2 e^{ax} \, dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2) + C$$

$$\int \ln x \, dx = x \ln x - x + C$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax + C$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax + C$$

$$\int \tan ax \, dx = \frac{1}{a} \ln \sec ax + C = -\frac{1}{a} \ln \cos ax + C$$

$$\int \sec ax \, dx = \frac{1}{a} \ln (\sec ax + \tan ax) + C$$

$$\int \sin^2 ax \, dx = \frac{x}{2} - \frac{\sin 2ax}{4a} + C$$

$$\int \cos^2 ax \, dx = \frac{x}{2} + \frac{\sin 2ax}{4a} + C$$

$$\int x \sin ax \, dx = \frac{1}{a^2} (\sin ax - ax \cos ax) + C$$

$$\int x \cos ax \, dx = \frac{1}{a^2} (\cos ax + ax \sin ax) + C$$

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

$$\int \sin ax \sin bx \, dx = \frac{\sin (a - b)x}{2(a - b)} - \frac{\sin (a + b)x}{2(a + b)} + C, \quad a^2 \neq b^2$$

$$\int \sin ax \cos bx \, dx = -\frac{\cos (a - b)x}{2(a - b)} - \frac{\cos (a + b)x}{2(a + b)} + C, \quad a^2 \neq b^2$$

$$\int \cos ax \cos bx \, dx = \frac{\sin (a - b)x}{2(a - b)} + \frac{\sin (a + b)x}{2(a + b)} + C, \quad a^2 \neq b^2$$

$$\int \sinh ax \, dx = \frac{1}{a} \cosh ax + C$$

$$\int \cosh ax \, dx = \frac{1}{a} \sinh ax + C$$

$$\int \tanh ax \, dx = \frac{1}{a} \ln \cosh ax + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$\int \frac{x dx}{x^2 + a^2} = \frac{1}{2} \ln (x^2 + a^2) + C$$

$$\int \frac{x^2 dx}{x^2 + a^2} = x - a \tan^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{x^2 - a^2} = \begin{cases} \frac{1}{2a} \ln \frac{x-a}{x+a} + C, & x^2 > a^2 \\ \frac{1}{2a} \ln \frac{a-x}{a+x} + C, & x^2 < a^2 \end{cases}$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln(x + \sqrt{x^2 \pm a^2}) + C$$

$$\int \frac{xdx}{\sqrt{x^2 + a^2}} = \sqrt{x^2 + a^2} + C$$

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x/a^2}{\sqrt{x^2 + a^2}} + C$$

$$\int \frac{xdx}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}} + C$$

$$\int \frac{x^2 dx}{(x^2 + a^2)^{3/2}} = \ln \left(\frac{\sqrt{x^2 + a^2}}{a} + \frac{x}{a} \right) - \frac{x}{\sqrt{x^2 + a^2}} + C$$

$$\int \frac{dx}{(x^2 + a^2)^2} = \frac{1}{2a^2} \left(\frac{x}{x^2 + a^2} + \frac{1}{a} \tan^{-1} \frac{x}{a} \right) + C$$

A.9 DEFINITE INTEGRALS

$$\int_0^\pi \sin mx \sin nx \, dx = \int_0^\pi \cos mx \cos nx \, dx = \begin{cases} 0, & m \neq n \\ \pi/2, & m = n \end{cases}$$

$$\int_0^\pi \sin mx \cos nx \, dx = \begin{cases} 0, & m + n = \text{even} \\ \frac{2m}{m^2 - n^2}, & m + n = \text{odd} \end{cases}$$

$$\int_0^{2\pi} \sin mx \sin nx \, dx = \int_{-\pi}^\pi \sin mx \sin nx \, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

$$\int_0^\infty \frac{\sin ax}{x} \, dx = \begin{cases} \pi/2, & a > 0, \\ 0, & a = 0 \\ -\pi/2, & a < 0 \end{cases}$$

$$\int_0^\infty \frac{\sin^2 x}{x} \, dx = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\sin^2 ax}{x^2} dx = |a| \frac{\pi}{2}$$

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} e^{-(ax^2+bx+c)} dx = \sqrt{\frac{\pi}{a}} e^{(b^2-4ac)/4a}$$

$$\int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2}$$

$$\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

A.10 VECTOR IDENTITIES

If **A** and **B** are vector fields while *U* and *V* are scalar fields, then

$$\nabla(U + V) = \nabla U + \nabla V$$

$$\nabla(UV) = U \nabla V + V \nabla U$$

$$\nabla \left[\frac{U}{V} \right] = \frac{V(\nabla U) - U(\nabla V)}{V^2}$$

$$\nabla V^n = n V^{n-1} \nabla V \quad (n = \text{integer})$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A} + \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A})$$

$$\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \cdot (V\mathbf{A}) = V \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla V$$

$$\nabla \cdot (\nabla V) = \nabla^2 V$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A} (\nabla \cdot \mathbf{B}) - \mathbf{B} (\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B}$$

$$\nabla \times (V\mathbf{A}) = \nabla V \times \mathbf{A} + V(\nabla \times \mathbf{A})$$

$$\nabla \times (\nabla V) = 0$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{A} \cdot d\mathbf{S}$$

$$\oint_L V d\mathbf{l} = - \int_S \nabla V \times d\mathbf{S}$$

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{A} dv$$

$$\oint_S V d\mathbf{S} = \int_v \nabla V dv$$

$$\oint_S \mathbf{A} \times d\mathbf{S} = - \int_v \nabla \times \mathbf{A} dv$$